Universität Bielefeld

Technische Fakultät

Masterarbeit

Intelligent Interactive Systems / Master of Science

Pretraining for Efficient Reinforcement Learning-Based Chlorination Control in Drinking Water Distribution Systems

vorgelegt von

Amadeus Beckmann

1. Gutachter: Dr. André Artelt
2. Gutachter: Paul J. Stahlhofen, M. Sc.

Bielefeld, 7. Oktober 2026

Abstract

Adaptive chlorination in drinking-water distribution networks must maintain suitable residual concentrations while responding to contamination. Delayed transport and changing demand complicate control when measurements do not reveal the complete network state. Reinforcement learning (RL) offers a way to learn feedback control, but training from scratch requires repeated environment interaction. This thesis examines whether model predictive control (MPC)-guided pretraining improves the online sample efficiency and final physical control performance of a recurrent chlorination policy.

A multistep long short-term memory (LSTM) forecaster supports an offline MPC teacher. Five-fold scenario-wise cross-fitting generates teacher actions with forecasters trained on other hydraulic scenarios. Behavior cloning uses these actions to initialize a recurrent actor, which is then fine-tuned with recurrent proximal policy optimization (RecurrentPPO). Forecasting and MPC are confined to offline preparation. The deployed policy receives only chlorine measurements, its own dosing, and time records. The comparison uses the simulated Hanoi network with 31 monitored demand nodes, varying hydraulic conditions, and arsenite (As(III)) contamination. Five matched pairs of training runs compare this initialization with training from scratch under a common nominal budget of one million online interactions.

The pretrained approach achieved higher mean validation return at every evaluated checkpoint and first reached the baseline's one-million-interaction endpoint return at 300 000 nominal interactions. After both approaches completed the common budget, policies selected by validation return were evaluated on 100 separate test cases. The selected pretrained policies had 17.9 % lower mean transported As(III) burden and reached persistent clearance about 25 minutes earlier on average in events where both policies exceeded the clearance threshold. Both differences favored pretraining in all five seed pairs. Chlorine use increased by 1.6 %, and the normal-operation chlorine excess fraction increased by 0.108 percentage points. An exploratory analysis located almost all additional excess within 12 h after clearance in the zero-chlorine reference runs. The shortfall difference remained uncertain, and neither approach exceeded the 5.0 mg/L event-time chlorine ceiling in applicable observations.

With fresh preparation, the summed stage runtimes across all five seeds, including shared trajectory generation and control evaluation, were about 4.8 times the baseline total. Pretraining also required 2.4 million offline simulation steps for training trajectories. The results support improved online sample efficiency and contamination treatment under the modeled As(III) oxidation reaction, at the cost of additional preparation and chlorine use.

Contents

1 Introduction 1
1.1 Motivation 1
1.2 Research Question 2
1.3 Contributions and Study Scope 2
1.4 Thesis Structure 2

2 Related Work 3

3 Foundations 4
3.1 Mathematical Foundations 5
3.1.1 Discrete-Time Dynamical Systems and Trajectories 5
3.1.2 Time-Series Representations 5
3.2 Sequential Decision-Making 6
3.2.1 Markov Decision Processes 6
3.2.2 Partial Observability 6
3.3 Hydraulic and Water-Quality Simulation 7
3.3.1 Water Distribution Networks and Hydraulic States 7
3.3.2 Solute Transport, Mixing, and Mass-Rate Sources 7
3.3.3 Free-Chlorine Decay and Chlorine–Arsenite Reactions 8
3.3.4 EPANET and EPANET-MSX 8
3.4 Neural Models for Sequential Data 8
3.4.1 Parameterized Models and Gradient-Based Learning 8
3.4.2 Recurrent Neural Networks and Backpropagation Through Time 9
3.4.3 Long Short-Term Memory Networks 9
3.5 Reinforcement Learning 10
3.5.1 Policies, Rewards, and Returns 10
3.5.2 Actor–Critic Learning 11
3.5.3 Proximal Policy Optimization 11
3.5.4 Recurrent Proximal Policy Optimization 13
3.6 Neural Time-Series Forecasting 13
3.6.1 Supervised Multistep Forecasting 13
3.6.2 Encoder–Decoder LSTMs 13
3.6.3 Forecasting Objectives and Error Measures 14
3.7 Predictive Control and Imitation Learning 14
3.7.1 Model Predictive Control 15
3.7.2 Cross-Entropy Method 15
3.7.3 Behavior Cloning and Transfer Learning 15

4 Dataset 16
4.1 Hydraulic Base Scenarios 16
4.1.1 Hanoi Water Distribution Network 17
4.1.2 Demand Patterns 18
4.1.3 Network Parameter Uncertainties 18
4.2 Arsenic Contamination Events 19
4.2.1 Event Scheduler 20
4.2.2 Event Parameter Sampling 21
4.3 Simulation Framework and Recorded Variables 21
4.4 Offline Dataset Construction 23
4.4.1 Chlorine Injection Profiles 23

4.4.2 Fixed Arsenic Schedule Sets ... 23
4.4.3 Trajectory Construction and Stored Variables ... 23
4.5 Online Interaction Data ... 23
4.6 Training, Validation, and Test Data ... 24
4.7 Reproducibility ... 25
4.8 Dataset Summary ... 25
5 Methodology ... 26
5.1 Chlorination-Control Problem ... 27
5.1.1 Operational Objectives and Partial Observability ... 27
5.1.2 Control Action and Actuator Constraints ... 28
5.2 Reward Function ... 28
5.2.1 Chlorine Tracking ... 29
5.2.2 Wasted Chlorine Movement ... 29
5.2.3 Action Change ... 30
5.2.4 Recovery Mechanism ... 31
5.2.5 Reward Components and Parameters ... 32
5.2.6 Parameter Selection and Optimization ... 33
5.3 Policy Observations ... 33
5.4 Recurrent Policy Learning and Baseline ... 35
5.4.1 RecurrentPPO Architecture and Baseline ... 35
5.4.2 Training Scenario and Contamination Sampling ... 36
5.4.3 Hyperparameter Optimization ... 37
5.5 Cross-Fitted Teacher Generation ... 37
5.5.1 Data Allocation and Scenario-Wise Cross-Fitting ... 37
5.5.2 History-Window and Forecast-Horizon Selection ... 39
5.5.3 Forecasting Data and Causal Inputs ... 39
5.5.4 Explicit Multistep Chlorine Forecaster ... 40
5.5.5 MPC Teacher and CEM Optimization ... 41
5.5.6 Query-State Sampling and Teacher-Dataset Construction ... 43
5.6 Behavior Cloning and PPO Fine-Tuning ... 44
5.6.1 Policy Initialization ... 44
5.6.2 Transfer of Actor and Normalization State ... 45
5.6.3 RecurrentPPO Fine-Tuning ... 45
5.7 Principal Experimental Comparison ... 45
5.7.1 Additional Comparison Variants ... 46
5.8 Implementation and Reproducibility ... 46
6 Evaluation ... 46
6.1 Evaluation Protocol ... 47
6.1.1 Episode Structure ... 47
6.1.2 Policy Comparison and Evaluation Procedure ... 47
6.2 Learning Curves and Sample Efficiency ... 48
6.3 Control Performance Metrics ... 49
6.3.1 Reference-Event Windows ... 49
6.3.2 Common Notation ... 50
6.3.3 Chlorine Shortfalls and Excesses ... 52
6.3.4 Arsenic Burden ... 52
6.3.5 Mean Arsenic Clearance Lead ... 52
6.3.6 Chlorine Ceiling Violation ... 53

6.3.7 Chlorine Use 53
6.3.8 Cross-Window Aggregation 54
6.4 Statistical Analysis 54
6.5 Training Time and Resource Requirements 54
6.6 Additional Variant Comparisons 55
6.7 Supporting Evaluation of the Forecasting Model 55
7 Results 55
7.1 Online Sample Efficiency 56
7.2 Final Physical Control Performance 57
7.2.1 Normal Operation 57
7.2.2 Contamination Response and Clearance 58
7.2.3 Chlorine Use 59
7.3 Computational Cost 60
7.4 Additional Variant Comparisons 60
7.5 Supporting Case-Level Evidence 61
7.6 Supporting Forecasting Results 63
8 Discussion 64
8.1 Answer to the Research Question 64
8.2 Online Learning and Final Control 64
8.3 Physical Control Trade-offs 65
8.4 Interaction Cost and Computational Cost 66
8.5 Meaning of the Supporting Forecast Evaluation 68
8.6 Limitations 69
8.7 Overall Interpretation 70
9 Conclusion 71
9.1 Answer to the Research Question 71
9.2 Main Contribution and Scope 71
9.3 Future Work 72
References 73
A Dataset Reproduction Artifacts 78
A.1 Arsenic-Schedule Generation 78
A.2 Chlorine-Profile Generation 78
B Simulation Configuration 79
B.1 Local Source Correction for Transported Arsenic 79
C Network-Delay Diagnostic for Temporal-Window Selection 80
D Policy and Reward Parameter Selection 81
D.1 Policy-Training Parameters 82
D.2 Reward Parameters 82
D.3 Selection Criteria and Supporting Diagnostic 83
E Numerical Results 83
E.1 Validation Return and Selected Policy Checkpoints 83
E.2 Paired Physical Metric Differences 85
E.3 Forecast Errors 85
F Submission Repository 87

Acronyms

As(III): arsenite
BPTT: backpropagation through time
CEM: cross-entropy method
CI: confidence interval
DAgger: dataset aggregation
EPANET-MSX: EPANET Multi-Species Extension
GAE: generalized advantage estimation
KL: Kullback–Leibler divergence
LSTM: long short-term memory
MAE: mean absolute error
MDP: Markov decision process
MPC: model predictive control
MSE: mean squared error
PETS: probabilistic ensembles with trajectory sampling
POMDP: partially observable Markov decision process
PPO: proximal policy optimization
RL: reinforcement learning
RMSE: root mean squared error
RNN: recurrent neural network
SD: standard deviation
WHO: World Health Organization

1 Introduction

Safe drinking water is essential for human health and daily life [55]. Drinking water safety depends on more than the quality of the water at its source. Risks must be managed throughout the entire supply chain, from the water source through the treatment plant to storage, distribution, and use by consumers [55]. As water moves through the distribution network, it may be exposed to various microbial and chemical contaminants [55].

Protecting drinking-water quality involves preventing contamination, monitoring water quality, and responding to contamination events [55]. Prevention seeks to stop contamination from entering the supply and to preserve water quality during distribution. Maintaining a disinfectant residual is one established measure for limiting microbial growth within a network [55]. Monitoring then provides evidence of changes in water quality [55]. It may combine conventional measurements such as laboratory tests [55] with biological early-warning approaches. Freshwater mussels, for example, have been proposed as sentinels for drinking-water safety [16] and have been used to monitor raw water before treatment at facilities in Warsaw and Skawina, Poland [27, 57]. If contamination is suspected or confirmed, utilities must react to limit its spread and consequences [54]. Depending on the incident, available responses include increasing the disinfectant dose for microbial contamination, flushing the system, diverting flows, and isolating affected areas [50, 55].

Maintaining a chlorine residual is a well-established measure for preserving water quality after treatment, as it can provide partial protection against low-level microbial contamination while water travels through the distribution network [55]. The residual that reaches different locations, however, varies with conditions in the network and changes over time [42]. A fixed dose at the point of injection therefore cannot be expected to maintain the same residual concentration throughout the system at all times. When residual concentrations can be monitored, chlorination can respond to variation through feedback [53]. It thereby becomes a control problem in which dosing decisions must be adapted to the observed condition of the network.

1.1 Motivation

Adaptive chlorination must balance several objectives. Too little residual chlorine can leave parts of a distribution network without the intended protection [55], while excessive chlorination can affect the taste of the water [55] and increase chlorination costs [53]. Finally, when contamination can be addressed through chlorination, a controller must respond quickly and effectively, and find a compromise between effective treatment and excessively chlorinating unaffected parts of the network [50]. A useful controller must therefore preserve residual protection while avoiding unnecessary dosing and still retain the ability to respond appropriately when contamination occurs.

This balance becomes a difficult task as the transport of water and dissolved substances depends on changing flow conditions in the network [42]. Changes in demand patterns affect the path that water takes, where it mixes, and how long it remains in the network [42, 53]. During transport, chlorine reacts with substances in the water and biofilm on pipe walls, so longer travel times can reduce the residual that reaches downstream locations [42]. A dosing change at the source may therefore affect monitored locations only after a delay, and its effect can vary across the network. Control decisions must account for consequences that are both spatially distributed and delayed [51, 53].

The combined effects of changing demand, flow conditions, local chlorine consumption, contamination, and preceding transport determine how chlorine concentrations develop across a distribution network [42, 45, 53]. When the available measurements do not uniquely determine the underlying network state, chlorination control becomes a partially observable problem [24].

Reinforcement learning (RL) offers a way to develop a chlorination controller that learns from the consequences of its actions [49]. Direct training from scratch, however, requires repeated interaction with the environment before useful control behavior can emerge [12, 29, 30]. At the same time, predictive models and model-based control methods can provide prior information about the consequences of dosing decisions [12, 29, 53].

Predictive control and RL have both been investigated for drinking-water chlorination, including data-enabled predictive control and recurrent policy learning [1, 35]. Learned surrogate models have also been used to support the optimization of neural chlorination controllers [28]. In other control domains, imitation of model predictive control (MPC) policies has provided an initialization for subsequent model-free RL [29]. These findings motivate investigating whether offline MPC supervision can improve learning in partially observable chlorination control. This study therefore examines how MPC-guided pretraining affects the online sample efficiency and final physical control performance of a recurrent policy under a fixed interaction budget. The additional computational cost of pretraining is assessed separately.

1.2 Research Question

Based on the preceding motivation, this thesis addresses the following research question:

> **How does MPC-guided pretraining affect the online sample efficiency and final physical control performance of a chlorination policy trained with RecurrentPPO compared with training from scratch?**

Online sample efficiency is assessed through learning curves over a common online interaction budget. This examines whether pretraining allows the policy to reach comparable performance with fewer online interactions. Final control performance is assessed on separate test cases using the policy selected through validation after completion of the common training budget. The comparison distinguishes an early learning advantage from better final control, as the baseline may catch up within the shared budget. Offline preparation and computational cost are accounted for separately, since fewer online interactions do not necessarily imply lower total training effort.

1.3 Contributions and Study Scope

This thesis develops a pretraining pipeline that transfers predictive control decisions into a recurrent chlorination policy. A learned model of chlorine dynamics supports an offline MPC teacher, whose demonstrations are used to initialize the policy through behavior cloning. RecurrentPPO then fine-tunes the policy through simulator interaction. The resulting controller uses the recurrent policy alone, with forecasting and MPC used only during offline preparation. During training and deployment, the policy receives only measured chlorine concentrations and information derived from the controller's own dosing actions and time records.

The empirical contribution is a comparison with RecurrentPPO trained from scratch under the same online training conditions and interaction budget. It examines learning progress and final control quality alongside the additional preparation cost. The study is limited to adaptive chlorination in the simulated Hanoi drinking-water distribution network. The modeled reaction oxidizes arsenite (As(III)) but does not establish removal of total arsenic or drinking-water safety. Its conclusions apply to the examined demand and contamination scenarios and training budgets.

1.4 Thesis Structure

Chapter 2 places the study within existing research on chlorination control and policy pretraining. Chapter 3 introduces the simulation and learning concepts needed for the study, and Chapter 4

describes the Hanoi network, hydraulic and contamination scenarios, and data partitions. Building on this setting, Chapter 5 defines the control problem and develops the baseline and pretraining pipeline. Chapter 6 establishes how their learning progress, physical control performance, and costs are compared.

Chapter 7 reports these measurements, and Chapter 8 interprets them in relation to the research question and the study's limitations. Chapter 9 summarizes the answer and identifies directions for further work. The appendices provide dataset reproduction details, simulation settings, network-delay diagnostics, parameter-selection details, and supplementary numerical results.

2 Related Work

Research on chlorination control in water distribution networks has developed along several lines. Early work treated dosing as a scheduling problem over a known hydraulic and water-quality model. More recent approaches use feedback control, data-driven prediction, or RL. A second body of work studies how learned dynamics and expert demonstrations can reduce the interaction required by model-free RL. The study at hand connects these lines by using a learned chlorine forecaster and MPC only to generate teacher demonstrations. The resulting control behavior is transferred to a recurrent policy that is fine-tuned online.

Boccelli et al. formulated periodic booster-disinfection scheduling as a deterministic linear program that minimizes injected disinfectant mass while enforcing lower and upper residual bounds [8]. The formulation assumes known periodic hydraulics, linear decay kinetics, and fixed booster locations. This established chlorine dosing as a network-wide optimization problem, but produced an open-loop schedule rather than a learned feedback policy. Wang et al. later derived a time-varying state-space model for disinfectant transport and developed both an MPC formulation and an analytical plug-and-play control law for real-time water-quality regulation [53]. Their approach remains dependent on the water-quality model, state estimates, and hydraulic forecasts, whereas the deployed controller in this study uses none of these components.

Subsequent work has addressed uncertainty and reduced dependence on explicit state-space models. Vrachimis et al. computed robust open-loop dosing schedules for bounded, flow-induced input-delay uncertainty [51]. Feedback from continuing chlorine measurements is not part of their method and is identified as a future extension. Perelman and Ostfeld instead applied data-enabled predictive control to pressure management and chlorine disinfection using recorded input-output trajectories rather than an identified physical model [35]. Their chlorine case repeatedly optimizes three booster inputs online from measurements at three sensor nodes. In contrast to this online predictive optimization, the present study uses predictive control only to generate offline teacher actions.

The AI for Drinking Water Chlorination Challenge introduced a recent sensor-based benchmark in which controllers regulate several chlorine injectors under unobserved contamination and limited measurements [3]. Its reports include approaches based on proximal policy optimization (PPO), as well as rule-based, predictive, and evolutionary approaches. The report by Kowalski and Schilin is particularly relevant because it separately investigated PPO, a controller combining long short-term memory (LSTM) forecasting with MPC, and a rule-based controller within the same challenge [25]. These alternatives were not combined into one pipeline and were only partially evaluated. Their MPC setup used the learned predictor to optimize action sequences online at every control step, but the computational cost prevented the evaluation of complete MPC episodes. In the same challenge, Abzanov and Haße investigated PPO variants including an LSTM-based recurrent policy [1]. The challenge also differs from the single-source Hanoi problem studied here through its multiple injectors and its sensor and contamination setup. The methodological connection therefore concerns the use of a forecaster, MPC, and PPO rather than the complete control problem. The present study restricts

predictive optimization to offline teacher generation and evaluates whether the resulting actions can be transferred to a recurrent policy through pretraining, so neither the forecaster nor the optimizer is required afterward.

Monsia et al. used an LSTM surrogate to support multi-objective neuroevolution of neural chlorination controllers, updating the surrogate with trajectories from simulator evaluations [28]. Their approach places the learned dynamics within the evolutionary controller search, whereas the present study uses them to generate MPC teacher actions.

Outside the domain of water distribution networks and water-quality control, model-based RL provides established precedents for combining learned dynamics with online planning. The probabilistic ensembles with trajectory sampling (PETS) algorithm uses probabilistic ensemble models, trajectory sampling, and the cross-entropy method (CEM) to select actions while accounting for model uncertainty [12]. This supports the use of sample-based optimization over a learned dynamics model, although the forecaster in the present study is deterministic. Nagabandi et al. took a related approach. They trained neural dynamics models, generated trajectories with MPC, cloned the MPC policy, and then fine-tuned it with model-free RL [29]. Their experiments concern fully observed locomotion and iteratively query the MPC teacher through dataset aggregation (DAgger). The present study examines the same general transfer from predictive control to model-free learning in a partially observable chlorination task, using a fixed cross-fitted teacher dataset, a recurrent actor, and RecurrentPPO fine-tuning.

Work on MPC imitation clarifies the trade-off introduced by a fixed dataset. Ahn et al. analyze imitation of constrained linear MPC and propose an on-policy procedure because ordinary behavior cloning may encounter states outside the teacher distribution [2]. DAgger addresses the same sequential distribution shift by repeatedly collecting learner trajectories and requesting expert labels for the visited states [41]. These procedures reduce mismatch between training and deployment states, but require continued access to the expert. The pipeline studied here deliberately separates teacher generation from policy learning. After offline teacher generation, subsequent PPO fine-tuning allows the policy to adapt through reward feedback without further MPC queries or expert annotation.

Nair et al. combined off-policy RL with demonstration replay and an auxiliary behavior-cloning loss to support exploration in sparse-reward manipulation tasks [30]. Their method continues to use demonstrations during RL, whereas demonstration supervision in the present study ends before online learning begins.

Prior work therefore covers the main components of the proposed pipeline, including chlorine optimization, learned predictive control, MPC imitation, and RL from demonstrations. The experiment in this thesis focuses on their combination under partial observability and asks whether cross-fitted offline MPC supervision improves RecurrentPPO under the same online interaction budget as training from an ordinary initialization.

3 Foundations

Water and dissolved substances take time to move through a distribution network [42]. Their paths and travel times depend on the hydraulic conditions, so a change in solute injection may affect downstream concentrations only after a transport delay. A controller must therefore account for consequences that unfold over time. When, on top of that, the complete system state is not directly observable, it must additionally base its decisions on partial observations [24].

This problem draws on dynamical systems, water-quality simulation, machine learning, and control theory, and the underlying foundations are introduced in this chapter. It first introduces trajectories and observations, followed by the models used for sequential decisions and forecasting. It then

presents predictive control and imitation learning, while specific choices concerning the network, data allocation, model configuration, and control objective are deferred to later chapters.

3.1 Mathematical Foundations

Prediction and control can be described in terms of the evolution of a physical system. The state describes the system at one instant, while a trajectory orders successive states over time. Probability represents variation among possible trajectories, and time-series notation organizes the measurements obtained from them. These concepts provide a common notation for the evolving system state, the observations derived from it, and the uncertainty affecting its development.

3.1.1 Discrete-Time Dynamical Systems and Trajectories

A dynamical system can be represented by a state that evolves over time [40]. At discrete instants $t = 0, 1, \ldots, T$, a state transition with a control input and a disturbance can be written as follows.

$$\boldsymbol{x}_{t+1} = f(\boldsymbol{x}_t, u_t, \boldsymbol{w}_t), \tag{1}$$

where $\boldsymbol{x}_t$ denotes the system state, u_t a control input, and $\boldsymbol{w}_t$ an external disturbance or unresolved source of variation. The transition function f maps these quantities to the next state. Random disturbances make this a stochastic model [40]. A Markov state contains the past information needed to determine the next-state distribution given the control input [49]. Its components need not be directly observable.

A simulation run or system interaction generates an ordered sequence of states and controls [49]. We write this trajectory as

$$\tau = (\boldsymbol{x}_0, u_0, \boldsymbol{x}_1, u_1, \ldots, u_{T-1}, \boldsymbol{x}_T). \tag{2}$$

The temporal order is part of the trajectory. Rearranging otherwise identical values generally describes a different evolution because controls, transport delays, and accumulated disturbances affect later states.

An observation $\boldsymbol{o}_t = h(\boldsymbol{x}_t, \boldsymbol{v}_t)$ represents a measurement of the state, where $\boldsymbol{v}_t$ accounts for measurement effects. This notation includes the additive-noise model $\boldsymbol{o}_t = h_0(\boldsymbol{x}_t) + \boldsymbol{v}_t$ as a special case [40]. If the measurement uniquely determines the state, it provides full state information. Otherwise, several observations and the intervening actions may help infer the current condition [24].

3.1.2 Time-Series Representations

A univariate time series is an ordered sequence $y_1, y_2, \ldots, y_T$ indexed by time [7]. In a multivariate series, each entry is a vector $\boldsymbol{y}_t \in \mathbb{R}^m$, where m is the number of variables. With a fixed sampling interval Δt, an index difference of k corresponds to a physical duration of $k\,\Delta t$. Successive entries can be statistically dependent [7].

At a forecast origin t, a history window of length H contains the observations available up to that instant, while a prediction horizon of length K contains the subsequent targets,

$$\begin{aligned} \boldsymbol{X}_t &= (\boldsymbol{y}_{t-H+1}, \ldots, \boldsymbol{y}_t), \\ \boldsymbol{Y}_t &= (\boldsymbol{y}_{t+1}, \ldots, \boldsymbol{y}_{t+K}). \end{aligned} \tag{3}$$

Moving the forecast origin along a trajectory yields multiple forecasting windows [7]. Adjacent windows generally overlap and share historical or future values. They therefore do not constitute independent repetitions of the underlying trajectory.

Periodic calendar variables can be represented with sine and cosine terms [22]. For phase q with period P, the encoding

$$\varphi(q) = \left(\sin\left(2\pi\frac{q}{P}\right), \cos\left(2\pi\frac{q}{P}\right)\right) \tag{4}$$

maps the phase to the unit circle. The pair identifies the phase modulo P and keeps the end and beginning of the cycle adjacent. Separate pairs can represent daily, weekly, and annual cycles without the jump in a scalar index at each cycle boundary.

3.2 Sequential Decision-Making

A forecast describes how a system may evolve under given inputs, whereas a controller selects inputs that change this evolution [40]. Actions can affect both future outcomes and the conditions under which subsequent decisions are made. Sequential decision-making captures this feedback. Markov decision processes provide the basic model, and partial observability accounts for measurements that reveal only part of the physical state [24].

3.2.1 Markov Decision Processes

A Markov decision process (MDP) describes sequential interaction through states, actions, transition probabilities, and rewards [49]. Including the discount factor, we write

$$\mathcal{M} = (\mathcal{S}, \mathcal{A}, P, R, \gamma). \tag{5}$$

Here, $\mathcal{S}$ is the state space and $\mathcal{A}$ the action space. In a finite-action problem, $\mathcal{A}$ contains a discrete set of choices, while in continuous control, it is instead a subset of $\mathbb{R}^d$ [49]. The transition kernel $P(s'|s,a)$ describes how the current state and action determine the next-state distribution, and $R(s,a,s')$ specifies the expected immediate reward. The discount factor $\gamma \in [0,1]$ determines the relative weight of future rewards, with $\gamma = 1$ permitted for finite episodic returns. The Markov property requires the current state to contain all past information relevant to the next-state and reward distribution [49]. A policy $\pi(a|s)$ maps states to action probabilities or densities. Together with an initial-state distribution, the policy and transition dynamics induce a distribution over trajectories.

This formulation assumes that the controller has access to a state representation that satisfies the Markov property [24]. When the available observations reveal only part of that state, the decision process must instead account for partial observability.

3.2.2 Partial Observability

A partially observable Markov decision process (POMDP) extends an MDP with an observation space and an observation model [24]. We denote the observation space by $\mathcal{O}$. For the special case in which observations depend only on the resulting state, we use the model $\Omega(o|s)$. The environment evolves through a latent state S_t, but the agent receives only an observation O_t. Even if the latent state is Markovian, the current observation need not be. The same measurement can arise from several hidden physical conditions with different future consequences [24].

The observation-action history available at time t can be written as $\mathcal{H}_t = (o_0, a_0, \ldots, a_{t-1}, o_t)$. Given an initial belief, the history determines a belief state over the latent states [24]. For discrete states,

$$b_t(s) = \Pr(S_t = s \mid \mathcal{H}_t). \tag{6}$$

This belief state is a probability distribution over latent states [24]. Exact Bayesian belief updates require the transition and observation models. Decisions may instead be conditioned on the observa-

tion-action history or on a compact representation learned from it. Recurrent models, introduced later in this chapter, provide one way to learn such a representation [32].

Such representations can retain evidence that is absent from the current observation alone [32]. A learned history representation need not equal the Bayesian belief state. Moreover, identical observation-action histories yield the same representation when the initial recurrent state and deterministic update rule are fixed, so they cannot resolve the underlying state ambiguity.

3.3 Hydraulic and Water-Quality Simulation

A change in solute injection can produce a delayed concentration response at downstream locations [42]. Water follows paths determined by the current flows, transports dissolved substances downstream, and mixes at junctions while chemical reactions continue [31]. A node's concentration therefore depends on the chemical inputs and on the hydraulic history that transported the water there. Hydraulic and water-quality simulation represents these processes as coupled parts of the same evolving system.

3.3.1 Water Distribution Networks and Hydraulic States

A water distribution network can be represented as a graph $\mathcal{G} = (\mathcal{V}, \mathcal{E})$ [42]. Reservoirs, tanks, and demand junctions form the nodes $\mathcal{V}$, while pipes, pumps, and valves form the links $\mathcal{E}$. Elevation, storage, pipe geometry, roughness, and component settings supplement this topology with the properties required for hydraulic calculation.

At each junction, conservation of water requires the incoming flow to equal the outgoing flow plus external demand [42].

$$\sum_{e \in \mathcal{E}_v^{\text{in}}} q_e(t) = \sum_{e \in \mathcal{E}_v^{\text{out}}} q_e(t) + d_v(t), \tag{7}$$

where $\mathcal{E}_v^{\text{in}}$ and $\mathcal{E}_v^{\text{out}}$ contain the links whose reference orientations point into and out of junction v, respectively. The signed volumetric flow through link e is $q_e(t)$, and $d_v(t)$ denotes the external demand flow at the junction. Hydraulic head combines elevation and pressure head, while link equations relate head differences to flow through pipes, pumps, and valves. Solving these coupled continuity and energy relations yields nodal heads, pressures, and link flows for the specified demand model [42].

A fixed reference orientation defines the sign of each computed link flow without requiring water to move permanently in that direction [42]. A negative value indicates flow opposite to the reference orientation. As demands, storage levels, and active components change, flow magnitude and direction may change as well [42]. The time-varying hydraulic conditions therefore determine the effective downstream paths and travel times of transported substances even though the physical topology remains fixed.

3.3.2 Solute Transport, Mixing, and Mass-Rate Sources

Advection transports a dissolved species with the bulk water flow [45]. In a pipe with cross-sectional area A_e and volumetric flow $q_e(t)$, the mean velocity is $v_e(t) = \frac{q_e(t)}{A_e}$. When longitudinal dispersion is neglected, the concentration field $c_e(x, t)$ follows an advection–reaction equation of the form [42]

$$\frac{\partial c_e}{\partial t} + v_e(t)\frac{\partial c_e}{\partial x} = r(c_e). \tag{8}$$

Here, x denotes position along the pipe and t denotes continuous physical time. The function r describes reactions that consume or produce the species. Through $v_e(t)$, the time-varying hydraulic conditions determine the transport direction and residence time [42].

At an ideal junction without storage, incoming streams are assumed to mix instantaneously [42]. Conservation of solute mass yields a single junction concentration from the mass carried by the incoming flows, any externally added mass, and the total flow leaving the junction. This concentration applies to all water leaving the junction, including local demand and flow entering downstream links. Mixing thus combines concentrations from different transport paths and carries their preceding reaction histories into the downstream network.

A mass-rate source specifies injected solute mass per unit time, whereas concentration denotes mass per unit water volume [42]. If a source adds mass at rate $\dot{m}$ to water flowing at a positive rate q, its concentration contribution under ideal mixing is $\frac{\dot{m}}{q}$. A fixed mass rate therefore does not imply a fixed concentration. Higher flows dilute the input more strongly, and concentrations at downstream locations also depend on mixing, reaction, and transport delay [42].

3.3.3 Free-Chlorine Decay and Chlorine–Arsenite Reactions

Free-chlorine loss is commonly represented by a first-order decay model, whose rate is proportional to the local chlorine concentration [42, 45]. In this model, the concentration in a given volume of water decreases with increasing residence time. Transport, dilution, and mixing determine its concentration profile along each flow path [42].

Free chlorine also oxidizes As(III) to arsenic(V) where both species coexist [13]. The reaction consumes both reactants in a 1:1 molar ratio, and its rate depends on their concentrations. The local reaction extent also depends on contact time. Reactant concentrations and contact times depend on upstream inputs, flow-dependent dilution, mixing, and travel time [42]. The chemical response therefore remains coupled to the hydraulic trajectory.

3.3.4 EPANET and EPANET-MSX

EPANET is a simulator for extended-period hydraulic and water-quality analysis of pressurized pipe networks [42]. It solves the network equations over a sequence of hydraulic conditions and tracks head, pressure, flow, demand, tank level, and dissolved-species concentration. Reservoirs provide hydraulic boundary conditions, while time-varying junction demands influence the resulting flow distribution. Its water-quality model uses the hydraulic state to represent pipe transport and mixing at junctions, while configured reactions determine changes in concentration [42]. Hydraulic and water-quality calculations may be advanced at different time resolutions.

EPANET Multi-Species Extension (EPANET-MSX) extends this formulation to multiple interacting species [45]. For kinetic reactions, a concentration vector $\boldsymbol{c}$ and a vector of kinetic parameters $\boldsymbol{k}$ define a vector-valued reaction function $r(\boldsymbol{c}, \boldsymbol{k})$, where r contains the species-specific reaction rates, whose components may depend on several species. Transport and reaction can thus be simulated jointly under the same time-varying hydraulic state [45].

3.4 Neural Models for Sequential Data

Neural networks approximate mappings from inputs to outputs by learning from examples [19]. A feedforward model receiving only the current measurement has no internal memory. Historical information can instead be supplied as a fixed input window or retained in a recurrent state. Recurrent networks use the latter approach by updating a recurrent state throughout the sequence.

3.4.1 Parameterized Models and Gradient-Based Learning

A neural network is a parameterized function $F_{\boldsymbol{\theta}}$ composed of affine transformations and nonlinear operations [19]. Its parameters $\boldsymbol{\theta}$ are learned weights and biases, while its inputs and outputs are the values it receives and computes. In supervised learning, a loss function $\mathcal{L}$ assigns a cost to each model prediction relative to its target. Averaging this loss over the training examples gives an

empirical training objective. When this objective is differentiable with respect to $\boldsymbol{\theta}$, its gradient guides the parameter updates. For N supervised examples $(\boldsymbol{X}^i, \boldsymbol{Y}^i)$, empirical parameter estimation can be written as [19]

$$\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \frac{1}{N} \sum_{i=1}^{N} \mathcal{L}\big(F_{\boldsymbol{\theta}}(\boldsymbol{X}^i), \boldsymbol{Y}^i\big). \tag{9}$$

Backpropagation applies the chain rule to differentiate this objective with respect to every parameter [19]. Minibatch optimization estimates the gradient from a subset of examples instead of evaluating the complete dataset for each update. This stochastic estimate reduces the computational cost per update and permits repeated parameter updates during training.

Large gradients can destabilize optimization [34]. Gradient-norm clipping rescales a gradient $\boldsymbol{g}$ when $\|\boldsymbol{g}\|$ exceeds a chosen threshold. This limits its magnitude while preserving its direction. Clipping mitigates exploding gradients.

Model parameters, auxiliary optimizer state, and fitted preprocessing statistics are distinct from individual model inputs and outputs and from the objective function.

3.4.2 Recurrent Neural Networks and Backpropagation Through Time

A simple recurrent neural network (RNN) processes a sequence by updating a hidden state [19],

$$\begin{aligned} \boldsymbol{h}_t &= \varphi(\boldsymbol{W}_x \boldsymbol{x}_t + \boldsymbol{W}_h \boldsymbol{h}_{t-1} + \boldsymbol{b}_h), \\ \hat{\boldsymbol{y}}_t &= g\big(\boldsymbol{W}_y \boldsymbol{h}_t + \boldsymbol{b}_y\big). \end{aligned} \tag{10}$$

Here, $\boldsymbol{x}_t$, $\boldsymbol{h}_t$, and $\hat{\boldsymbol{y}}_t$ are the input, hidden-state, and output vectors. The quantities $\boldsymbol{W}_x$, $\boldsymbol{W}_h$, and $\boldsymbol{W}_y$ are weight matrices, $\boldsymbol{b}_h$ and $\boldsymbol{b}_y$ are bias vectors, and φ and g are activation functions. The hidden state $\boldsymbol{h}_t$ summarizes the inputs processed through time t [19]. Reusing the same matrices and biases at every step imposes a common transition rule and allows the model to process sequences of different lengths.

Backpropagation through time (BPTT) unfolds the recurrence into a computation graph containing one instance of the recurrent cell per time step, where the instances share parameters [34]. Gradients from later losses pass through the intervening state transitions, and their contributions to each parameter are accumulated before the optimizer update.

This backward propagation multiplies chains of transition Jacobians [34]. Contracting products attenuate the contribution of early states to gradients of later losses, while amplifying products can make these gradients grow rapidly. These effects are known as vanishing and exploding gradients. Although gradient clipping mitigates the exploding case, it cannot restore long-range learning signals that have vanished [19]. Gated architectures provide more direct paths for information and gradient flow.

Full BPTT differentiates through the complete sequence [19]. Truncated BPTT instead operates on shorter consecutive fragments [36]. The final recurrent state of one fragment can initialize the next and thereby preserve information in the forward pass. Treating this boundary state as constant during differentiation prevents gradients from crossing the fragment boundary. For independent episodes or trajectories, the recurrent state is reinitialized at their boundaries. Otherwise, the recurrence could carry information from one trajectory into another.

3.4.3 Long Short-Term Memory Networks

LSTM is a gated recurrent architecture designed to preserve relevant information over long sequences [20]. Its memory cells provide paths for learning signals to persist across many time steps, aiming to help the network learn dependencies over long time intervals. A forget gate adds a learned multiplier

to the previous cell state [18]. The LSTM update can be written in vector form as follows, with tanh used to compute the candidate state and the hidden-state output [19]:

$$\begin{aligned}
\boldsymbol{i}_t &= \sigma(\boldsymbol{W}_i\,\boldsymbol{x}_t + \boldsymbol{U}_i\,\boldsymbol{h}_{t-1} + \boldsymbol{b}_i),\\
\boldsymbol{f}_t &= \sigma\big(\boldsymbol{W}_f\,\boldsymbol{x}_t + \boldsymbol{U}_f\,\boldsymbol{h}_{t-1} + \boldsymbol{b}_f\big),\\
\boldsymbol{o}_t &= \sigma(\boldsymbol{W}_o\,\boldsymbol{x}_t + \boldsymbol{U}_o\,\boldsymbol{h}_{t-1} + \boldsymbol{b}_o),\\
\boldsymbol{g}_t &= \tanh\big(\boldsymbol{W}_g\,\boldsymbol{x}_t + \boldsymbol{U}_g\,\boldsymbol{h}_{t-1} + \boldsymbol{b}_g\big),\\
\boldsymbol{c}_t &= \boldsymbol{f}_t \cdot \boldsymbol{c}_{t-1} + \boldsymbol{i}_t \cdot \boldsymbol{g}_t,\\
\boldsymbol{h}_t &= \boldsymbol{o}_t \cdot \tanh(\boldsymbol{c}_t).
\end{aligned} \tag{11}$$

Here, $\boldsymbol{i}_t$, $\boldsymbol{f}_t$, and $\boldsymbol{o}_t$ are the input, forget, and output gate vectors, $\boldsymbol{g}_t$ is the candidate cell-state vector, $\boldsymbol{c}_t$ is the cell-state vector, and $\boldsymbol{h}_t$ is the exposed hidden-state vector. The $\boldsymbol{W}$ and $\boldsymbol{U}$ quantities are weight matrices, and the $\boldsymbol{b}$ quantities are bias vectors. The logistic sigmoid σ and tanh act component-wise, and $\cdot$ denotes element-wise multiplication. The input gate controls what is written to the cell state, the forget gate what is retained, and the output gate what is exposed.

The additive cell-state update provides a direct path for retaining information [18]. Forget-gate values near one preserve the corresponding contributions from the previous cell state, whereas values near zero suppress them. The gates can therefore learn which information to retain. The hidden and cell states can provide a compact history representation for sequence forecasting and for decisions that cannot rely on the current observation alone [19, 20].

3.5 Reinforcement Learning

In many interactive control problems, representative examples of the correct action are difficult to obtain [49]. The outcome of an interaction can nevertheless be evaluated through a reward. RL uses this feedback to improve a policy. Because actions influence both later rewards and the states encountered next, learning must account for delayed outcomes when optimizing expected long-term return.

3.5.1 Policies, Rewards, and Returns

A stochastic policy $\pi_{\boldsymbol{\theta}}(a|s)$ defines an action distribution conditional on the available state representation [49]. During interaction, action A_t affects the next-state distribution, and the environment emits an immediate reward R_{t+1}. For an episode ending at T, the discounted return from time t is

$$G_t = \sum_{k=0}^{T-t-1} \gamma^k\, R_{t+k+1}, \tag{12}$$

where $\gamma \in [0, 1]$ determines the relative weight of delayed rewards. For this episodic formulation, the objective is the expected return over trajectories induced by the policy, $J(\boldsymbol{\theta}) = E_{\tau\sim\pi_{\boldsymbol{\theta}}}[G_0]$.

For continuous control, a common policy parameterization uses a Gaussian distribution with a learned mean and standard deviation [44, 49]. Sampling from this distribution provides exploration, while deterministic evaluation can use its mean. The mean determines the central action and the standard deviation controls the spread of sampled actions.

An action may influence rewards many steps later. Temporal credit assignment determines how these outcomes affect the earlier decisions that contributed to them [49]. For $0 < \gamma < 1$, discounting reduces the influence of distant rewards without removing this dependency. The reward should be designed so that maximizing expected return promotes the intended behavior [49].

A finite penalty discourages a constraint violation by reducing its reward, but does not exclude the violating action or outcome [40]. Hard constraints instead exclude inadmissible choices from the feasible set.

Expected return is the optimization objective used to train the policy. External evaluation criteria instead assess whether the resulting behavior satisfies the scientific or operational goals. A high training return indicates success under the encoded objective, which need not capture every intended outcome [49]. Separate outcome measures are therefore required for behavior that this scalar signal does not fully represent.

Sample efficiency concerns the amount of environment interaction needed to achieve a given level of performance [49]. Learning curves compare performance as this interaction budget increases. Sample efficiency measures data requirements, whereas computational efficiency concerns processing time and resource use [32].

3.5.2 Actor–Critic Learning

Actor–critic methods combine an action-selecting actor with a critic that estimates expected return [49]. For a policy π, the state-value and action-value functions are

$$\begin{aligned} V^{\pi}(s) &= E_{\pi}[G_t \mid S_t = s], \\ Q^{\pi}(s,a) &= E_{\pi}[G_t \mid S_t = s, A_t = a]. \end{aligned} \tag{13}$$

Their difference $A^{\pi}(s,a) = Q^{\pi}(s,a) - V^{\pi}(s)$ is the advantage [43]. It indicates whether an action is expected to perform better or worse than the policy's typical behavior in the same state. The actor can increase the probability of actions with positive estimated advantage and decrease it for actions with negative estimated advantage.

The critic approximates the value function from sampled trajectories [43]. A one-step temporal-difference residual is

$$\delta_t = R_{t+1} + \gamma\, V_{\boldsymbol{\varphi}}(S_{t+1}) - V_{\boldsymbol{\varphi}}(S_t). \tag{14}$$

The bootstrap value is zero at a true terminal state [49]. The residual compares the critic's current estimate with the observed reward plus its estimate at the successor state. Residuals from several time steps can be combined to estimate advantages [43].

Actor and critic may use separate feature extractors or share an intermediate representation with distinct output heads [32]. With otherwise matched feature extractors, sharing avoids duplicate parameters and feature computation but also lets both objectives update the same representation. Separate representations avoid this interaction between their feature-learning gradients [32].

3.5.3 Proximal Policy Optimization

PPO is an on-policy actor–critic method that alternates between data collection and optimization [44]. The current policy first collects a rollout, which is divided into minibatches and reused for several optimization epochs before being discarded. This reuse improves sample utilization within an update. However, large parameter updates can cause the new policy to diverge substantially from the policy that generated the rollout [44].

PPO measures this change for each sampled action using the probability ratio between the current policy and the policy that generated the rollout

$$r_t(\boldsymbol{\theta}) = \frac{\pi_{\boldsymbol{\theta}}(A_t|S_t)}{\pi_{\boldsymbol{\theta}_{\text{old}}}(A_t|S_t)}. \tag{15}$$

Given an advantage estimate $\hat{A}_t$, the clipped surrogate objective is [44]

$$L^{\mathrm{CLIP}}(\boldsymbol{\theta}) = E_t\Big[\min\Big(r_t(\boldsymbol{\theta})\,\hat{A}_t,\, \mathrm{clip}\big(r_t(\boldsymbol{\theta}), 1-\varepsilon, 1+\varepsilon\big)\,\hat{A}_t\Big)\Big]. \tag{16}$$

Here, E_t denotes the empirical average over the sampled rollout steps. Once the ratio crosses the clipping interval in a direction that would improve the unclipped objective, clipping removes the incentive to move the sampled action probability farther [44]. It discourages abrupt policy changes but does not impose a hard constraint on every parameter or action probability.

Generalized advantage estimation (GAE) combines temporal-difference residuals over several future steps [43]. For a trajectory fragment ending at T, its truncated form is [44]

$$\hat{A}_t^{\mathrm{GAE}}(\gamma,\lambda) = \sum_{l=0}^{T-t-1} (\gamma\lambda)^l\, \delta_{t+l}. \tag{17}$$

Small values of λ emphasize short bootstrapped estimates, which depend more strongly on the critic and often have lower variance [43]. Larger values incorporate longer reward sequences and generally trade lower bootstrap bias for higher variance.

At the end of a rollout fragment, the environment may still be continuing [44]. The final temporal-difference residual then retains the critic's estimate of the successor state, whereas a true terminal state has no future return to bootstrap [49]. Treating a continuing fragment boundary as termination would omit this remaining value and can bias the advantage estimates.

The policy objective is optimized together with a regression objective for the critic [44]. A common combined loss is

$$\begin{aligned} L(\boldsymbol{\theta},\boldsymbol{\varphi}) &= -L^{\mathrm{CLIP}}(\boldsymbol{\theta}) + c_V\, L_V(\boldsymbol{\varphi}),\\ L_V(\boldsymbol{\varphi}) &= E_t\Big[\big(V_{\boldsymbol{\varphi}}(S_t) - \hat{V}_t\big)^2\Big], \end{aligned} \tag{18}$$

where $\hat{V}_t$ is a return target held fixed during the update and c_V weights the value loss. With GAE, this target can be formed by adding the estimated advantage to the value estimate used during rollout collection [43]. The combined loss provides optimization terms for both policy improvement and value estimation. An optional entropy bonus encourages a broader action distribution, where setting its coefficient to zero gives the objective above [44].

The Kullback–Leibler divergence (KL) measures the difference between two probability distributions [26]. For continuous densities p and q with common support,

$$D_{\mathrm{KL}}(p \,\|\, q) = \int p(a) \ln\!\left(\frac{p(a)}{q(a)}\right) \mathrm{d}a. \tag{19}$$

The divergence is non-negative and vanishes exactly when the distributions agree almost everywhere [26]. It is generally asymmetric, so the argument order matters. For PPO, taking p as the previous policy and q as the updated policy at a given state measures change relative to the distribution that generated the actions [44].

An estimated KL divergence averaged over sampled rollout states can serve as a diagnostic and stop further policy updates when a chosen threshold is exceeded [14]. Unlike the clipped sample-wise ratio, this statistic aggregates distributional change across sampled states. Neither clipping nor this stopping rule guarantees an exact bound outside the observed rollout.

3.5.4 Recurrent Proximal Policy Optimization

Recurrent PPO introduces a recurrent representation into the actor–critic architecture [36]. At time t, a recurrent policy updates its memory as $\boldsymbol{h}_t = f_{\boldsymbol{\theta}}(\boldsymbol{h}_{t-1}, \boldsymbol{o}_t)$ and conditions the action distribution on $\boldsymbol{h}_t$. In a partially observable environment, this memory summarizes preceding observations that may distinguish latent situations with similar current measurements [32]. Past actions can also inform the memory when they are included in the recurrent inputs [36].

Training must preserve the sequential dependencies on which this memory depends [36]. Rollout fragments are processed as ordered sequences rather than independently shuffled transitions. Each fragment receives the recurrent state associated with its first time step. Backpropagation can stop at fragment boundaries while the forward state continues into the next fragment.

For independent episodes, the recurrent state is reset at each episode boundary [21]. Otherwise, information from a completed trajectory could influence decisions in the next one. Applying the same reset convention during data collection, optimization, and evaluation gives the recurrent state a consistent meaning.

3.6 Neural Time-Series Forecasting

A forecasting model approximates future values from information available at the forecast origin, which may include an observation history and known future inputs [22]. Constructing such a forecast therefore requires a strict distinction between information available at that origin and values observed only afterward. The model architecture determines how available information is processed to generate future predictions.

3.6.1 Supervised Multistep Forecasting

Supervised multistep forecasting pairs the history available at a forecast origin with the subsequent target sequence. Examples derived from simulated trajectories must obey the same information restriction as deployment, so inputs may only contain information available when the prediction is made [22]. Moving the origin along a trajectory produces multiple history–future pairs, including the overlapping examples defined above.

With M target variables and K future steps, the task is multivariate, multi-horizon regression. A parameterized forecaster can be written as

$$\hat{\boldsymbol{Y}}_t = F_{\boldsymbol{\theta}}(\boldsymbol{X}_t, \boldsymbol{Z}_t), \tag{20}$$

where $\boldsymbol{X}_t$ is the observed history and $\boldsymbol{Z}_t$ contains future covariates known or assumed at time t [22]. If future actions affect the target, including a planned action sequence in $\boldsymbol{Z}_t$ makes the forecast action-conditioned. Its output then applies to that sequence rather than to arbitrary future actions.

Observed histories, known future covariates, and prediction targets have distinct roles. Historical target values may be valid inputs because they are available at the forecast origin [22]. At prediction time, future target values are unavailable. They serve as labels for training or evaluation. Future covariates are valid inputs only if they are known in advance or explicitly assumed as part of the conditional forecast [22].

3.6.2 Encoder–Decoder LSTMs

An encoder–decoder architecture maps an input sequence to an output sequence of possibly different length [11]. An LSTM encoder can compress the input history into a fixed-dimensional representation that initializes an LSTM decoder [48]. This representation provides the context for generating the future sequence.

At each forecast step, the decoder updates its recurrent state using this context and any step-specific input [19]. Such inputs can include known future covariates or a planned control action. Conditioning each step on the corresponding future information allows predictions to depend on those inputs.

The decoder may predict absolute target levels or deltas between successive target values. For delta prediction, absolute predictions are reconstructed from the final observation $\boldsymbol{y}_t$ by accumulating the predicted deltas,

$$\begin{aligned} \Delta \boldsymbol{y}_{t+k} &= \boldsymbol{y}_{t+k} - \boldsymbol{y}_{t+k-1}, \\ \hat{\boldsymbol{y}}_{t+k} &= \boldsymbol{y}_t + \sum_{j=1}^{k} \widehat{\Delta \boldsymbol{y}}_{t+j}. \end{aligned} \tag{21}$$

Delta prediction anchors the sequence at the final observed level, but an error in an early increment propagates to every later reconstructed value.

A decoder can produce the full prediction horizon without feeding its predictions back as later inputs [19]. An autoregressive decoder instead conditions each step on its preceding output [6]. This makes dependencies between successive forecasts explicit but allows prediction errors to accumulate through feedback. If training uses preceding ground-truth targets, inference introduces an additional mismatch because only model-generated outputs are then available.

3.6.3 Forecasting Objectives and Error Measures

Mean squared error (MSE) averages squared prediction errors [23]. For N examples, K forecast steps, and M target variables, let $y_{i,k,m}$ and $\hat{y}_{i,k,m}$ denote the target and prediction for example i, forecast step k, and variable m. Averaging over all three indices gives

$$\mathrm{MSE} = \frac{1}{N\,K\,M} \sum_{i=1}^{N} \sum_{k=1}^{K} \sum_{m=1}^{M} \left(\hat{y}_{i,k,m} - y_{i,k,m} \right)^2. \tag{22}$$

MSE is differentiable and penalizes large residuals quadratically, making it a common regression objective [19]. Weights can change the relative importance of variables or horizons. In that case, the optimized quantity is the specified weighted objective rather than an unweighted average error.

When the target variables share a common unit, mean absolute error (MAE) and root mean squared error (RMSE) summarize prediction errors in that unit [23]. They are defined as

$$\mathrm{MAE} = \frac{1}{N\,K\,M} \sum_{i=1}^{N} \sum_{k=1}^{K} \sum_{m=1}^{M} \left| \hat{y}_{i,k,m} - y_{i,k,m} \right|, \tag{23}$$

$$\mathrm{RMSE} = \sqrt{\frac{1}{N\,K\,M} \sum_{i=1}^{N} \sum_{k=1}^{K} \sum_{m=1}^{M} \left(\hat{y}_{i,k,m} - y_{i,k,m} \right)^2}. \tag{24}$$

MAE weights absolute residuals linearly, whereas RMSE gives large errors more influence through squaring [23]. Targets may be normalized during training so that variables with different scales contribute comparably to optimization. To report errors in physical units, predictions and targets are first transformed back to those units. A normalized training loss and a physical-unit evaluation metric therefore have different numerical meanings.

3.7 Predictive Control and Imitation Learning

In predictive control, a predictive model makes the expected consequences of alternative actions comparable [40]. MPC performs this comparison repeatedly over a finite horizon. If gradients through the predictive model are unavailable, sampling-based search can select the action sequence [12].

The resulting actions can also provide demonstrations for behavior cloning [2]. A cloned policy can subsequently be fine-tuned through RL [30].

3.7.1 Model Predictive Control

MPC chooses a finite sequence of future actions by optimizing their predicted outcomes over a planning horizon [40]. Given the current information I_t, a candidate control sequence $\boldsymbol{U}_t = (u_t, ..., u_{t+K-1})$ produces a forecast $\hat{\boldsymbol{Y}}_t(\boldsymbol{U}_t) = (\hat{\boldsymbol{y}}_{t+1}, ..., \hat{\boldsymbol{y}}_{t+K})$. The argument $\boldsymbol{U}_t$ makes the dependence on the candidate controls explicit. We write the resulting optimization as

$$\boldsymbol{U}_t^* = \arg\min_{\boldsymbol{U} \in \mathcal{U}^K} C\Big(\hat{\boldsymbol{Y}}_t(\boldsymbol{U}), \boldsymbol{U}\Big), \tag{25}$$

where C encodes the control cost and $\mathcal{U}$ is the feasible action set. State or output constraints can additionally restrict the candidate sequences when the predictive model represents the required quantities [40].

In its standard discrete-time formulation, MPC follows the receding-horizon principle, where only the first action of $\boldsymbol{U}_t^*$ is applied [40]. When the next observation arrives, the optimization is solved again using the updated information. Replanning reduces reliance on an open-loop sequence and lets new measurements inform subsequent predictions [40].

Control quality depends on the predictive model, planning horizon, and optimizer [40]. Model error makes future trajectories uncertain. The horizon determines which delayed effects appear explicitly in the predicted trajectory. Online optimization must fit within the available computation time [2]. Receding-horizon feedback alone does not guarantee robustness to model error [40]. It also continues to optimize the specified objective, even if that objective omits a relevant outcome.

3.7.2 Cross-Entropy Method

The CEM searches for high-performing candidate vectors by sampling and updating a proposal distribution [9]. Applied to predictive control, it searches action sequences without differentiating through the objective or predictive model [12]. For a scalar control problem with horizon K, one choice is a multivariate Gaussian with mean $\boldsymbol{\mu} \in \mathbb{R}^K$ and diagonal covariance matrix $\boldsymbol{\Sigma} \in \mathbb{R}^{K \times K}$ [10].

Each iteration samples a finite population of candidate sequences, evaluates their objective values, and retains a fixed top fraction as the elite set [9]. For the diagonal Gaussian, the empirical coordinate-wise means and variances of these elites determine the proposal parameters [10]. The means and standard deviations may then be smoothed. Repeated updates direct sampling toward regions favored by the evaluated elites. The algorithm returns either the final mean or the best evaluated candidate as an approximate solution [10].

For bounded controls, candidates can be drawn from a truncated Gaussian distribution [10]. A finite population and iteration budget do not guarantee that CEM finds the global optimum. Premature shrinking of the proposal distribution can stop exploration, motivating smoothing or the injection of additional variance [10].

3.7.3 Behavior Cloning and Transfer Learning

Behavior cloning treats observation–action demonstrations as supervised examples [41]. For N demonstrations, a teacher produces action a_i^{T} for observation or history $\boldsymbol{o}_i$, where T denotes the teacher. A student action predictor $m_{\boldsymbol{\theta}}$ learns to reproduce these actions, for example by minimizing

$$\hat{\boldsymbol{\theta}} = \arg\min_{\boldsymbol{\theta}} \frac{1}{N} \sum_{i=1}^{N} \mathcal{L}_{\mathrm{BC}}\Big(m_{\boldsymbol{\theta}}(\boldsymbol{o}_i), a_i^{\mathrm{T}}\Big). \tag{26}$$

For continuous actions, $\mathcal{L}_{\mathrm{BC}}$ may be a regression loss, while for discrete actions, it may be a classification loss [41]. This objective fits the student on the demonstration distribution. Exact reproduction additionally requires that the student inputs contain the information needed to choose the demonstrated actions.

For a Gaussian policy, $m_{\boldsymbol{\theta}}$ can represent the action mean introduced in Section 3.5.1. When the standard deviation is parameterized separately, a squared-error loss on the mean does not train it.

Closed-loop use can create distribution shift [41]. A student error may change the next state and lead to observations that were rare or absent in the teacher data. Further errors can then accumulate as the student acts outside its supervised training distribution. Low one-step imitation error is therefore insufficient evidence of closed-loop policy quality.

Behavior cloning can serve as pretraining rather than as the final policy-learning stage [30]. Its parameters initialize a policy for a downstream objective, providing a starting point shaped by the demonstrations. Fine-tuning then adapts them using the downstream signal. The benefit of transferring learned features depends on their suitability for the target task [56]. Task-specific features can limit that benefit. Transferring a policy also requires compatible input and action meanings, or a corresponding adaptation of its interface.

4 Dataset

The data used in this study are based on hydraulic base scenarios for the Hanoi water distribution network and arsenic contamination events. These components form the common basis for the precomputed offline dataset and the interaction data generated during online policy training.

For the offline dataset, additional predefined chlorine injection profiles are generated. Each profile is combined with a hydraulic base scenario and an arsenic schedule to compute and store the complete network response in advance as a trajectory for pretraining. During online policy training, chlorine inputs are instead generated by the policy as it interacts with the simulator.

Figure 1 summarizes how the shared hydraulic scenarios, arsenic schedules, and chlorine inputs form the precomputed offline trajectories and online interaction data.

Before any derived data are generated, the hydraulic base scenarios are assigned to separate training, validation, and test partitions. Every precomputed offline trajectory, online policy-training episode, and fixed control case inherits the partition of its base scenario. The exact allocation is reported in Section 4.6.

The chapter first introduces the hydraulic base scenarios and arsenic contamination events. It then describes the simulation framework, offline trajectories, and online interaction data, followed by the dataset partitions and fixed control cases, before concluding with reproducibility details and a dataset summary.

4.1 Hydraulic Base Scenarios

The study uses 100 hydraulic base scenarios, labeled `scenario-0` to `scenario-99`. Each scenario combines the Hanoi network model with generated 25-day demand patterns for each junction and a fixed set of sampled network parameters. The network model, demand patterns, and parameter uncertainties are described in the following subsections. Together, they define the hydraulic conditions of each scenario, which remain fixed whenever it is reused.

The collection is limited to 100 scenarios to keep trajectory generation, model training, and repeated validation within the computational budget while maintaining separate scenario-level training, validation, and test partitions.

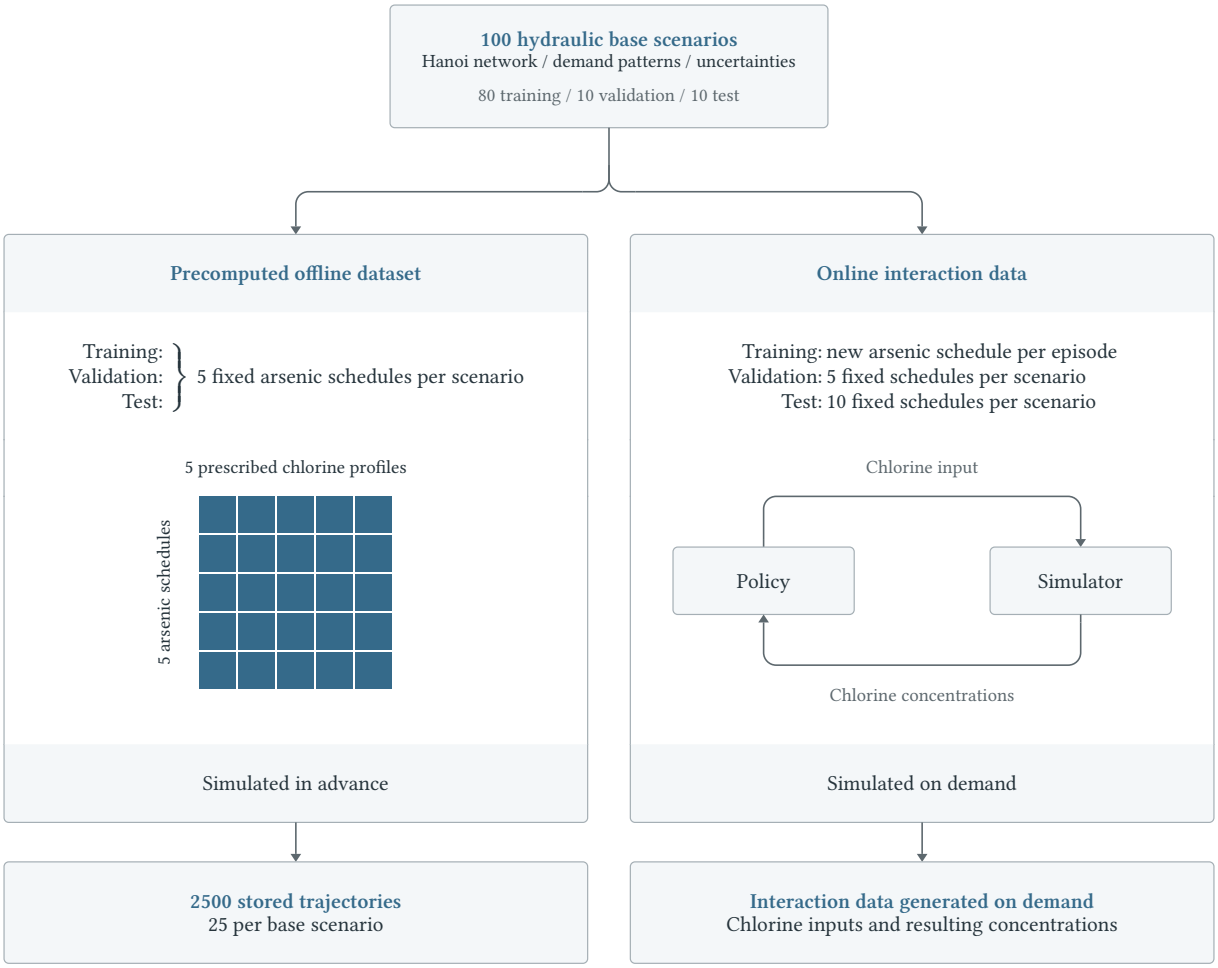


Figure 1: Construction of offline trajectories and online interaction data from shared hydraulic base scenarios. Each cell in the offline matrix represents one pairing of a chlorine profile and an arsenic schedule. Online policy training uses only the training split.

4.1.1 Hanoi Water Distribution Network

The Hanoi water distribution network was introduced by Fujiwara and Khang [17] as a mathematical model based on a planned trunk distribution system for Hanoi, Vietnam. It served as a test case for the cost-optimal design of a looped water distribution network. In this study, it provides the physical basis for the hydraulic and water-quality simulations used to generate the dataset.

The network comprises one fixed-head reservoir, 31 demand junctions, and 34 pipes arranged in three loops, with two terminal junctions. The reservoir is denoted as node 1 and supplies the network at a fixed hydraulic head of 100 m. Junctions 2–32 represent the locations where water is withdrawn to meet consumer demand. The topology and node numbering are shown in Figure 2.

In the classical Hanoi network optimization problem, pipe diameters are treated as design variables. In this study, all scenarios share the topology and pipe diameter configuration of the model provided through EPyT-Flow [4]. Hydraulic variation instead arises from differences in nodal demand patterns, base demand, pipe length, and pipe roughness.

This study uses reservoir node 1 as the only chlorine-injection location. Chlorine is introduced there as a mass-rate source and transported through the pipes with the hydraulic flow to the demand junctions.

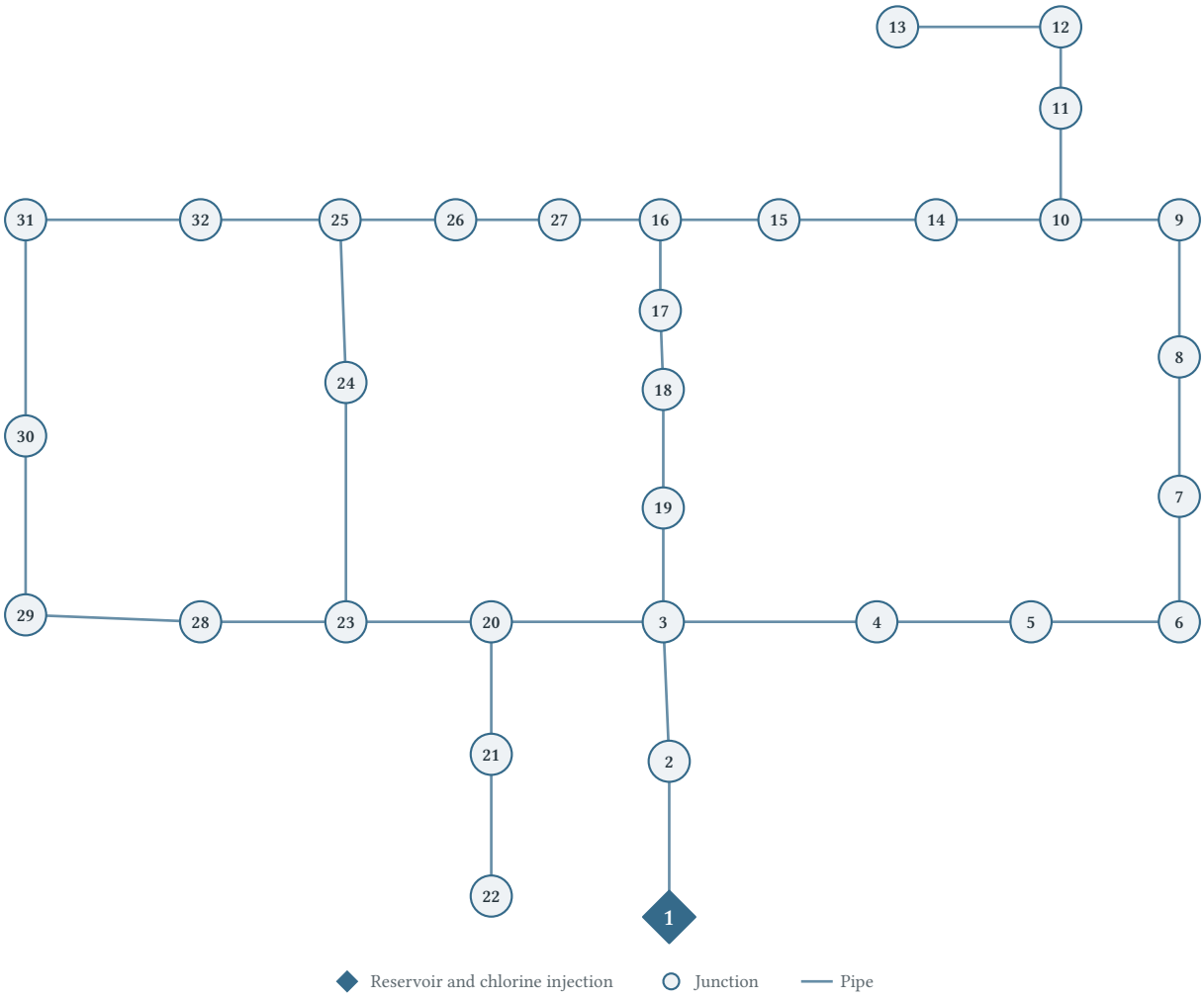


Figure 2: Topology and node numbering of the Hanoi water distribution network used in this study. The diamond denotes reservoir node 1, circles denote the 31 demand junctions, and lines denote the 34 pipes. The layout is schematic, and line lengths do not represent physical pipe lengths.

4.1.2 Demand Patterns

Nodal demand patterns describe how water demand varies over time. For each hydraulic base scenario, we generate a distinct set of these patterns using the LeakDB demand-pattern generator integrated into EPyT-Flow [4]. LeakDB was developed as a benchmark dataset for leakage diagnosis in water distribution networks [52]. Its demand patterns combine weekly and seasonal variation with random fluctuations. Scenario-specific random seeds make the patterns reproducible. Their assignment is documented in Appendix A.

The generator produces demand patterns covering 365 days. For this study, we use the first 25 consecutive days, beginning at day 0, so all scenarios cover the same calendar period. Restricting the demand period to 25 days limits the computational cost. The selected demand sequence remains fixed whenever a base scenario is reused. Figure 3 shows the nodal water demand for one example scenario over this period, with weekly and daily excerpts showing variation at shorter timescales.

4.1.3 Network Parameter Uncertainties

In addition to the scenario-specific demand patterns, this study varies base demand, pipe length, and pipe roughness. Each affected value is multiplied by a factor sampled uniformly between 0.8 and 1.2,

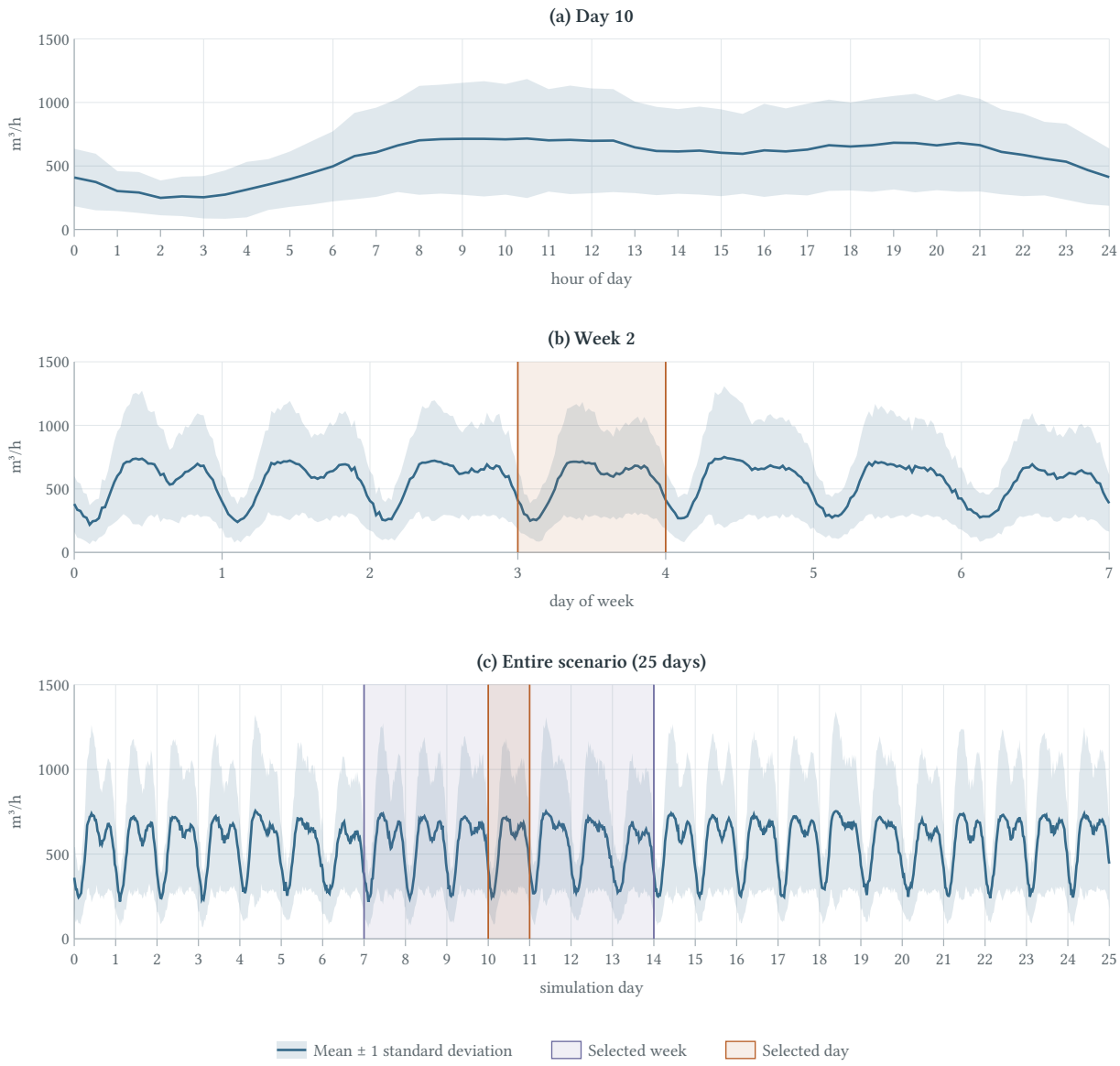


Figure 3: Nodal water demand over (a) day 10, (b) week 2, and (c) the selected 25-day period for `scenario-0`. Lines show the mean demand across junctions in m³/h, with bands indicating one population standard deviation. Demands are computed by the hydraulic model for this base scenario. All panels use 30-minute values. Colored intervals identify the selected week and day.

permitting deviations of up to ± 20 %. The demand-pattern multipliers also receive additive Gaussian perturbations with a mean of zero and a standard deviation of 0.1.

These perturbations are sampled once for each scenario using a scenario-specific random seed as documented in Appendix A. The resulting uncertainty realization remains fixed whenever the scenario is reused. Together with the demand patterns, it determines the scenario's hydraulic conditions.

4.2 Arsenic Contamination Events

While demand patterns and network parameter uncertainties affect the hydraulic conditions over time and between scenarios, we introduce arsenic events as chemical disturbances to which the chlorine controller must respond.

Each event introduces As(III) at one source junction for a predetermined duration with a constant rate. The event is represented as a mass-rate source, meaning that the injected arsenic strength is specified

as a mass flow rate in mg/min rather than directly as a concentration. The injected arsenite creates a local concentration contribution that varies with nodal outflow, as described in Section 3.3.2. In this simulation, reactions with chlorine occur only in downstream pipes. Figure 4 illustrates downstream transport for a relatively mild and a relatively strong event, followed by temporally overlapping source events.

Multiple arsenic events with varying start time, duration, location, and severity may occur during a single simulation run. All contamination events assigned to one specific run are specified by an arsenic schedule, which includes each event's start time, duration, source junction, and injection rate.

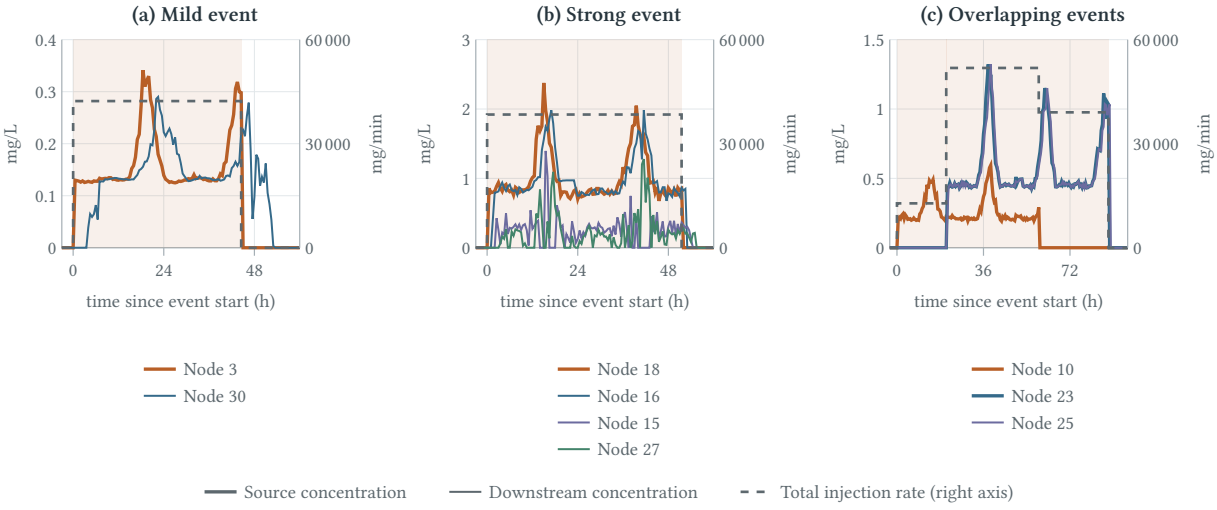


Figure 4: Illustrative As(III) concentration histories from the fixed control-validation cases with zero chlorine injection and a chlorine-free initial state. (a, b) A relatively mild and a relatively strong event show delayed downstream transport. Source-node peaks lie in the lower and upper quartiles, respectively, of the isolated events eligible for the transport illustration. (c) Two source events overlap in time. The examples use `scenario-86` with `arsenic-4`, at 55.5 h (a), `scenario-89` with `arsenic-1`, at 252 h (b), and `scenario-83` with `arsenic-2`, at 205 h (c) into the 25-day simulation. Schedule indices start at zero, and the generation seeds are specified in Appendix A. Time zero denotes the start of the first arsenic injection shown in each panel. Solid lines show nodal concentrations on the left axes, with thicker lines at source nodes and thinner lines at downstream nodes. Gray dashed steps show the combined mass injection rate of all active source events on the right axes. Orange shading marks the injection intervals. Flow-dependent dilution, downstream mixing, and travel times shape the concentration histories even under constant mass injection. Concentration ranges differ between panels, while all injection-rate axes use the same range.

4.2.1 Event Scheduler

A common scheduler defines all arsenic event schedules used throughout the study. The scheduler produces finite sets of fixed schedules for the precomputed offline dataset, generates a new schedule for each use of a scenario during online policy training, and produces separate fixed schedule sets for control validation and testing. The same event-generation rules therefore apply to all three uses. Fixed arsenic schedules are labeled `arsenic-<index>`, where the index starts at zero within each base scenario and schedule set.

Each schedule comprises 1–4 arsenic events, with all event counts having equal probability. Each event introduces arsenic at a randomly selected eligible junction at a constant, randomly sampled mass injection rate. Events at different junctions may overlap in time, but the same junction cannot be affected by two events at the same time.

4.2.2 Event Parameter Sampling

Arsenic contamination events are restricted to the middle 21 days of each 25-day simulation run, leaving two injection-free days at both the beginning and the end. Both injection-free periods are part of the evaluation protocol described in Section 6.1.1. Event durations are sampled in 30-minute increments between 2 and 72 h. Start times lie on the same 30-minute grid and are distributed across the weekly cycle within the 21-day event window.

Arsenic sources are selected from junctions 2–32, excluding the two terminal junctions 13 and 22, leaving 29 eligible source locations. Reservoir node 1 is also excluded as an injection source. After the duration and weekly start position have been sampled, the scheduler considers all eligible junction–week combinations. Sampling the weekly start position first aims to give each weekday an equal probability of an event start, despite duration constraints at the end of the final week. A combination remains eligible only if the complete event fits within the 21-day window, does not overlap another event at the same junction, and the local outflow permits at least one injection rate that satisfies the concentration and mass-rate limits described below. The junction and week are then sampled from these eligible combinations, and the exact event strength is determined afterward.

The event strength is selected according to the water outflow at the affected junction over the event interval. The resulting immediate local arsenic contribution remains between 0.1 and 10 mg/L throughout the event, while the mass injection rate is limited to 50 000 mg/min per event. These bounds are intended to produce disturbances that affect chlorine concentrations while remaining manageable through adjustments to chlorine injection. Both limits apply to each event separately. Simultaneous events may therefore exceed the mass-rate limit in total, and arsenic transported from another event may add to the concentration observed at an affected node.

Figure 5 summarizes the event characteristics and spatial reach of the fixed control-validation cases described in Section 4.6, with all cases simulated without chlorine. For the spatial-reach analysis, transported As(III) excludes the estimated immediate local contribution of newly injected arsenic at source nodes. Source events are grouped into contamination clusters when they are not separated by persistent clearance of the transported concentration sum across monitored nodes at 0.01 mg/L. The complete sampling procedure is documented in Appendix A.1.

4.3 Simulation Framework and Recorded Variables

Rather than collecting network responses from a physical water distribution system, this study generates them through simulation. Each simulation run combines one hydraulic base scenario, one arsenic schedule, and one chlorine input sequence. The resulting ordered sequence of hydraulic and water-quality states forms a trajectory.

The hydraulic states are computed using EPANET [42], while EPANET-MSX uses these states to transport two bulk species, free chlorine and As(III), and to model their reactions within the pipes [45]. Chlorine undergoes first-order decay and also reacts with As(III), while longitudinal dispersion is not included. Only free chlorine and As(III) are tracked, while the oxidation product As(V) is not represented explicitly. Chlorine-induced oxidation of As(III) therefore does not represent removal of total arsenic from the water. The pipe reaction rates are given in Equation 60, and the simulator settings are documented in Appendix B.

Hydraulic states and network outputs are computed and recorded at 30-minute intervals, while EPANET-MSX advances water-quality transport and reactions using an internal five-minute time step. Each simulation run covers 25 days, corresponding to 1200 30-minute intervals. The simulation records the initial state at 0 h and the state after each interval, giving 1201 samples through the 600 h endpoint. At the start of each simulation, chlorine injection and the concentrations of chlorine and As(III) throughout the network are initialized to zero.

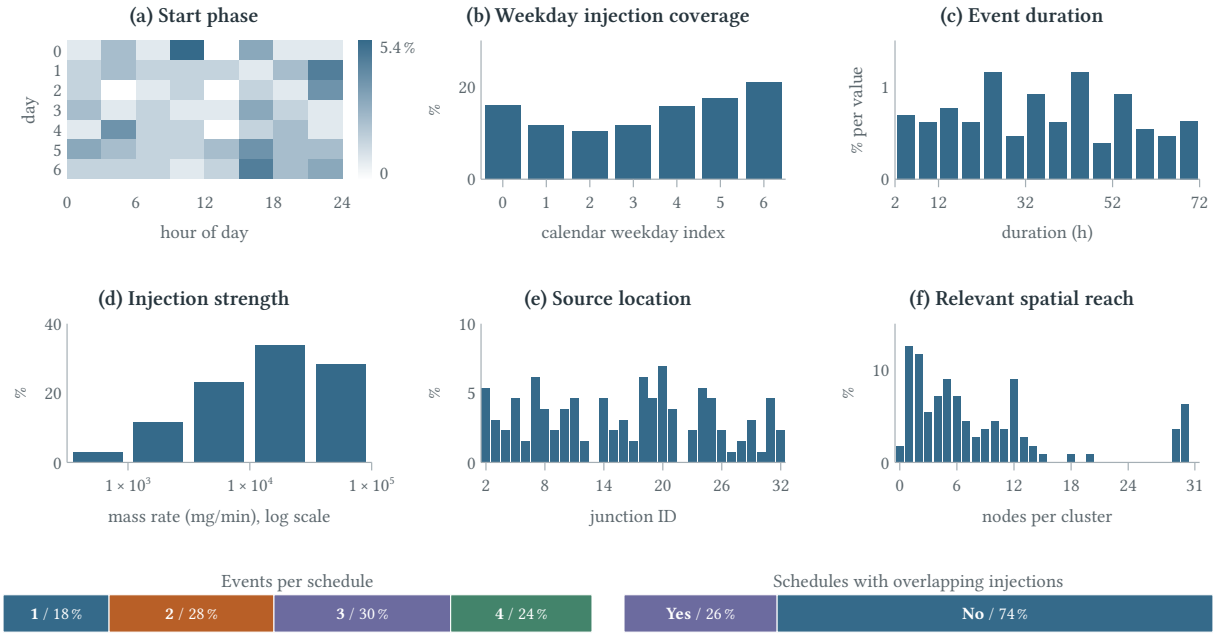


Figure 5: Arsenic-event statistics for the fixed control-validation cases, comprising 130 events in 50 schedules (5 per validation hydraulic scenario, seed 8129). All panels use these same cases, simulated without chlorine. Transported As(III) excludes the estimated immediate local contribution of newly injected arsenic at source nodes. (a) Colors show event-start shares by simulation day modulo seven and three-hour interval. (b) For each calendar weekday index (0–6), active-injection steps are divided by all observed 30-minute steps of that weekday across complete episodes. Weekday indices follow the recorded calendar times. (c) Five-hour classes span 2–72 h. Each event percentage is divided by the number of admissible half-hour duration values in its class, including 72 h in the final class. (d, e) Bars show event percentages. Source junctions 13 and 22 are excluded, as is reservoir 1. (f) The 111 contamination clusters show the number of nodes exceeding 0.01 mg/L transported As(III) at any sampled time, including the clearance tail. Clusters group source events that are not separated by persistent clearance of the transported concentration sum across monitored nodes at 0.01 mg/L, retaining zero-reach clusters. The bottom bars show the percentages of schedules containing one to four events (left) and with or without overlapping injections (right). Local source correction follows Appendix B.1.

Chlorine and As(III) concentrations are reported in mg/L, while source mass injection rates are expressed in mg/min. At an injection node, newly introduced As(III) is mixed with the outgoing water, but no chlorine–arsenic reaction is modeled at the junction itself. The reaction begins after the water enters a pipe.

The recorded network variables used in this study are demand, chlorine concentration, and As(III) concentration at junctions 2–32, together with the flow through all 34 pipes. Node 1 remains the reservoir and chlorine source and is therefore not included among the monitored demand junctions. Recorded values are exact simulator outputs without any artificially added noise.

The stored hydraulic results of each base scenario are reused to determine nodal water outflow from demand and pipe flows. This outflow is used to set the arsenic mass injection rates described in Section 4.2. Only a subset of the recorded variables is used as model inputs.

4.4 Offline Dataset Construction

The precomputed offline trajectories provide the fixed simulation data used by the pretraining pipeline. Each trajectory combines a fixed arsenic schedule with a predefined chlorine injection profile and stores the resulting hydraulic and water-quality response of the network. The aim is to provide examples from which a forecasting model can learn how chlorine concentrations evolve under different injection sequences and operating conditions. For this purpose, the injection profiles vary in their temporal structure to generate a range of network responses. The following sections describe the chlorine injection profiles, fixed arsenic schedule sets, and resulting trajectories.

4.4.1 Chlorine Injection Profiles

Prescribed chlorine injection profiles are generated in advance. For each of the 100 hydraulic base scenarios, five such profiles are generated, resulting in 500 profiles across the complete offline dataset. Each profile contains 1200 actions, one for each executed 30-minute interval. Profiles are labeled `chlorine-<index>`, with indices from 0 to 4 within each base scenario.

Profile generation begins by dividing the 25-day period into five contiguous segments of equal length. The segments represent constant, periodic, piecewise-constant, ramp, and random-walk inputs, and their order varies between profiles. Across all segment types, injection rates remain between 0 and 500 000 mg/min, and the change between two consecutive 30-minute values does not exceed 50 000 mg/min.

These profiles are used exclusively in the precomputed offline dataset, and their complete generation procedure is documented in Appendix A.2.

4.4.2 Fixed Arsenic Schedule Sets

A finite set of arsenic schedules is generated for the precomputed offline trajectories using the event-generation procedure described in Section 4.2. Five schedules are generated for each hydraulic base scenario.

4.4.3 Trajectory Construction and Stored Variables

The complete pairing of five prescribed chlorine profiles and five fixed arsenic schedules makes a total of 25 trajectories per scenario and 2500 trajectories across the 100 hydraulic base scenarios. Every complete trajectory contains 1201 reported samples, including the initial state and the final state after 25 days, generated by the coupled EPANET and EPANET-MSX simulation.

Each trajectory contains the time series of the network variables described in Section 4.3, together with the applied chlorine injection profile and the corresponding arsenic schedule. The selection of model inputs and prediction targets is described in Section 5.5.3.

Figure 6 illustrates the chlorine and arsenic injections and the resulting nodal concentrations over a complete 25-day trajectory.

4.5 Online Interaction Data

Each online policy-training episode combines one hydraulic base scenario from the training partition defined in Section 4.6 with a newly generated arsenic schedule. The policy determines the chlorine sequence as the simulation advances, and the simulator computes the network response defined in Section 4.3. The resulting observations, actions, and network responses provide the interaction data for policy training.

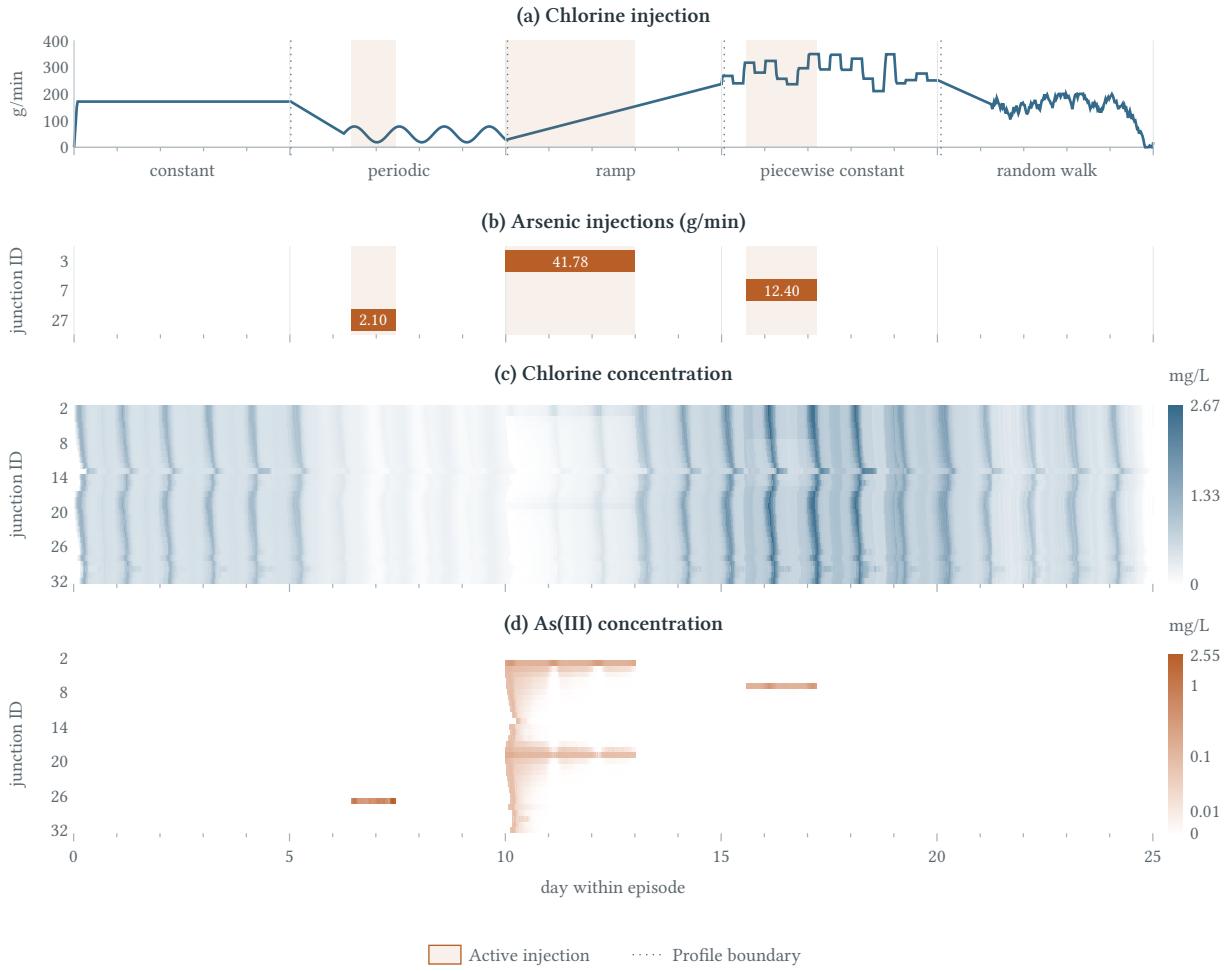


Figure 6: Example trajectory from the offline validation split for `scenario-85`, beginning at calendar day 0, with arsenic schedule `arsenic-4` (seed 6104) and chlorine profile `chlorine-3` (seed 7104). All panels share the episode time axis. Dotted lines and labels in (a) mark the chlorine-profile segments. Source bars identify injection nodes and rates. Heatmaps show individual junctions in ascending order and all 30-minute records. The As(III) color scale uses $\ln\left(1 + \frac{a}{\hat{a}}\right)$, where a is the As(III) concentration and $\hat{a} = 0.01\,\text{mg/L}$.

4.6 Training, Validation, and Test Data

The hydraulic base scenario is the indivisible split unit. Every prescribed chlorine profile, arsenic schedule, precomputed offline trajectory, online policy-training episode, and fixed control case inherits its scenario's assignment. Keeping each hydraulic realization within one partition prevents training and evaluation data from sharing demand, hydraulic, and chemical histories. The 100 base scenarios are divided into 80 training, ten validation, and ten test scenarios.

The precomputed offline trajectories are distributed across all three partitions. The training partition provides data for fitting forecasting models and scenarios for online policy training. Validation supports model development, while the test partition is reserved for final evaluation on unseen hydraulic realizations.

For control validation and testing, we generate separate collections of fixed arsenic schedules, referred to as catalogs, using the procedure described in Section 4.2. These catalogs are separate from the schedule sets used for the offline dataset. Each validation or test scenario is combined with individual schedules to form fixed control cases. Validation scenarios are paired with five schedules each, while

test scenarios are paired with ten. Table 1 summarizes the scenario allocation and the associated offline trajectories, online training use, and fixed control cases.

Table 1: Scenario-level partitions and their associated data. The 'Control schedules' column gives the number of fixed arsenic schedules per scenario for control evaluation, or indicates resampling per training episode. The 'Control cases' column gives the total number of fixed control cases. Offline trajectories use separate schedule sets.

Partition	Scenario IDs	Offline trajectories	Control schedules	Control cases
Training	0–79	2000	Resampled	–
Validation	80–89	250	5	50
Test	90–99	250	10	100

4.7 Reproducibility

All components of the dataset are generated deterministically. This includes the demand patterns, hydraulic parameter realizations, chlorine profiles, and arsenic schedules. Regenerating the dataset from the same configuration therefore reproduces the same hydraulic base scenarios, offline trajectories, and fixed control cases.

The offline dataset and control runs reuse the same hydraulic base scenarios and parameter realizations, but use separate seeds for their arsenic schedules. Each precomputed offline trajectory can be traced to its hydraulic base scenario, fixed arsenic schedule, and prescribed chlorine profile. The fixed control-validation and control-test schedules are likewise associated with their respective base scenarios. The seeds, generation procedures, and artifacts required for reproduction are documented in Appendix A.

4.8 Dataset Summary

The hydraulic scenarios, arsenic schedules, and chlorine inputs together define the conditions under which the learning and control methods are studied. Table 2 summarizes these components in a compact overview of the network, temporal resolution, offline trajectories, and online interaction data.

The precomputed offline trajectories provide the data for forecasting and for generating MPC teacher demonstrations for behavior cloning. The online interaction data comprise the observations, actions, and network responses generated during PPO training. The next chapter explains how each data source is prepared for the respective models and incorporated into their training.

Table 2: Summary of the dataset composition and online interaction data.

Item	Contents
Shared components	
Network	Hanoi: 32 nodes, including one reservoir, and 34 pipes.
Base scenarios	100 generated Hanoi realizations, scenario IDs `scenario-0` to `scenario-99`, using the LeakDB demand-pattern generator and fixed scenario-specific model uncertainty realizations.
Simulation period	25 consecutive days, starting at calendar day 0 for all scenarios.
Spatial coverage	All 31 demand junctions and all 34 pipes.
Temporal resolution	Network variables recorded at 30-minute intervals.
Simulation outputs	Exact simulator outputs without added measurement noise.

Table 2 (continued)

Item	Contents
Precomputed offline dataset	
Chlorine profiles	Five mixed-family profiles per base scenario, giving 500 across the complete collection.
Offline arsenic schedules	Five fixed schedules per base scenario. The schedules remain separate across dataset partitions and are used only in the offline dataset.
Precomputed trajectories	A complete pairing of five chlorine profiles and five fixed arsenic schedules produces 25 trajectories per scenario and 2500 in total.
Stored simulation records	Time series of demand, pipe flow, chlorine and As(III) concentrations, with applied chlorine inputs and associated arsenic schedules.
Control training and evaluation	
Training arsenic schedules	One newly generated schedule for each policy-training episode, separate from all fixed schedule sets.
Validation catalog	Five fixed arsenic schedules per validation scenario, 50 cases in total.
Testing catalog	Ten fixed arsenic schedules per test scenario, 100 cases in total.

5 Methodology

This chapter describes the two training approaches compared in this study. Both policies use the same recurrent architecture and chlorination-control setting but differ in their initialization. The baseline approach trains a policy from an ordinary initialization. The pretrained approach initializes a policy from teacher demonstrations generated using a prediction model and then trains it online under the same conditions. We refer to this initialization procedure as MPC-guided policy pretraining. The comparison assesses how this procedure affects online sample efficiency and final control performance under a common interaction budget.

Our pretraining pipeline first trains a multistep forecaster to predict the network's chlorine response to candidate injection sequences. MPC uses these forecasts to evaluate the candidate sequences and select teacher actions. Behavior cloning trains the recurrent actor to reproduce these actions, and PPO subsequently fine-tunes the initialized policy through interaction with the simulator. The forecaster and MPC optimizer are used only during pretraining and are not part of PPO fine-tuning, evaluation, or deployment.

In this chapter, we first define the chlorination-control problem, including its partial observability, operational objectives, and action constraints. We then describe the reward function and policy observations before presenting the shared recurrent policy-learning framework, the baseline approach, and the online training procedure. Next, we describe cross-fitted teacher generation, followed by behavior-cloned policy initialization and PPO fine-tuning. Finally, we define the principal experimental comparison and document the implementation and reproducibility requirements. Both approaches use the Hanoi network, the generated hydraulic base scenarios, and arsenic schedules sampled using the procedure described in Section 4.2. The prescribed chlorine injection profiles are used to generate the offline trajectories for pretraining.

5.1 Chlorination-Control Problem

First, we establish the chlorination-control problem shared by both approaches. This includes defining the operational objectives and partial observability of the task and specifying the control action and actuator constraints.

5.1.1 Operational Objectives and Partial Observability

We begin by considering what the controller can influence within the testbed. Chlorine is injected at a single central source, whereas its concentration is monitored at multiple locations throughout the network. At every control step, the policy must therefore select one source action that maintains suitable chlorine concentrations across all monitored nodes while also responding to arsenic events that may originate at different junctions. This creates a network-wide control problem, since a decision taken at one location must account for responses that occur at different locations and at different times.

The controller does not, however, have direct access to the complete physical condition of the network. We consequently formulate the task as a POMDP. Its latent state comprises the hydraulic and water-quality conditions, current demands, contamination events, uncertain physical properties, and the transport history that connects earlier conditions to the present network response. By contrast, the final deployed policy receives only the restricted observation defined later in this chapter.

This distinction is important because the controller's current observation does not uniquely identify the underlying state. Similar chlorine patterns can result from different demand conditions, arsenic events, pipe properties, or previous transport dynamics. A single measurement snapshot is therefore insufficient to determine which process produced the observed concentrations. A sequence of observations provides additional temporal evidence, which motivates the use of a recurrent policy that can combine current measurements with information retained from previous time steps.

With this information constraint in place, we can define the operational objectives for the chlorine controller. During normal operation, chlorine concentrations at all monitored nodes should remain within the range of 0.2–0.4 mg/L. These bounds are adopted from the IJCAI 2025 AI for Drinking Water Chlorination Challenge [3], which posed a similar control task on a larger network with wastewater contamination involving pathogens and organic matter. Its controller was required to respond to contamination events while maintaining node-level chlorine concentrations within this range. The same interval is used here to define the normal-operation objective. During arsenic events, however, the objective shifts to effective treatment through a rapid, but appropriate response. For this, the upper bound at monitored nodes is temporarily lifted from 0.4 to 5.0 mg/L to allow for higher injections and effective treatment. This ceiling follows the health-based guideline value for free chlorine in drinking water published by the World Health Organization (WHO) [55]. Within this enlarged operating range, the controller should maintain or restore sufficient chlorine residuals at affected nodes while limiting the controllable As(III) burden. The policy-independent periods over which normal operation and contamination response are evaluated are defined in Section 6.3.1.

Maintaining a positive chlorine residual also supports both arsenic treatment and indirect event detection. It provides free chlorine that can react with As(III) after contaminated water enters the pipe network. A sufficient chlorine residual can also make disturbances visible through changes in chlorine concentration to which the controller can respond. Such variations may indicate ongoing arsenic contamination, but they can also result from changing demand patterns and the corresponding change of flow through the network. The observed chlorine response therefore does not uniquely identify its underlying cause, reinforcing the partial observability of the control problem.

5.1.2 Control Action and Actuator Constraints

As introduced in Section 4.1.1, node 1 is the only chlorine-injection location, and the corresponding actuator is both bounded and rate-limited. We now formalize how the policy interacts with this actuator.

At each 30-minute control step t, the policy produces a scalar normalized action $a_t \in [-1, 1]$ from the observation after step $t-1$. The resulting observation is recorded after step t. This action does not specify an absolute injection rate. Instead, it determines the direction and magnitude of the change relative to the previous step's injection. An action of 1 requests the maximum increase, an action of -1 requests the maximum decrease, and an action of 0 leaves the injection rate unchanged.

Let u_t denote the physical chlorine injection rate in mg/min. Let $\Delta u_{\max}$ = 50 000 mg/min denote the maximum permitted change per control step and $u_{\max}$ = 500 000 mg/min the upper rate limit. The normalized action is scaled by $\Delta u_{\max}$ and added cumulatively to the previous rate. The resulting value is then clipped to the admissible interval:

$$u_t = \mathrm{clip}(u_{t-1} + \Delta u_{\max}\, a_t, 0, u_{\max}), \quad a_t \in [-1, 1]. \tag{27}$$

Clipping the injection rate ensures that the actuator cannot request a negative injection rate or exceed the configured upper bound. When the previous rate is close to either bound, the realized change can therefore be smaller than the change represented by the normalized action. Each control episode starts with an injection rate u_0 of 0 mg/min, so the policy establishes dosing from the first control step. Both policy approaches use this same action representation, initial value, and actuator constraints, so their comparison is not affected by differences in how control actions are applied to the network.

Both limits are practical choices rather than empirically optimized actuator parameters. The upper bound keeps offline profiles, MPC plans, and online policy training within the injection range used to generate the forecaster's training data. This alone does not ensure coverage of all resulting sequences or network states. The rate-change limit balances dosing precision against response speed: smaller values could delay contamination response, while larger values would amplify small variations in normalized policy output into larger physical dosing changes, potentially complicating fine adjustments and promoting oscillations.

5.2 Reward Function

Both training approaches use the same reward design to maintain adequate chlorine levels throughout the network. The reward also discourages unnecessary concentration fluctuations and abrupt changes in the requested dosing increments. When chlorine levels fall, as can occur during contamination, the weighting of these objectives shifts to give greater priority to restoring sufficient residuals.

To express these objectives, the reward combines three penalty components: chlorine tracking, wasted chlorine movement, and changes in the normalized action. The reward is the negative of their sum,

$$r_t = -\Big(P_{\mathrm{track},t} + P_{\mathrm{move},t} + P_{\mathrm{action},t}\Big). \tag{28}$$

Its maximum value is zero, and larger penalties produce more negative rewards. A continuous recovery factor $\beta_t \in [0, 1]$ adjusts the tracking and movement costs in response to low chlorine residuals. At $\beta_t = 0$, these costs use their ordinary weighting, while $\beta_t = 1$ favors recovery from low chlorine residuals.

Let N be the number of monitored nodes, indexed by $v = 1, \ldots, N$. Let $c_{v,t}$ denote the chlorine concentration at monitored node v after control step t. Let c^* = 0.30 mg/L denote the target concentration and c_s = 0.01 mg/L the reference concentration scale.

5.2.1 Chlorine Tracking

The tracking penalty $P_{\text{track},t}$ averages the squared, scaled deviations of chlorine concentrations from the target c^* across all monitored nodes. It is defined as

$$P_{\text{track},t} = L_t + f_t\, U_t. \tag{29}$$

Here, L_t measures deviations below the target and U_t measures deviations above it, while f_t weights the upper contribution. Evaluating these contributions separately allows the pressure to restore a low residual to be maintained while the penalty for concentrations above the target is temporarily reduced during recovery. The two contributions are defined as

$$\begin{aligned} L_t &= \frac{1}{N}\sum_{v=1}^{N}\left(\frac{\max\bigl(c^* - c_{v,t},\,0\bigr)}{c_s}\right)^2, \\ U_t &= \frac{1}{N}\sum_{v=1}^{N}\left(\frac{\max\bigl(c_{v,t} - c^*,\,0\bigr)}{c_s}\right)^2. \end{aligned} \tag{30}$$

The reference scale c_s sets the magnitude of the tracking contributions relative to the other reward components. Deviations smaller than this scale make relatively small contributions, while larger deviations receive increasingly strong penalties through the quadratic scaling. This weighting emphasizes larger concentration errors while remaining sensitive to smaller deviations. Averaging across all monitored nodes gives each location equal weight.

The recovery factor β_t determines the upper weighting through

$$f_t = (1 - \beta_t) + 0.0001\,\beta_t. \tag{31}$$

At $\beta_t = 0$, both sides of the target receive equal weight. As β_t increases, f_t decreases linearly. At $\beta_t = 1$, the upper contribution is reduced to one ten-thousandth of its ordinary value, while the lower contribution remains unchanged. The upper weighting therefore remains positive but is substantially reduced during recovery. This allows increased chlorine levels at some nodes to correct a low residual elsewhere in the network.

5.2.2 Wasted Chlorine Movement

The wasted-movement penalty $P_{\text{move},t}$ evaluates changes in chlorine concentrations between two consecutive steps. It penalizes the part of a concentration change that does not reduce the absolute error relative to the target. The penalty is defined as

$$P_{\text{move},t} = 0.1\left[(1 - \beta_t)\, M_t + \beta_t\, \widetilde{M}_t\right]. \tag{32}$$

Here, M_t averages the wasted-movement costs equally across all monitored nodes, while $\widetilde{M}_t$ weights them according to each node's previous chlorine concentration. The recovery factor β_t controls the transition between these averages. The coefficient 0.1 sets the contribution of this penalty relative to the other reward components.

To distinguish corrective movement from wasted movement, let $m_{v,t}$ denote the total concentration change at node v and $d_{v,t}$ the reduction in its absolute deviation from the target,

$$\begin{aligned} m_{v,t} &= \bigl|c_{v,t} - c_{v,t-1}\bigr|, \\ d_{v,t} &= \max\Bigl(\bigl|c_{v,t-1} - c^*\bigr| - \bigl|c_{v,t} - c^*\bigr|,\,0\Bigr). \end{aligned} \tag{33}$$

The remaining movement is $w_{v,t} = \max\big(m_{v,t} - d_{v,t}, 0\big)$. This wasted movement is zero for a correction that approaches the target without crossing it. It penalizes movement away from the target, as well as changes that pass through the target and continue beyond it. The corresponding node-level cost is

$$W_{v,t} = \left(\frac{w_{v,t}}{c_s}\right)^2. \tag{34}$$

The reference scale c_s follows the scaling used for chlorine tracking. Small amounts of wasted movement therefore make relatively small contributions, while larger amounts receive increasingly strong penalties through squaring.

During recovery, the weighting shifts toward nodes whose previous concentration was not already high. This allows concentrations at other nodes to rise while low residuals are being restored. To obtain a gradual transition between full and zero node weight, we use the clipped smoothstep function

$$S(x) = z^2\,(3 - 2z), \quad z = \mathrm{clip}(x, 0, 1). \tag{35}$$

Let c_g = 0.35 mg/L be the full-weight threshold and Δc = 0.05 mg/L the transition width. The node weight is then

$$g_{v,t} = 1 - S\left(\frac{c_{v,t-1} - c_g}{\Delta c}\right). \tag{36}$$

It is one at or below 0.35 mg/L, decreases smoothly between 0.35 and 0.40 mg/L, and is zero at or above 0.40 mg/L. During recovery, this reduces the penalty for further increases at nodes with already high concentrations, giving priority to restoring low residuals elsewhere. Using the previous concentration prevents an increase from lowering its own weight during the step in which that increase is evaluated. The ordinary and weighted averages are

$$M_t = \frac{1}{N}\sum_{v=1}^{N} W_{v,t}, \quad \widetilde{M}_t = \frac{\sum_{v=1}^{N} g_{v,t}\,W_{v,t}}{\sum_{v=1}^{N} g_{v,t}}. \tag{37}$$

When every node weight is zero, $\widetilde{M}_t$ is defined as zero. Otherwise, normalization by the sum of the weights prevents its scale from decreasing merely because fewer nodes contribute. At $\beta_t = 0$, the penalty uses the ordinary average M_t. At $\beta_t = 1$, it uses the weighted average $\widetilde{M}_t$, while intermediate values blend the two.

5.2.3 Action Change

The action-change penalty $P_{\text{action},t}$ penalizes differences between consecutive normalized actions. It discourages abrupt changes in the requested dosing increments and is defined as

$$P_{\text{action},t} = 0.1\left(\frac{a_t - a_{t-1}}{0.05}\right)^2. \tag{38}$$

As defined in Section 5.1.2, a_t requests an injection increment of $\Delta u_{\max}\,a_t$. The difference $a_t - a_{t-1}$ therefore measures how this requested increment changes between consecutive steps. Repeating the same action gives zero action-change penalty, even when that action continues to increase or decrease the injection rate.

The reference scale of 0.05 corresponds to a change of 2500 mg/min in the requested increment, or 5 % of the maximum permitted increment of 50 000 mg/min. Smaller differences make relatively small contributions, while larger differences receive increasingly strong penalties through quadratic scaling. A weight of 0.1 balances the action-change cost against the tracking and movement penalties.

The penalty uses the requested actions rather than the realized injection changes. Actuator clipping therefore does not reduce the penalty when the injection rate approaches its lower or upper bound.

5.2.4 Recovery Mechanism

The recovery factor β_t adjusts the tracking and movement penalties when low chlorine residuals require corrective action. Its value depends on the severity of the observed depletion, the time since the latest recovery trigger, and the time elapsed since the start of the episode. These contributions are combined as

$$\beta_t = v_t \, \eta_t \, S(\ell_t). \tag{39}$$

Here, η_t stores the recovery demand at the latest trigger, ℓ_t controls how long its effect persists, and v_t gradually activates recovery at the beginning of an episode. The function S is the clipped smoothstep function defined above.

Recovery demand is determined by the lowest measured chlorine concentration, so depletion at an individual node can affect the reward weighting. Let c_r = 0.20 mg/L be the recovery trigger concentration. Using the transition width Δc defined above, the instantaneous demand is

$$\rho_t = S\left(\frac{c_r - \min_{1 \le v \le N} c_{v,t}}{\Delta c}\right). \tag{40}$$

It is one when the minimum is at or below 0.15 mg/L and zero when the minimum is at or above 0.20 mg/L. Between these thresholds, the demand increases smoothly as the minimum concentration falls. The interval of 0.05 mg/L determines the concentration range over which this transition occurs.

To maintain the recovery weighting beyond an individual low measurement, each trigger starts a hold period followed by a gradual release. Let k_t denote the number of steps since the latest trigger, capped at 16. The release remainder is

$$\ell_t = 1 - \mathrm{clip}\left(\frac{k_t - 4}{12}, 0, 1\right). \tag{41}$$

It remains one for four steps after a trigger and then decreases linearly to zero over twelve further steps. Applying S to this remainder gives a smooth reduction in the recovery weighting. Without another trigger, the product $\eta_t \, S(\ell_t)$ therefore remains constant during a two-hour hold and then decays to zero over a six-hour release.

A new trigger occurs when ρ_t is positive and at least as large as $\eta_{t-1} \, S(\ell_{t-1})$, the previous stored demand after applying the release function. The trigger sets $\eta_t = \rho_t$ and restarts the hold period at $k_t = 0$. Otherwise, the stored value is held and k_t increases by one until it reaches 16, when η_t is set to zero. Persistent or renewed depletion can therefore restart the recovery period before the previous response has decayed completely.

At the beginning of an episode, recovery remains inactive for the first twelve executed control steps. During this period, $\eta_t = 0$ and $k_t = 16$, giving the policy an initial period to establish dosing before low residuals activate recovery. From step 13 onward, the state follows the update rule above. The startup activation v_t introduces its effect linearly over twelve steps,

$$v_t = \mathrm{clip}\left(\frac{t - 12}{12}, 0, 1\right). \tag{42}$$

Startup activation reaches its full value at step 24, after which $\beta_t = \eta_t \, S(\ell_t)$. The initial delay and gradual activation are intended to prevent chlorine concentrations, which are initialized to zero at

episode start, from immediately encouraging excessive injection through the recovery mechanism. The initial simulation sample does not advance this schedule.

5.2.5 Reward Components and Parameters

Table 3 summarizes the three penalty components and their roles. The movement and action-change penalties compare consecutive executed control steps and are therefore zero at the first executed control step of an episode.

Table 4 lists the selected reference scales, penalty weights, and concentration thresholds, together with the timing parameters of the recovery mechanism. The reference scales and weights set the relative contributions of the penalties, while the thresholds define transitions in node weighting and recovery demand. The timing parameters specify the startup delay and gradual activation, together with the hold and release periods following a recovery trigger.

Table 3: Components of the reward function.

Component	Penalty	Role
Chlorine tracking	$P_{\mathrm{track},t} = L_t + f_t\, U_t$	Penalizes deviations from 0.30 mg/L across all monitored nodes. Recovery reduces the weighting above the target while preserving the full penalty below it.
Wasted chlorine movement	$P_{\mathrm{move},t} = 0.1\left[(1-\beta_t)\, M_t + \beta_t\, \widetilde{M}_t\right]$	Penalizes concentration changes that do not reduce the absolute error relative to the target. Recovery shifts the weighting toward nodes with lower previous concentrations.
Action change	$P_{\mathrm{action},t} = 0.1\left(\frac{a_t - a_{t-1}}{0.05}\right)^2$	Penalizes changes in the requested injection increment between consecutive steps.

Table 4: Parameters of the reward function. Each control step corresponds to 30 min.

Parameter	Value
Chlorine tracking	
Target concentration c^*	0.30 mg/L
Tracking reference scale	0.01 mg/L
Minimum upper-tracking factor	0.0001
Wasted chlorine movement	
Movement reference scale	0.01 mg/L
Movement penalty weight	0.1
Full-weight threshold	0.35 mg/L
Zero-weight threshold	0.40 mg/L
Action change	
Action-difference reference scale	0.05
Action-change penalty weight	0.1

Table 4 (continued)

Parameter	Value
Recovery mechanism	
Threshold for $\rho_t = 1$	0.15 mg/L
Threshold for $\rho_t = 0$	0.20 mg/L
Hold period after a trigger	4 steps / 2 h
Release duration after the hold	12 steps / 6 h
Startup delay	12 steps / 6 h
Startup activation period	12 steps / 6 h

5.2.6 Parameter Selection and Optimization

The reward parameters combine heuristic design choices with a focused empirical comparison. The target of 0.30 mg/L is the midpoint of the normal-operation range of 0.2–0.4 mg/L. The movement-weight thresholds of 0.35 mg/L and 0.40 mg/L were chosen through inspection of chlorine trajectories. The recovery thresholds follow the assumption that concentrations below the lower operating bound may indicate contamination-related depletion. Recovery demand therefore begins below 0.20 mg/L and reaches its maximum at 0.15 mg/L. These thresholds express a heuristic response to low residuals, whose possible causes were discussed above.

The reference scales and penalty weights were chosen during exploratory development of several reward formulations. We kept values that produced useful control behavior in these preliminary runs, without a systematic comparison of the scales or weights. The startup delay and activation duration were chosen to bring recovery into full effect within the first day of an episode. These choices were held fixed during the subsequent parameter search.

We subsequently compared alternative hold periods, release durations, and minimum upper-tracking factors while keeping the selected policy-training hyperparameters fixed. Selection considered contamination response and chlorine control during normal operation on separate validation cases. Physical control performance was used because changing the reward parameters also changes the scale and meaning of the return. The tested configurations and selection procedure are documented in Appendix D. The comparison led to the reward configuration in Table 4.

5.3 Policy Observations

Both policies receive the same 43-dimensional observation, comprising chlorine measurements, dosing information, calendar features, and the state variables of the recovery mechanism. Node-level chlorine measurements describe the spatial distribution of residuals, while the current injection and its realized change describe the applied dosing. The calendar features indicate the daily, weekly, and annual phases of the periodic demand patterns.

During stochastic training, the action is sampled after the actor's recurrent state has been updated, so that state does not directly record the sampled value. The observation returned after step t therefore includes the selected action a_t, clipped to $[-1, 1]$, together with the applied injection u_t, and is used to select a_{t+1}. Including both the action and the realized injection change distinguishes the requested increment from the change achieved under the actuator constraints in Section 5.1.2.

The most recent action also serves as the reference for the action-change penalty at the next control step. Because actuator clipping can map different normalized actions to the same realized injection change, that change alone does not determine the penalty for a given subsequent action.

The recovery factor β_t, release remainder ℓ_t, and startup activation v_t provide the temporal context of the reward weighting. These values depend on earlier chlorine depletion and elapsed control steps, so current chlorine measurements alone are insufficient to determine them. Including them in the observation allows the policy to account for the current weighting of tracking and movement costs and the progress of the recovery mechanism.

All observation components can be obtained from chlorine measurements and the controller's own action and time records. Arsenic, demand, pressure, pipe flow, and forecasts are excluded from the policy inputs. The deployed controller therefore uses only a subset of the recorded simulation variables and operates independently of the forecasting model used during pretraining. Table 5 lists the raw observation components. Their normalization is described in the following section on recurrent policy learning.

Table 5: Policy observation for the 31 monitored junctions of the Hanoi network. Time indices refer to the completed control step. Vector indices are zero-based and follow the order of the observation components.

Observation component	Entries	Indices	Definition and purpose
Node chlorine	31	0–30	Chlorine concentration in mg/L at each monitored junction, ordered according to the node list. These entries preserve the spatial distribution available to the controller.
Current injection	1	31	Physical injection rate u_t in mg/min applied during the completed control step. It is zero in the initial observation.
Realized injection change	1	32	Difference $u_t - u_{t-1}$ in mg/min, including the effect of actuator clipping. It is zero in the initial observation.
Time of day	2	33–34	Sine and cosine encoding of the daily phase.
Time of week	2	35–36	Sine and cosine encoding of the weekly phase.
Time of year	2	37–38	Sine and cosine encoding of the annual phase.
Most recent action	1	39	Normalized action $a_t \in [-1, 1]$ applied in the completed control step to produce u_t. It is zero in the initial observation.
Recovery factor	1	40	Applied recovery factor β_t, including startup activation. It controls the weighting of the tracking and movement costs.
Release remainder	1	41	Fraction ℓ_t, equal to one at a trigger and throughout the hold period, decreasing linearly during release, and zero when the release is complete.
Startup activation	1	42	Factor v_t, equal to zero during the first twelve executed control steps, increasing linearly to one over the next twelve steps, and remaining one thereafter.

5.4 Recurrent Policy Learning and Baseline

We next introduce the recurrent policy-learning framework shared by both approaches, which builds on the control problem, reward design, and policy information defined above.

5.4.1 RecurrentPPO Architecture and Baseline

We use RecurrentPPO from SB3-Contrib [46], with its `MlpLstmPolicy` implementation, to process the 43-dimensional observation defined above. It extends PPO [44] within the Stable-Baselines3 ecosystem [39]. A feedforward multilayer perceptron supplied with a fixed window of past observations could also use this temporal context, although the window length would determine which past observations are directly available. Here, we use the LSTM mechanism to learn a compact representation of the observation history that is updated at each control step, without specifying a fixed input window for the policy. This choice provides a common recurrent architecture for evaluating MPC-guided pretraining, while a comparison with history-stacked multilayer perceptron policies lies outside the scope of the study.

At each time step, the observation vector is processed by two separate single-layer LSTMs with 128 hidden units, one for the actor and one for the critic. The actor and critic therefore learn their own recurrent representations rather than sharing a single memory state. Each LSTM output is subsequently processed by two fully connected layers of 64 units with hyperbolic-tangent activation functions. The actor uses the resulting representation to parameterize a probability distribution over the normalized control action, whereas the critic estimates the value of the current observation history.

Training proceeds in four parallel control environments. Each environment contributes 2048 consecutive transitions to a rollout, giving 8192 transitions per PPO update cycle. These transitions are divided into minibatches of 512 samples and reused for ten optimization epochs. Table 6 summarizes the remaining optimization settings. For a nominal training budget of 1 000 000 environment interactions and 8192 transitions per rollout, the implementation reaches 1 007 616 interactions before stopping.

Both observations and rewards are normalized online using statistics accumulated from the training environments. Each observation component is standardized by its running mean and variance and then clipped to the interval $[-10, 10]$. Rewards are scaled using the running variance of discounted returns, with the same discount factor of $\gamma = 0.99$, and are likewise clipped to $[-10, 10]$. Later, during validation and testing, the learned observation statistics are held fixed and reward normalization is disabled. This preserves the input scale learned during training while allowing policy performance to be assessed using the original reward values.

The architecture and training procedure described above define the RecurrentPPO setup shared by both policy approaches. The baseline approach trains the recurrent policy from an ordinary random initialization. The behavior-cloned policy introduced later uses the same architecture and PPO configuration. Consequently, the two approaches differ in how the actor and its observation normalization are initialized, not in the representational capacity available during PPO training. The following subsection defines the hydraulic scenarios and arsenic schedules used to train both approaches under the common interaction budget.

Table 6: RecurrentPPO configuration used by both policy approaches.

Configuration item	Value	Role
Learning rate	1×10^{-4}	Sets the step size of the Adam optimizer.
Discount factor γ	0.99	Weights future rewards when estimating discounted returns.
GAE parameter λ	0.95	Controls the bias–variance trade-off of GAE.

Table 6 (continued)

Configuration item	Value	Role
PPO clipping range	0.2	Clips the surrogate objective to discourage excessively large policy updates.
Target KL divergence	0.02	Provides an early-stopping safeguard against excessively large policy updates.
Value-loss coefficient	0.5	Weights the critic loss within the joint PPO objective.
Entropy coefficient	0	Adds no separate entropy bonus beyond the stochasticity of the learned action distribution.
Maximum gradient norm	0.5	Clips the global gradient norm to stabilize optimization.
Rollout size	8192	Number of transitions collected per rollout.
Minibatch size	512	Number of transitions per optimization minibatch.
Optimization epochs	10	Number of passes over each rollout during a PPO update.

5.4.2 Training Scenario and Contamination Sampling

Both MPC-guided pretraining and policy learning use the same 80 hydraulic scenarios in the training partition, corresponding to scenario identifiers `scenario-0` to `scenario-79` in Table 1, defined in Section 4.6. This keeps the hydraulic scenario collection consistent across the two stages, although their action and state distributions may still differ. Diversity between the individual learning stages is instead maintained through the scenario-wise cross-fitting for MPC-guided pretraining described in Section 5.5.1 and through newly generated contamination schedules for policy training.

Each of the four parallel PPO environments receives its own deterministic permutation of the 80 training scenarios. This permutation is derived from the policy-training seed together with the environment index, so the environments generally encounter the same scenario collection in different orders. At every environment reset, the next scenario in the corresponding permutation is selected. After all 80 scenarios have been used once, the environment returns to the beginning and repeats the same order without reshuffling between cycles. This procedure gives every environment regular coverage of the complete training split while reducing the synchronization of identical hydraulic conditions across the parallel rollouts.

Whenever an environment selects a scenario, it also generates a new arsenic schedule. Unlike forecasting, validation, and testing, policy training does not draw from a finite set of schedules. Each training environment instead draws from an unbounded deterministic stream. Every generated schedule contains between one and four events and follows the common procedure for sampling event times, locations, durations, and strengths defined in Section 4.2. The policy therefore encounters new contamination conditions after every reset, even when it returns to a hydraulic scenario seen in an earlier cycle.

The stream must provide varied schedules while remaining reproducible. For this purpose, each parallel environment extends the dedicated control-training schedule seed of 8128 with its environment index. Within an environment, a separate episode counter is maintained for every hydraulic scenario. A schedule is therefore determined by the schedule seed, the environment-specific seed path, the scenario identifier, and the number of earlier episodes generated for that scenario. As a result, repeated visits to a scenario produce different schedules, while a repeated run with the same configuration reconstructs the same sequence.

The schedule seed is intentionally separate from the policy-training seed. Changing the latter can therefore alter the neural-network initialization and the environment-specific scenario permutations without redefining the contamination stream associated with a given environment, scenario, and episode counter. This separation is important for controlled comparisons between policy initializations because it prevents changes to the learning seed from implicitly changing how individual arsenic schedules are generated.

5.4.3 Hyperparameter Optimization

The PPO configuration was selected through a staged search using ordinarily initialized policies, with the recurrent architecture and reward parameters held fixed. An initial screening varied the learning rate, rollout length, minibatch size, and entropy coefficient one at a time, using 500 000 environment interactions per candidate. Selected settings and targeted combinations were then compared with budgets of up to one million interactions, followed by a comparison of discount factors and GAE parameters at the same budget.

Selection considered learning progress, contamination response, and chlorine control during normal operation on separate validation cases. The test scenarios were excluded from parameter selection. The resulting settings in Table 6 are shared by the baseline approach and the pretrained approach. The search used one training seed and a limited set of candidates, providing a practical configuration for this comparison. The tested configurations, budgets, and selection procedure are documented in Appendix D.

5.5 Cross-Fitted Teacher Generation

Having defined the online training conditions shared by both policy approaches, we now turn to the offline teacher-generation stage used to initialize the second policy.

Generating the teacher dataset requires estimating how candidate action sequences affect future chlorine concentrations, since MPC compares these sequences through their predicted cumulative rewards. Evaluating every candidate directly with the hydraulic and water-quality simulator would require repeated, computationally expensive simulations. We therefore train multistep forecasters to approximate the network's chlorine response and use them within MPC to generate teacher actions. Scenario-wise cross-fitting keeps these actions out of sample with respect to forecaster training. The resulting teacher actions form the demonstration dataset used for behavior cloning.

Figure 7 summarizes the five-fold data flow from offline trajectories to teacher demonstrations.

5.5.1 Data Allocation and Scenario-Wise Cross-Fitting

As MPC teacher actions are generated using a pretrained multistep forecaster rather than online simulator evaluations, the data used at each stage must be selected carefully. MPC uses the forecaster to search for action sequences with high predicted returns. If this search were performed on scenarios that the forecaster had already encountered during training, the selected actions could benefit from scenario-specific model fit or exploit prediction errors present in those trajectories. The resulting action sequences could consequently provide an overly optimistic representation of the teacher's performance on unseen hydraulic conditions.

One way to avoid this overlap would be to construct two independently simulated datasets. The first would be used to train the forecaster, while the second would provide previously unseen query states for the MPC teacher. However, constructing such datasets with comparable coverage would require a substantial number of simulator interactions. The offline training dataset already contains 2000 trajectories, corresponding to 2.4 million simulator time steps. Generating an equivalent collection

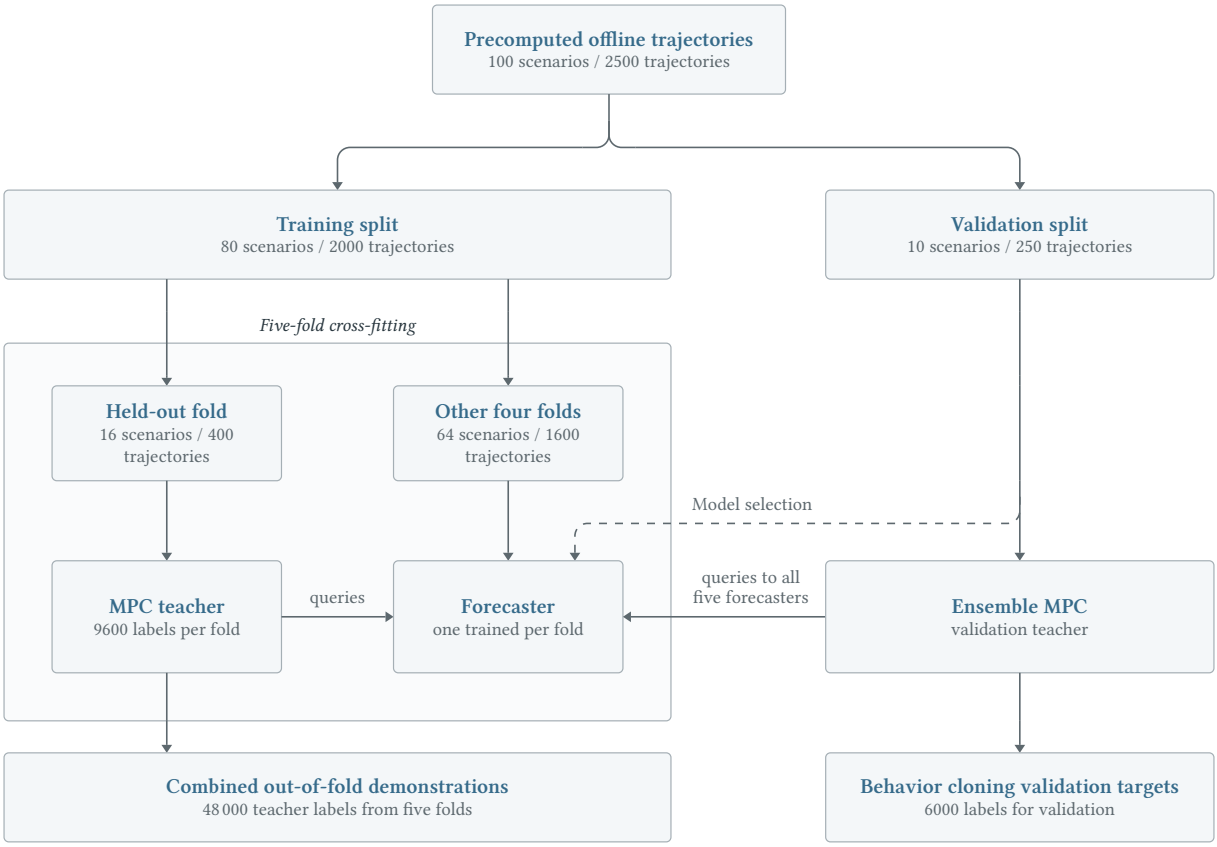


Figure 7: Scenario-wise cross-fitted teacher generation. In each of five folds per training seed, MPC draws query states from one held-out fold and queries a forecaster trained on the other four folds. The resulting teacher labels are combined for behavior cloning training. The same five trained forecasters are used together by ensemble MPC to label query states from the separate validation trajectories for behavior cloning validation. The test split of 10 scenarios or 250 trajectories is omitted because it is not used for teacher generation.

of trajectories for the MPC teacher would require another 2.4 million simulation steps and therefore double the simulation effort before behavior cloning begins.

To avoid this additional simulation effort, the pipeline instead uses five-fold cross-fitting across the 80 scenarios assigned to the forecasting-training split. These scenarios are partitioned in stable round-robin order into five non-overlapping folds of 16 scenarios. For each fold, one forecaster is trained on the 64 scenarios contained in the other four folds. All trajectories associated with the 16 held-out scenarios, including their different chlorine profiles and arsenic schedules, remain excluded from that forecaster's training data. The precomputed offline trajectories of the held-out fold then provide query states for the corresponding MPC teacher without requiring further simulator runs.

The held-out role rotates across the five folds. Each scenario is therefore used to train four forecasters and supplies teacher query states for the remaining forecaster. Cross-fitting requires five predictive models instead of one, but allows all 80 scenarios to contribute both to model fitting and to out-of-fold teacher generation.

The ten global forecasting-validation scenarios are used to select the forecasting configuration and checkpoints, and later to construct ensemble teacher labels and select the behavior-cloning checkpoint. This reuse makes the intermediate validation losses model-selection criteria, not independent estimates of teacher-action quality. Final policy quality is assessed on the separate test cases.

Once a fold-specific forecaster has been trained, MPC uses it to evaluate candidate injection sequences for query states from its held-out fold. For each query state, the first action of the sequence with the highest predicted return is retained as the teacher target. Repeating this process for all five folds produces five teacher datasets, which are subsequently combined into a single out-of-fold dataset for behavior cloning. Every forecasting-training scenario contributes to this training dataset exactly once. Separate teacher labels are generated for the existing forecasting-validation trajectories using all five forecasters as an ensemble. These labels provide the validation targets for early stopping and checkpoint selection during behavior cloning.

Cross-fitting reduces the risk of optimistic teacher supervision, but it does not make the selected actions equivalent to simulator-derived ground truth. MPC continues to evaluate candidate actions using predicted chlorine trajectories. The quality of its decisions therefore depends on how accurately each fold-specific forecaster represents the dynamics of its held-out scenarios. The procedure ensures that this accuracy is assessed outside the scenarios used for fitting, but it cannot remove errors caused by imperfect generalization.

5.5.2 History-Window and Forecast-Horizon Selection

The forecasting model uses a 48-step history and a 16-step prediction horizon, corresponding to 24 h of past context and eight hours of planned future injections at the 30-minute sampling interval. The prediction horizon also defines the action sequence evaluated by MPC.

We checked the timing of the selected windows by comparing the network response to a brief increase in chlorine injection with a constant-injection reference. Extending the history from 20 to 24 h covers most remaining detectable chlorine effects, while longer histories add little coverage. The 24-hour window also spans a complete daily cycle. By the eight-hour horizon, almost all tested pulse effects have begun and peaked, and most have settled. Longer horizons cover only small additional shares of cases while requiring longer forecasts and evaluation of more predicted steps for each MPC plan. Both lengths are practical choices for this study. The diagnostic supports this choice but does not establish an empirically optimal horizon, forecasting accuracy, or controller performance. Its protocol, results, and limitations are reported in Appendix C.

5.5.3 Forecasting Data and Causal Inputs

Each cross-fitted forecasting model is trained on the precomputed offline trajectories. Its 64 training scenarios each combine five prescribed chlorine-injection profiles with five fixed arsenic schedules, yielding 25 trajectories per scenario and 1600 training trajectories per fold-specific model.

The complete pairing varies the chlorine input and contamination schedule while leaving the demand pattern and physical network parameters fixed. The prescribed profiles include both near-steady and changing segments. Starting from zero injection, the profile generator limits every change between consecutive 30-minute injection rates to 50 000 mg/min, matching the actuator constraint in Section 5.1.2. This exposes the forecaster to sustained operating conditions and delayed responses after changes in chlorine injection, even though the action sequences do not necessarily correspond to realistic behavior. Multiple arsenic schedules also prevent a hydraulic scenario from being associated with only one contamination pattern. The choice of five profiles and five schedules is a computational-budget decision rather than an empirically optimized configuration.

Every trajectory covers 25 days at the 30-minute reporting interval defined in Section 4.3, resulting in 1201 ordered samples from the initial state to the final state. Supervised examples are extracted by moving a forecast origin through this sequence in increments of one sample. For each origin, the preceding 48 samples provide a 24-hour causal history, while the following 16 samples form an eight-

hour prediction horizon. A complete trajectory yields 1138 overlapping windows because examples are retained only where both the full history and the full horizon are available.

The information within one window is divided according to when it is available. The historical input contains the measured chlorine concentration at all 31 monitored nodes, the chlorine injection that was applied at the source, and sine–cosine encodings of the daily, weekly, and annual phases. Together, these quantities form 38 features at each of the 48 historical steps. The future input contains only information that is known when a forecast is requested, namely the planned source-injection sequence and the same six calendar features. This makes seven features for each of the 16 future steps. The prediction target is the chlorine concentration at all 31 monitored nodes over these 16 steps. Table 7 summarizes this causal arrangement.

Before forecaster training, the forecasting inputs and targets are standardized to account for their different numerical scales. Historical and future-known inputs are standardized feature by feature, while separate statistics are used for each node-level chlorine target. All means and standard deviations are estimated exclusively from training data and subsequently applied unchanged to held-out data. The normalization statistics are computed across the extracted forecasting windows. Each sample contributes once for every window in which it appears, so the resulting scaling reflects the frequency with which it is encountered during forecaster training.

Advancing the forecast origin by one sample creates overlapping and therefore strongly correlated windows. An example-level split could place nearly identical histories and targets in both training and validation data, causing evaluation performance to reflect this overlap rather than generalization to unseen hydraulic conditions. We therefore preserve the scenario-level partitions defined in Section 4.6 and keep all 25 trajectories and their derived windows together. This prevents information leakage and supports the cross-fitting design defined in Section 5.5.1.

Table 7: Causal input and target structure of one supervised forecasting window.

Window component	Steps	Features	Contents and role
History	48	38	Chlorine at 31 monitored nodes, the applied source injection, and six cyclic calendar features. It describes the observed 24 h preceding the forecast horizon.
Planned injection	16	7	The planned source injection and six cyclic calendar features for the next eight hours. During MPC, the injection entries are replaced by a candidate control sequence.
Prediction target	16	31	Future chlorine levels at all monitored nodes. These values supervise the forecaster but are not available as inputs.

5.5.4 Explicit Multistep Chlorine Forecaster

The forecasting task defined above requires a model that can represent the action-dependent network response over the selected horizon. For this purpose, we use the sequence-to-sequence encoder–decoder LSTM to approximate how the simulator would respond to a planned chlorine-injection sequence. The encoder learns a compact representation of the observed history, while the decoder combines this context with the planned injections and calendar features at each future step. This structure allows MPC to compare the predicted chlorine responses to different injection sequences under the same historical conditions. We adopt this architecture as a practical forecasting model for

evaluating MPC-guided policy pretraining, while a systematic comparison with alternative forecasting architectures lies outside the scope of the study.

The encoder processes the normalized 48×38 historical input and summarizes the preceding 24 h in the hidden and cell states of a single-layer LSTM with 192 hidden units. These states initialize a single-layer LSTM decoder with the same hidden size. The decoder then processes the 16×7 future-known input, which combines a candidate injection sequence with the corresponding calendar features. A shared linear output head maps the decoder state at each future step to the chlorine concentrations of all 31 monitored nodes. The resulting prediction therefore has shape 16×31 and represents absolute chlorine levels at each junction over the next eight hours.

Training minimizes the MSE between predicted and simulated chlorine levels after independently standardizing the target of each monitored node with statistics fitted on the training split. The loss gives equal weight to all 16 horizon steps and 31 node targets. After every epoch, predictions are transformed back to mg/L and the overall validation RMSE is computed across nodes and horizon steps. The checkpoint with the lowest validation RMSE is retained, and training stops early after ten consecutive epochs without an improvement. The model is optimized with AdamW for at most 100 epochs, using a batch size of 256, a learning rate of 7.5×10^{-4}, and a weight decay of 1×10^{-5}. These hyperparameters were fixed without systematic optimization, as a broader hyperparameter search was considered outside the scope of this study.

The architecture and optimization procedure define the forecasting configuration used throughout the teacher-generation stage. To generate teacher labels, a forecaster is trained separately within each scenario-wise cross-fitting fold, as described in Section 5.5.1. Consequently, the MPC teacher does not evaluate a query state with a forecaster trained on the corresponding hydraulic scenario.

5.5.5 MPC Teacher and CEM Optimization

Next, we use MPC together with the fold-specific forecasters to generate teacher actions. MPC searches over the same normalized rate-change actions available to the control policy, while the forecasters predict the resulting chlorine trajectories. These trajectories are scored using the policy-training reward, and the first action of the highest-scoring plan is retained as the teacher action.

Each teacher action is generated for a query state representing a selected time point in one of the precomputed trajectories. The query state provides the recent forecasting history and the current control and reward context from which MPC begins its search.

For each query state, MPC evaluates candidate action sequences over the forecaster's complete 16-step horizon, equivalent to eight hours. Directly optimizing one independent action at every step results in a 16-dimensional search space. We therefore represent each sequence using eight control points at zero-based horizon offsets 0, 1, 2, 3, 5, 8, 11, and 15 and obtain the remaining actions through linear interpolation. The control points are concentrated near the beginning of the horizon and spaced more coarsely at later steps. This gives the beginning of the plan, including the action retained as the teacher label, greater temporal resolution while reducing the dimension of the search. All interpolated actions remain within $[-1, 1]$.

Each interpolated action sequence is first converted into a feasible physical injection sequence using the actuator update in Equation 27. For a given query state, all candidates share the same normalized forecasting history and future calendar features and differ only in their planned injection sequences. The fold-specific forecaster evaluates these candidates as one batch and predicts a 16×31 chlorine trajectory for each plan, expressed in mg/L after reversing the target normalization. Before scoring, the predictions are checked for finite values and negative concentrations are set to zero. A nonfinite prediction causes the complete MPC query to fail.

These trajectories are evaluated sequentially using the same stateful reward mechanism as policy training. Every candidate begins from the same reward state at the query time, ensuring that all plans share identical recovery and temporal context. For candidate m, the MPC objective is

$$J_t^{(m)} = \sum_{h=1}^{16} \gamma^{h-1} r_{t+h}^{(m)}, \quad \gamma = 0.99. \tag{43}$$

Thus, earlier predicted outcomes receive slightly greater weight, while delayed network responses within the eight-hour horizon still contribute to the comparison. No separate terminal value is added after the final forecast step.

For behavior-cloning validation, every candidate plan is predicted by all five fold-specific forecasters. Each predicted chlorine trajectory is scored separately from the same initial reward state, yielding one planned return $J_t^{(m,k)}$ per model k. The ensemble objective is their arithmetic mean,

$$\overline{J}_t^{(m)} = \frac{1}{5} \sum_{k=1}^{5} J_t^{(m,k)}. \tag{44}$$

CEM optimizes this mean return, and the first action of the best plan becomes the validation teacher label. Chlorine predictions are not averaged before reward evaluation, and independently selected actions are not averaged.

The continuous control-point vector is optimized with the CEM. CEM maintains an independent Gaussian sampling distribution for each of the eight control points. The initial mean is zero and the initial standard deviation is one, after which sampled values are clipped to the admissible action interval. In every iteration, 63 plans are drawn from this distribution and combined with one fixed all-zero plan. The latter represents holding the current injection constant throughout the horizon and provides the optimizer with the same reference candidate in every iteration. All 64 plans are expanded, forecast, and scored as described above.

With 64 candidates, the 10 % elite fraction is rounded up to seven, retaining those with the highest predicted returns. Their component-wise mean and standard deviation define the next sampling distribution. Each standard deviation is bounded below by $\frac{1}{60}$ to prevent complete collapse of the search distribution. After four iterations, the optimizer returns the highest-scoring candidate across all populations. Table 8 summarizes the planning configuration.

At each recorded query state, MPC optimizes a complete 16-step plan and retains only its first normalized action as the teacher label. The selected action is stored as a teacher label but is not applied to the simulator to generate subsequent trajectory states. The remaining actions serve only to evaluate the longer-term implications of that decision and are discarded when the behavior-cloning example is formed. The predicted chlorine trajectories and the forecaster's internal representations likewise remain confined to MPC scoring.

Table 8: CEM configuration used to optimize MPC teacher actions.

Configuration item	Value	Role
Planning horizon	16 steps / 8 h	Matches the explicit multistep forecast horizon.
Action parameterization	8 control points	Uses offsets 0, 1, 2, 3, 5, 8, 11, and 15 and linearly interpolates the remaining actions.
Candidate population	64	Evaluates 63 sampled plans and one fixed hold plan per iteration.

Table 8 (continued)

Configuration item	Value	Role
CEM iterations	4	Repeatedly refines the control-point sampling distribution.
Elite set	7 candidates	Uses the highest-scoring 10 % of the population to update the distribution.
Initial distribution	$\mu = 0, \sigma = 1$	Starts from zero-mean independent Gaussian control points before clipping to $[-1, 1]$.
Minimum standard deviation	1/60	Maintains non-zero sampling variation in every control-point dimension.
Reward discount	$\gamma = 0.99$	Weights the predicted rewards over the planning horizon.

5.5.6 Query-State Sampling and Teacher-Dataset Construction

Within each held-out fold, query states are sampled from the corresponding precomputed trajectories. A query time must leave sufficient data for both the preceding history and the subsequent planning horizon. The 48-step causal history includes the query itself as its final sample, while the 16-step planning horizon starts at the following sample.

Applying MPC at all valid positions would produce more labels than required for pretraining, many of which would be closely related. The sampling procedure therefore selects 24 query times from each trajectory. It divides the valid range into 24 consecutive strata of approximately equal length and samples one query from each stratum. The resulting labels span the complete usable portion of the 25-day episode instead of being concentrated in a small number of periods.

The temporal strata distribute queries across each trajectory, but do not necessarily balance their positions within the daily demand cycle. Each candidate query corresponds to one of 48 half-hour slots within a day. When selecting from a stratum, the sampler identifies the slots currently least represented across the fold and samples from their candidate positions. The teacher dataset consequently covers both episode progression and the complete daily cycle without imposing fixed query intervals within individual trajectories. Trajectory and stratum order are randomized deterministically so that this balancing does not introduce systematic associations with trajectory characteristics or position within the episode.

For each selected query time, the recorded trajectory is replayed from the beginning of the episode up to and including the query. This reconstructs the reward state under the temporal context in which it originally evolved and provides the initial state for evaluating candidate action sequences. The 48 causal samples ending at the query form the forecasting history defined in Table 7, which the fold-specific forecaster uses to predict the chlorine response to each candidate sequence.

After the query state has been reconstructed, the fold-specific MPC teacher applies the optimization procedure described above. The first normalized action of the highest-scoring sequence becomes the teacher target at that query time. Repeating this step for all sampled queries produces one set of query-level targets for each held-out fold. The five sets are then combined into a single scenario-wise out-of-fold teacher dataset. The same query-sampling and reward-state reconstruction procedure is applied to the 250 existing forecasting-validation trajectories, producing 24 ensemble-teacher labels per trajectory. If all MPC queries succeed, the 24 query times per trajectory yield 48 000 training and 6000 validation teacher labels per seed.

The resulting dataset associates each complete episode with the teacher actions generated at its sampled query times. During behavior cloning, the recurrent actor processes the complete sequence of policy observations and is supervised only at these labeled positions. The forecaster and MPC optimizer therefore supply the action targets but do not become part of the actor's inputs.

5.6 Behavior Cloning and PPO Fine-Tuning

The teacher dataset constructed above provides the action targets used to initialize the policy of the second training approach. This section explains how we first train the recurrent actor to reproduce these actions and then transfer the resulting actor and observation-normalization state to the policy before online fine-tuning begins.

5.6.1 Policy Initialization

Behavior cloning uses the actions from the MPC demonstration dataset to initialize the recurrent actor for the second policy approach. The cloned actor has the same architecture as the actor defined for RecurrentPPO above, comprising a single-layer LSTM with 128 hidden units, two fully connected layers with 64 units each, and a scalar action output. At each labeled query state, the actor LSTM is trained to reproduce the first normalized action selected by the MPC teacher.

Each teacher action was generated from a 48-step forecasting history, but presenting only this window to the actor would reset its recurrent state before every query. Behavior cloning therefore processes each complete 1201-sample episode as a chronological sequence of normalized 43-value policy observations. The actor's recurrent state is initialized to zero only at the beginning of an episode. To limit the computational cost, training uses truncated BPTT with 48-step chunks. After each chunk, the recurrent state is detached from the gradient graph and passed to the next chunk. The chunk length was selected as a practical configuration rather than through systematic optimization and is therefore not assumed to be optimal. All observations, including those in chunks without query labels, contribute to the evolving recurrent state, while the cloning loss is evaluated only at the 24 labeled query positions in each episode.

Behavior cloning uses the combined demonstrations from all 80 training scenarios, comprising 2000 trajectories. Early stopping and checkpoint selection use the separate validation demonstrations described in Section 5.5.1. Each of the ten validation scenarios retains all 25 trajectories and all 24 labels per trajectory. Complete validation episodes reconstruct the actor's recurrent state in the same way as training episodes, while validation MSE is computed only at labeled query positions. Validation data contribute neither gradients nor observation-normalization statistics.

Observation normalization is fitted before optimization using every time step from the 2000 complete trajectories covering all 80 training scenarios. The ten validation scenarios remain excluded from these statistics. Each observation component is standardized and clipped to $[-10, 10]$, matching the preprocessing used by RecurrentPPO. The fitted normalization state is held fixed during behavior cloning and saved together with the policy for subsequent fine-tuning.

At each labeled query, the MSE measures the difference between the actor mean and the teacher action. Adam optimizes the actor LSTM, its two fully connected layers, and the action-output layer with a learning rate of 3×10^{-4}. Each minibatch contains 16 complete trajectories, and actor gradients are cleared before its first chunk. For each chunk with labels, squared errors at the labeled queries are summed and divided by the total number of labels in the minibatch before backpropagation. The resulting gradients accumulate across chunks, are clipped to norm 0.5, and are applied in one Adam step after the final chunk. The deterministic teacher actions provide no target for the action-distribution standard deviation. We set this standard deviation to the median, across training queries,

of the final CEM elite standard deviations for the first action. This is a practical initialization rather than a value optimized for PPO.

Training runs for at most 100 epochs. After each epoch, the MSE is evaluated at the labeled queries in the 250 complete trajectories from the ten separate validation scenarios. The actor with the lowest validation loss is retained, and optimization stops after ten consecutive epochs without improvement. The selected behavior-cloning result supplies the actor parameters, initial action standard deviation, and observation-normalization state used in the following initialization stage. The critic remains ordinarily initialized.

5.6.2 Transfer of Actor and Normalization State

Before PPO fine-tuning, the second RecurrentPPO policy receives the actor trained through behavior cloning. Because both stages use the same actor architecture, the parameters of the actor LSTM, its two fully connected layers, and the action-output layer can be transferred directly. The action-distribution standard deviation specified during behavior cloning is transferred as well. The actor therefore begins online training with parameters already fitted to reproduce the MPC teacher actions.

The critic is not transferred because the teacher dataset provides action labels but no return or value targets. Its LSTM and value network retain their ordinary random initialization and learn from returns collected during PPO training. Behavior cloning thus initializes only the policy component that is directly supervised by the MPC teacher.

The observation-normalization state fitted on the behavior-cloning training trajectories, including its running count, is transferred with the actor. During PPO training, these statistics continue to incorporate observations from the control-training environments and can adapt to states visited by the fine-tuned policy. Behavior cloning provides no reward-normalization state, so these statistics are accumulated from online rewards in the same way as in the baseline approach.

5.6.3 RecurrentPPO Fine-Tuning

After the transfer, PPO fine-tunes the recurrent policy through interaction with the EPANET-based hydraulic and water-quality simulator, using a newly initialized optimizer. It updates the transferred actor and its action-distribution parameters while the ordinarily initialized critic is trained from the collected returns. The actor can therefore adapt the behavior learned from the MPC teacher using rewards observed during closed-loop control.

Fine-tuning uses the same nominal budget of one million environment interactions and the RecurrentPPO configuration in Table 6. It also follows the control-training scenario and contamination-sampling procedure defined in Section 5.4.2. The baseline approach and the pretrained approach consequently use the same PPO update budget, reward signal, and online training conditions. Their intended methodological difference is the actor parameters, initial action standard deviation, and observation-normalization state available when PPO begins.

Neither the chlorine forecaster nor the MPC optimizer participates in fine-tuning, evaluation, or deployment. These stages use only the recurrent policy and the deployable 43-value observation defined in Table 5. After fine-tuning, both approaches therefore produce policies with the same architecture and inference inputs. The following comparison can thus assess the effect of policy initialization without changing the controller inputs or architecture.

5.7 Principal Experimental Comparison

The principal experiment compares ordinarily initialized RecurrentPPO with MPC-guided policy pretraining under a fixed online interaction budget. Pretraining supplies the actor parameters, initial action standard deviation, and observation-normalization state, while the critic remains ordinarily ini-

tialized. Both approaches share the architecture, reward, training scenarios, contamination sampling, and online budget, so the comparison assesses the effect of initialization. The additional variants below examine the contributions of a larger online budget and the cloned actor weights.

5.7.1 Additional Comparison Variants

Two additional variants supplement the principal comparison. The extended-budget baseline trains ordinarily initialized RecurrentPPO for four million nominal online interactions. This budget corresponds to the three million simulator steps used to generate the precomputed offline trajectories across the training, validation, and test partitions, plus the pretrained approach's one million nominal online interactions. It examines whether additional online training allows the baseline approach to reach the pretrained approach's validation return. Separate simulator checks and control evaluations are excluded from this budget comparison.

The variant without actor weights receives the observation-normalization state and initial action standard deviation from the same behavior-cloning result as its paired pretrained run, while its actor remains ordinarily initialized. It trains for one million nominal online interactions to assess the contribution of the cloned actor weights conditional on these two transferred components. Both variants retain the ordinarily initialized critic and the common PPO settings, training scenarios, and contamination sampling. Observation-normalization statistics continue to update during online training. Their evaluation is defined in Section 6.6.

5.8 Implementation and Reproducibility

For each training seed listed in Evaluation, the five cross-fitted forecasters are trained separately, training and ensemble-validation MPC teacher labels are regenerated using that seed's five forecasters, and behavior cloning and both PPO runs are repeated. The precomputed offline trajectories and fixed evaluation cases are shared across seeds.

The experiments use Python 3.13.3 [37] with NumPy 2.4.4 [33], PyTorch 2.11.0 [38], Gymnasium 1.2.3 [15], Stable-Baselines3 2.8.0 [47], SB3-Contrib 2.8.0 [46], and EPyT-Flow 0.17.0 [4]. EPyT-Flow calls EPANET-PLUS 0.3.1, which provides the interface to OWA-EPANET 2.3.6 and EPANET-MSX 2.0.0 [5].

The training computer used for the reported runtime measurements runs on Windows 11 and has an AMD Ryzen 7 9800X3D processor with eight physical cores and eight logical processors available to the operating system, 32 GiB of DDR5 memory configured at 6000 MT/s, and an NVIDIA GeForce RTX 3080 Ti with 12 GiB of dedicated GPU memory.

6 Evaluation

This chapter presents the evaluation of the two control approaches introduced in the previous chapter in order to address the research question investigated in this thesis.

For this comparison, we refer to the approach trained without pretraining as the baseline approach and to the approach initialized through pretraining as the pretrained approach. Both follow the common online training conditions.

The evaluation uses reward learning curves to assess learning progress during online training and six primary physical metrics to compare the control quality of the policies selected by validation return after the common training budget has been completed. Computational cost is assessed separately, including offline preparation, because fewer online interactions do not necessarily imply less computation overall. We first establish common evaluation cases, episode windows, and policy selection, then define how learning progress, final control quality, and computational cost are evaluated. Additional variant comparisons and a supporting evaluation of the forecasting model complement these primary analyses.

6.1 Evaluation Protocol

A fair comparison requires both policies to encounter the same evaluation conditions. The protocol establishes which parts of each episode contribute to scoring and how the policies are evaluated across fixed cases and repeated training runs.

6.1.1 Episode Structure

Each evaluation rollout combines one trained policy with one hydraulic scenario and one fixed arsenic schedule. The 25-day episodes defined in Section 4.3 are divided into a two-day warm-up, a 21-day main window, and a two-day clearance tail. For the remainder of this chapter, we denote the timesteps in the main window and clearance tail by $\mathcal{T}_{\mathrm{M}}$ and $\mathcal{T}_{\mathrm{T}}$, respectively. We write $\mathcal{T}_w$ for either set with $w \in \{\mathrm{M}, \mathrm{T}\}$.

The warm-up gives the policy time to establish chlorine concentrations throughout the network and is excluded from scoring. The main window provides an extended observation period with and without contamination. Its duration is a practical choice rather than a requirement of the evaluation method. The clearance tail allows us to evaluate the policy response while remaining arsenic clears from the network. The evaluation return, defined in Section 6.2, sums rewards over the main window and complete clearance tail, excluding the warm-up. Figure 8 illustrates this division.

The clearance tail captures responses to contamination events occurring near the end of the main window. Although all arsenic inputs stop before the end of the 21-day main window, transported arsenite can still remain in the network and affect chlorine levels and, consequently, the control behavior of the policy. The two additional arsenic-injection-free days allow these effects to be evaluated until a common clearance boundary.

Before policy evaluation, all fixed validation and test cases were checked using reference runs with a zero-chlorine injection profile. The clearance check applies a threshold of 0.01 mg/L to the sum of raw As(III) concentrations across all monitored nodes. This numerical value is taken from the WHO's provisional guideline for arsenic in drinking-water [55]. Its application to a concentration sum serves as an operational clearance criterion rather than a measure of compliance with that guideline. In every case, the raw concentration sum reached or fell below this threshold during the clearance tail and remained at or below it until the end of the episode. These checks confirmed that the two-day clearance tail was sufficient under this criterion. The criterion concerns the monitored nodes within the observed horizon and does not imply the complete absence of As(III) from the network.

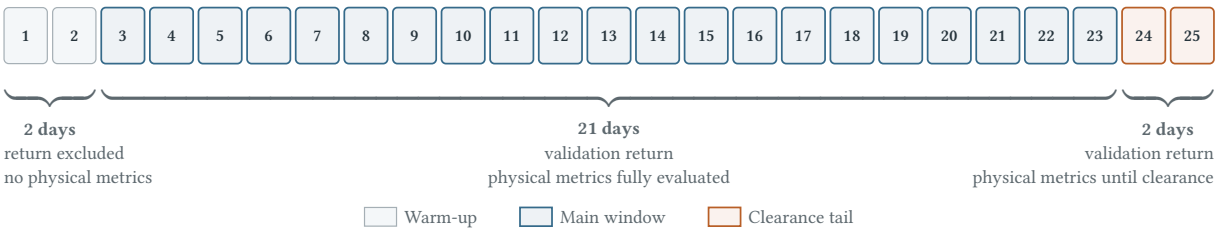


Figure 8: Episode windows in a 25-day evaluation rollout. The arsenic-injection-free warm-up is excluded from scoring. Arsenic events can only start in the main window. Validation return covers the main window and full clearance tail. Physical control metrics are evaluated as indicated.

6.1.2 Policy Comparison and Evaluation Procedure

Both policies are evaluated on the same fixed sets from Section 4.6. Each set comprises ten scenarios, with five arsenic schedules per validation scenario and ten per test scenario. The 50 validation cases track progress during training and determine which policy checkpoint is selected. The 100 test cases

provide the physical comparison of the selected policies. Within each case, hydraulics, demand timing, initial conditions, and the arsenic schedule are identical for the two policies. For each case, a single reference run without chlorine injection is evaluated to quantify arsenic contamination in the absence of dosing and compare it with contamination under each policy.

Five training runs per approach use the seeds 17, 42, 123, 1337, and 71026. Model fitting is repeated for each training seed, with shared offline trajectories and fixed evaluation cases. Corresponding runs encounter the same sequence of hydraulic scenarios and contamination schedules during training, following the sampling procedure described in Section 5.4.2. Pairing training runs by their seed allows for comparison under matched environmental conditions.

Both approaches complete the nominal one-million-interaction budget. For each approach and training seed, the checkpoint with the highest mean validation return is then selected from the 21 checkpoints defined in Section 6.2, including the zero-interaction checkpoint and the checkpoint at the nominal one-million-interaction threshold. The mean gives equal weight to scenarios after averaging their schedule-level returns. If several checkpoints attain the same maximum, the earliest is selected. Both approaches use the same candidate checkpoints, validation cases, and selection criterion.

During evaluation, action selection is deterministic, policy memory is reset at the start of each rollout, the observation-normalization statistics saved with the evaluated checkpoint remain fixed, and reward normalization is disabled. Evaluation follows the policy-input restrictions defined in Section 5.3, with arsenic measurements used only for scoring.

6.2 Learning Curves and Sample Efficiency

Training efficiency is assessed through reward learning curves, which track control performance under the common objective during online training. They show whether the policies differ in performance before online training and how their relative performance evolves during training.

The learning curves compare validation return before the first PPO update and at nominal thresholds of 50 000, 100 000, 150 000, and every further 50 000 interactions up to 1 000 000. At each threshold, we evaluate the first completed PPO update whose cumulative interaction count reaches or exceeds it. This rule applies to both approaches and all five training seeds. Actual interaction counts are recorded for each checkpoint. At zero interactions, the pretrained policy has completed behavior cloning, and neither policy has begun online training. The initial comparison measures the control behavior supplied by pretraining, while subsequent checkpoints show how the difference develops over time.

Each checkpoint is evaluated on the same 50 validation cases using the procedure above. Interaction counts include all parallel training environments and exclude evaluation interactions. Using the reward function defined in Section 5.2, the evaluation return for each rollout is the undiscounted sum of the original, unnormalized rewards over the main window and clearance tail,

$$R_{\text{eval}} = \sum_{t \in \mathcal{T}_{\text{eval}}} r_t. \tag{45}$$

Here $\mathcal{T}_{\text{eval}} = \mathcal{T}_{\text{M}} \cup \mathcal{T}_{\text{T}}$, using the window sets defined in Section 6.1.1. The reward definition and scoring duration are identical for both policies. Higher return indicates better performance under this objective. On the validation cases used to track learning progress, we refer to this quantity as the validation return.

At each checkpoint, returns are averaged across schedules within each scenario and then across scenarios for each seed. The mean and sample standard deviation are calculated across the five seeds. Pretrained-minus-baseline return differences are calculated within matched seed pairs and then averaged.

Because performance is evaluated only at discrete checkpoints, changes between checkpoints cannot be located more precisely, and apparent plateaus do not establish convergence beyond the evaluated budget. The learning curves are interpreted together with the final physical metrics to assess whether faster progress under the training objective is accompanied by acceptable control quality.

6.3 Control Performance Metrics

The evaluation uses six primary control metrics over the main and tail windows introduced in Section 6.1.1.

Two metrics assess chlorine compliance during normal operation. Arsenic burden, mean arsenic clearance lead, and chlorine ceiling violations characterize event response, while chlorine use measures chemical expenditure during both normal operation and event response. Table 9 provides an overview of the evaluation windows and reference regimes used by the six primary metrics.

Table 9: Evaluation windows and reference regimes used by the six primary control metrics. Clearance refers to persistent clearance in the zero-chlorine reference. The regime follows the definition introduced in Section 6.3.1.

Metric	Main window	Tail window	Regime
Chlorine shortfalls	Yes	No	Reference-normal
Chlorine excesses during normal operation	Yes	No	Reference-normal
Arsenic burden	Yes	Until clearance	Reference-event
Mean arsenic clearance lead	Yes	Until clearance	Reference-event
Chlorine ceiling violation	Yes	Until clearance	Reference-event
Chlorine use	Yes	Until clearance	Both regimes

6.3.1 Reference-Event Windows

The two policies must be scored over the same periods even if they clear arsenic at different rates. Reference-event windows are therefore determined from the zero-chlorine replay of each case and applied unchanged to both policies at every checkpoint. A window begins when the transported arsenic aggregate first exceeds the clearance threshold and ends with persistent clearance which is caused by outflow and dilution. The zero-chlorine replay provides the conservative upper bound on As(III) persistence, so every policy has cleared the event by this shared endpoint at the latest. Times outside these windows form the reference-normal regime.

The arsenic measure used to construct the windows is separate from the raw-concentration clearance check in Section 6.1.1 and accounts for the local mixing behavior described in Section 4.3. In this simulation model, chlorine reacts with arsenite only within pipes, so increasing chlorine injection cannot neutralize newly injected arsenite at the source node of the event itself. Including this immediate local contribution would therefore add a burden that the controller cannot reduce through chlorination. The evaluation excludes this contribution while continuing to account for arsenite transported through the network. At each monitored node v and timestep t, the estimated contribution of newly injected arsenic is subtracted from the reported As(III) concentration, and the remainder is clipped at zero. The calculation is specified in Appendix B.1. Denoting this transported concentration by $a_{v,t}$ gives the aggregate

$$A_{\Sigma}(t) = \sum_{v=1}^{N} a_{v,t}. \tag{46}$$

Clearance occurs when $A_\Sigma(t) \leq \varepsilon$, with $\varepsilon = 0.01\,\text{mg/L}$. This is the same operational threshold used for the clearance check in Section 6.1.1.

Source events are grouped into a contamination cluster whenever the zero-chlorine trajectory does not clear persistently between them. A reference-event window begins at the first threshold exceedance and ends at persistent clearance, which may precede source shutdown. The interval includes its start, but excludes its endpoint.

Clearance is persistent if the aggregate remains below or at the threshold until the next original source injection or the end of evaluation. Consequently, reference-event windows may differ from the original source-event intervals due to transport delays in the network. An event produces no reference-event window if the transported arsenic aggregate remains at or below the clearance threshold, as can occur when the arsenic event is too weak or when injected arsenic does not flow into adjacent nodes. Figure 9 illustrates how transported As(III) can persist after a single arsenic source event has ended.

During the clearance tail defined in Section 6.1.1, only samples within these reference-event windows contribute to the physical control metrics. The subsequent reference-normal part of the tail contributes only to the evaluation return defined in Section 6.2.

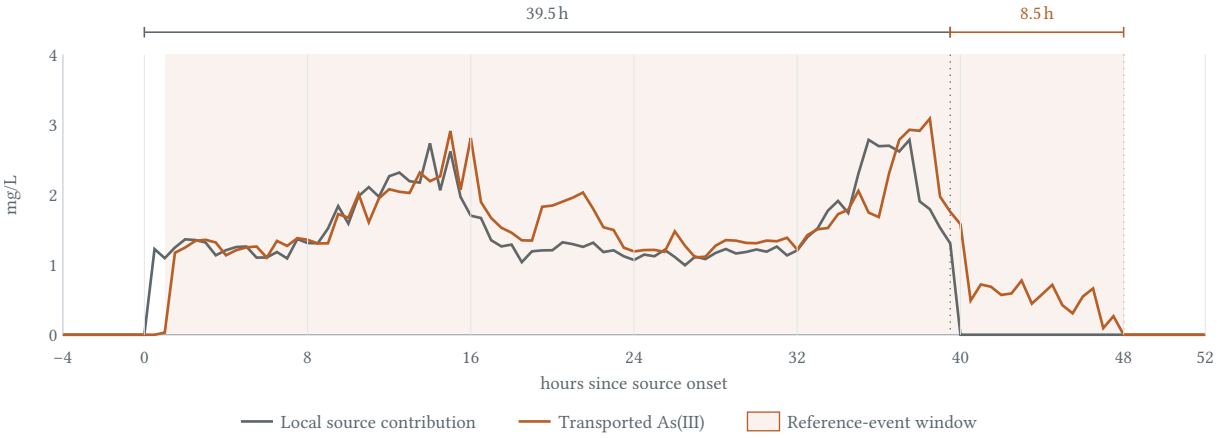


Figure 9: Zero-chlorine reference trajectory for one control-validation event in `scenario-85` with arsenic schedule `arsenic-2`. The plot starts four hours before injection and ends four hours after persistent clearance. Gray and orange lines show the sums of local source and transported As(III) concentrations across monitored nodes, respectively, in mg/L. The gray and orange spans above the curves mark the 39.5-hour source interval and the 8.5-hour interval from source shutdown to persistent clearance. Orange shading marks the reference-event window, which closes at persistent clearance, with $A_\Sigma(t)$ remaining at or below $\varepsilon = 0.01\,\text{mg/L}$ until the next source injection or episode end. Values are reported every 30 min.

6.3.2 Common Notation

We use N and E for the reference-normal and reference-event regimes defined in Section 6.3.1. They are represented by

$$I^{\mathrm{E}}(t) = \begin{cases} 1 & \text{if } t \text{ lies in a reference-event window} \\ 0 & \text{otherwise,} \end{cases}$$
$$I^{\mathrm{N}}(t) = 1 - I^{\mathrm{E}}(t). \tag{47}$$

Let N be the number of monitored nodes, indexed by $v = 1, \ldots, N$. The chlorine concentration at node v after control step t is $c_{v,t}$. The chlorine injection rate is denoted by u_t. The common reporting and

control interval is $\Delta t = 30\,\text{min}$ (0.5 h). It is expressed in hours for concentration-time metrics and clearance differences, and in minutes when multiplied by the injection rate in mg/min.

The chlorine limits follow the operational bounds established in Section 5.1.1. The lower and upper chlorine bounds for normal operation are $L = 0.2\,\text{mg/L}$ and $U = 0.4\,\text{mg/L}$, respectively. The absolute chlorine ceiling is $C_{\max} = 5.0\,\text{mg/L}$, chosen based on the WHO's health-based guideline value for free chlorine in drinking water [55]. Values on a bound are compliant, so only strict shortfalls or exceedances count as violations.

Using the evaluation-window timestep sets $\mathcal{T}_w$ introduced in Section 6.1.1, the number of eligible node–time observations for window w and regime $r \in \{\text{N}, \text{E}\}$ is

$$N_w^r = N \sum_{t \in \mathcal{T}_w} I^r(t). \tag{48}$$

Additive metrics are defined as zero when no timesteps are eligible for scoring. A fraction with a positive denominator is zero when its numerator is zero. If no eligible observations are present and its denominator is therefore zero, the fraction is recorded as not applicable.

Table 10 summarizes the recurring symbols and units used in the control metrics.

Table 10: Notation used in the control-performance evaluation. Counts, indicators, and fractions are dimensionless.

Symbol	Meaning	Unit
v	Monitored-node index	–
w	Evaluation-window index, M or T	–
r	Reference-regime index, N or E	–
$\mathcal{T}_w$	Timesteps in evaluation window w	–
N	Number of monitored nodes	–
$c_{v,t}$	Nodal chlorine concentration	mg/L
$a_{v,t}$	Nodal transported As(III) concentration	mg/L
$A_\Sigma(t)$	Sum of transported As(III) concentrations	mg/L
u_t	Chlorine injection rate	mg/min
L, U	Lower and upper normal chlorine bounds	mg/L
$C_{\max}$	Absolute chlorine ceiling	mg/L
ε	Clearance threshold for $A_\Sigma(t)$	mg/L
Δt	Common reporting and control interval	min or h
$I^{\text{N}}(t)$, $I^{\text{E}}(t)$	Reference-normal and reference-event indicators	–
N_w^r	Eligible node–time count in window w and regime r	–
$f_{\text{low}}^{\text{N}}$, $f_{\text{high}}^{\text{N}}$	Normal-operation shortfall and excess fractions	–
B^{E}	Integrated transported arsenic burden	mg h/L
Δ_{clear}	Mean arsenic clearance lead relative to the zero-chlorine reference	h
$f_{\text{cap}}^{\text{E}}$	Event-time ceiling-violation fraction	–
U_{eval}	Evaluated chlorine use	mg

6.3.3 Chlorine Shortfalls and Excesses

The two compliance metrics are measured only in the reference-normal regime of the main window. The fractions of eligible observations below the lower chlorine bound and above the normal upper bound are defined as

$$f^{\mathrm{N}}_{\mathrm{low}} = \frac{\sum_{t\in\mathcal{T}_{\mathrm{M}}}\sum_{v=1}^{N} I^{\mathrm{N}}(t)\,\mathbf{1}\left[c_{v,t} < L\right]}{N^{\mathrm{N}}_{\mathrm{M}}},$$
$$f^{\mathrm{N}}_{\mathrm{high}} = \frac{\sum_{t\in\mathcal{T}_{\mathrm{M}}}\sum_{v=1}^{N} I^{\mathrm{N}}(t)\,\mathbf{1}\left[c_{v,t} > U\right]}{N^{\mathrm{N}}_{\mathrm{M}}}. \tag{49}$$

Lower values indicate better compliance. Keeping the two fractions separate distinguishes insufficient residual chlorine from excessive concentrations during normal operation.

An exploratory analysis was added after the primary evaluation to examine when reference-normal chlorine excess occurs. For each reference-event window defined in Section 6.3.1, we consider the first 12 h after reference clearance. This period is shortened if a new source injection begins or the main window ends. The same periods apply to both policies.

We separate the exceedances contributing to $f^{\mathrm{N}}_{\mathrm{high}}$ into two groups: those recorded during these post-clearance periods and those recorded at other reference-normal times in the main window. For each rollout, each group's exceedance count is divided by the total number of reference-normal node–time observations, $N^{\mathrm{N}}_{\mathrm{M}}$. These contributions are then averaged using the weighting in Section 6.4. Their sum equals mean $f^{\mathrm{N}}_{\mathrm{high}}$. Dividing each mean contribution by this total gives the proportion of chlorine excess occurring in that group.

6.3.4 Arsenic Burden

Arsenic burden combines the magnitude and duration of transported arsenic concentrations in the reference-event regime. For each window $w \in \{\mathrm{M}, \mathrm{T}\}$, the component is

$$B^{\mathrm{E}}_{w} = \Delta t \sum_{t\in\mathcal{T}_w} I^{\mathrm{E}}(t)\, A_{\Sigma}(t), \tag{50}$$

where $A_{\Sigma}(t)$ is the sum of transported As(III) concentrations across all monitored nodes at timestep t, as defined in Equation 46.

The primary metric accumulates both components,

$$B^{\mathrm{E}} = B^{\mathrm{E}}_{\mathrm{M}} + B^{\mathrm{E}}_{\mathrm{T}}. \tag{51}$$

The tail component therefore contains only the part of a reference event whose clearance extends beyond the main window. Lower values indicate less monitored arsenic burden. The metric is a sum of concentration–time burdens across monitored nodes and is measured in mg h/L. It does not measure physical arsenic mass or population exposure.

Arsenic burden is also evaluated for the zero-chlorine injection profile. This provides an upper bound on untreated transported arsenic burden for each scenario and arsenic schedule combination, allowing us to compare policy performance with the absence of chlorine injection.

6.3.5 Mean Arsenic Clearance Lead

Mean arsenic clearance lead complements integrated arsenic burden by measuring how much earlier a policy reaches persistent clearance than the zero-chlorine reference.

Within a rollout of the policy being evaluated, let $\mathcal{W}$ be the set of reference-event windows from Section 6.3.1 in which $A_\Sigma(t)$ exceeds ε at least once. Each $W \in \mathcal{W}$ contains the step indices of one such window.

For each W, the clearance step $t_{\text{clear}}(W)$ is the first reported timestep after the final sample with $A_\Sigma(t) > \varepsilon$,

$$t_{\text{clear}}(W) = \max\{t \in W : A_\Sigma(t) > \varepsilon\} + 1. \tag{52}$$

For the zero-chlorine reference policy π_{ref}, $t_{\text{clear}}^{\pi_{\text{ref}}}(W)$ is the endpoint of the same reference-event window.

The primary metric for $|\mathcal{W}| > 0$ is the arithmetic mean of the clearance-step differences, multiplied by Δt expressed in hours,

$$\Delta_{\text{clear}} = \frac{\Delta t}{|\mathcal{W}|} \sum_{W \in \mathcal{W}} \left(t_{\text{clear}}^{\pi_{\text{ref}}}(W) - t_{\text{clear}}(W) \right). \tag{53}$$

Positive values indicate earlier clearance on average than the reference, and zero indicates no average timing difference.

6.3.6 Chlorine Ceiling Violation

Violations of the absolute chlorine ceiling are evaluated in the reference-event regime across both windows. For each window, the number of violations is

$$K_{\text{cap},w}^{\text{E}} = \sum_{t \in \mathcal{T}_w} \sum_{v=1}^{N} I^{\text{E}}(t)\, \mathbf{1}\left[c_{v,t} > C_{\max}\right]. \tag{54}$$

The primary violation fraction pools the eligible node–time observations from the main and tail windows,

$$f_{\text{cap}}^{\text{E}} = \frac{K_{\text{cap,M}}^{\text{E}} + K_{\text{cap,T}}^{\text{E}}}{N_{\text{M}}^{\text{E}} + N_{\text{T}}^{\text{E}}}. \tag{55}$$

The two window fractions are not averaged. Pooling their numerators and denominators gives every eligible node–time observation the same weight. Lower values indicate fewer ceiling violations. Concentrations between the normal upper bound U and $C_{\max}$ do not count as violations of this metric.

6.3.7 Chlorine Use

Chlorine use is evaluated throughout the main window, while in the tail, it is only measured during a reference event. The components for the eligible window–regime pairs $(w, r) \in \{(\text{M}, \text{N}), (\text{M}, \text{E}), (\text{T}, \text{E})\}$ are defined by

$$U_w^r = \Delta t \sum_{t \in \mathcal{T}_w} I^r(t)\, u_t. \tag{56}$$

The primary result is their sum,

$$U_{\text{eval}} = U_{\text{M}}^{\text{N}} + U_{\text{M}}^{\text{E}} + U_{\text{T}}^{\text{E}}. \tag{57}$$

The rate uses mg/min as established in Section 5.1.2. Chlorine use is calculated in mg and reported in tonnes (t). Injection in the tail after the final reference event has ended does not contribute to the metric. Chlorine use is interpreted alongside the concentration metrics to determine whether lower chemical use preserves control quality.

6.3.8 Cross-Window Aggregation

Arsenic burden and chlorine ceiling violations are first calculated separately for the main and tail windows and then combined into one primary result. Arsenic burden accumulates its two window components. The ceiling-violation fraction pools the corresponding numerators and denominators rather than averaging window-specific fractions. Chlorine use adds its main-normal, main-event, and tail-event components. These individual window components remain available as a transparent supplementary breakdown.

6.4 Statistical Analysis

Each primary metric is calculated once per rollout. For fraction-based metrics, a rollout with no eligible observations is retained as not applicable and does not enter the corresponding numerical average. Within each training run, rollout values are averaged across schedules within each scenario, excluding non-applicable values. The available scenario means are then averaged with equal weight. This produces one summary per metric and training seed. The numbers of applicable rollouts and the paired exclusions are reported alongside the results.

The five training seeds provide the replicates for estimating variability. Evaluation cases remain repeated measurements of their respective trained policy. For each metric, reporting includes the mean and sample standard deviation across seeds, together with the mean paired difference between the pretrained and baseline approaches and its 95 % Student-t confidence interval with four degrees of freedom. The difference is calculated as pretrained minus baseline. Positive differences favor pretraining for arsenic clearance lead. Negative differences favor pretraining for the other five primary metrics.

Individual paired differences are reported alongside the aggregate estimates to show the underlying variation. The confidence intervals quantify uncertainty in the estimated mean paired differences for the fixed Hanoi scenarios, evaluation catalogs, and training protocol. The learning curves in Section 6.2 and supplementary numerical summaries use the same scenario weighting. Return standard deviations are rounded to two significant figures and means to the corresponding precision. Other return values use up to three significant figures. All calculations use unrounded values.

Arsenic burden is also compared for each test case after averaging across the five training seeds. Each policy's mean arsenic burden is expressed as a percentage reduction relative to the mean zero-chlorine reference burden over the same test cases.

Paired comparisons of mean arsenic clearance lead use the intersection of the two policy-specific eligible-window sets $\mathcal{W}$ defined in Section 6.3.5. Both policies must therefore exceed the threshold within each included reference-event window. Therefore, a rollout pair without any common eligible windows is not applicable. Policy-specific no-exceedance counts and paired exclusions are reported alongside the results.

For a selected test case, nodal chlorine concentrations are summarized at each timestep by their mean and population standard deviation. Transported arsenic peaks are determined within individual reference-event windows, and case-level arsenic burdens are compared across the five training seeds and against the zero-chlorine reference.

6.5 Training Time and Resource Requirements

Time and resource measurements account for both online training and the offline preparation required by pretraining. Computational cost complements the interaction-based comparison in Section 6.2. Wall-clock times are measured separately for simulation-data generation, forecasting-model training, MPC-teacher demonstration generation, behavior cloning, PPO training, and control evaluation. Hardware, worker and device counts, and parallelism are recorded to put these metrics into perspective.

Runtimes repeated for each training seed are summarized by their mean and sample standard deviation across the five seeds. Forecasting-model training times are first summed across the five folds within each seed. Shared data-generation costs are counted once and excluded from per-seed totals. Overall costs sum the stage runtimes across all seeds and add shared data generation once. Online interaction counts and training runtimes cover the complete training budget, including any training after the selected checkpoint.

6.6 Additional Variant Comparisons

The variants defined in Section 5.7.1 are evaluated through validation return using the same five training seeds, 50 fixed validation cases, and evaluation procedure as the principal comparison. Checkpoints follow the same nominal 50 000-interaction spacing. The extended-budget baseline has 81 checkpoints from zero to 4 000 000 nominal interactions and completes 4 005 888 actual interactions. The variant without actor weights uses the 21 checkpoints through one million nominal interactions.

Endpoint comparisons use each variant's return at its full online budget and the pretrained approach's return at the one-million-interaction endpoint. Differences are calculated as variant minus pretrained within matched seed pairs and summarized using the statistical procedure defined above. To compare learning progress, we retrospectively identify the first observed checkpoint at which a variant's mean validation return reaches or exceeds the reference approach's mean return at one million interactions. The reference is the pretrained approach for the extended-budget baseline and the ordinary baseline for the variant without actor weights.

The supplementary runtime comparison reports PPO training time for both variants and checkpoint-validation time separately for the extended-budget baseline.

6.7 Supporting Evaluation of the Forecasting Model

The chlorine forecaster is evaluated separately on unseen hydraulic scenarios. For each of the five training seeds, each of the five fold-specific models used to generate the MPC teacher labels is applied to all valid forecast origins in the 250 offline trajectories of the ten forecasting test scenarios defined in Section 4.6. Every example uses the 48-step history, planned 16-step injection sequence, and calendar features specified in Section 5.5.3.

Predicted and simulated chlorine concentrations are compared after reversing the target normalization. Accuracy is measured using MAE and RMSE, both expressed in mg/L. Reporting both metrics in physical concentration units allows their magnitudes to be interpreted directly.

For each of the 16 forecast steps, both errors are aggregated across all valid forecast origins, trajectories, and monitored nodes within each test scenario. Applying the same aggregation across all forecast steps provides the MAE and RMSE over the complete eight-hour horizon. For both the step-wise and full-horizon metrics, the ten scenario values are averaged with equal weight for each model. Results are calculated separately for all 25 models. For each training seed, the five model errors are averaged. The overall summary then reports the mean and sample standard deviation across the five seed means.

Together, the step-wise and full-horizon errors answer the question of how accurately the fold-specific models predict chlorine concentrations over the next eight hours on unseen scenarios under prescribed injection profiles. However, a low prediction error alone does not establish the control quality of the demonstrations generated by the MPC teacher.

7 Results

This chapter reports the results obtained with the evaluation protocol. The principal comparison is between the baseline approach and pretrained approach under the same online interaction budget. Validation returns describe both policies before and during PPO training, while six physical metrics

summarize the test performance of the selected policies. Computational costs, additional variant comparisons, one matched test case, and forecasting errors complement this comparison. The chapter describes the observed values and their variation across training seeds. Their interpretation follows in Chapter 8.

In the final test evaluations, we refer to the checkpoints with the highest mean validation return for each approach and training seed as the selected baseline policy and selected pretrained policy. Selection considers the checkpoints evaluated within the nominal one-million-interaction budget and follows Section 6.1.2.

7.1 Online Sample Efficiency

Validation returns were evaluated before the first PPO update and at every nominal 50 000-interaction threshold up to one million, following Section 6.2. Both approaches reached 1 007 616 actual training interactions at the final checkpoint. Figure 10 shows two panels with the complete training range and the interval beginning at the nominal 500 000-interaction checkpoint. As defined in Equation 45, the validation return includes the main window and complete clearance tail and excludes the warm-up.

At zero online interactions, the baseline policy had its ordinary initialization and the pretrained policy had the initialization obtained through behavior cloning. The baseline mean validation return and sample standard deviation were −2 800 000 ± 4 700 000. The corresponding pretrained values were −95 000 ± 58 000. The mean pretrained-minus-baseline difference was approximately 2 700 000. With random untrained initialization, baseline seed 71026 started near −11.2 million, below the lower bound shown in panel (a).

The pretrained policy had a higher mean return at every checkpoint, while the mean gap narrowed overall despite fluctuations. The mean paired difference decreased from 283 000 at 100 000 nominal interactions to 31 400 at 500 000. The large negative baseline returns at the start dominate the vertical scale of panel (a), whereas panel (b) shows the later reduction in the gap on a separate scale.

By 900 000 nominal interactions, the difference between the mean returns had decreased to 5450. After that, the pretrained mean stayed near −11 010, while the baseline mean rose from −16 500 to −13 100. At the nominal one-million-interaction checkpoint, the baseline mean return and sample standard deviation were −13 100 ± 1800, compared with −11 010 ± 750 for the pretrained policy. The mean paired difference was 2090.

The return difference at this budget endpoint was positive in 4 matched seed pairs, with individual differences ranging from −66 to 4380. Table 19 in Appendix E lists the mean returns, sample standard deviations, and paired differences at all 21 checkpoints, together with the nominal and actual interaction counts.

At the nominal 300 000-interaction checkpoint, the pretrained mean validation return first reached or exceeded the baseline mean at the one-million-interaction endpoint. It remained above this level at all subsequent checkpoints. Across paired seeds, the first pretrained checkpoint reaching or exceeding the respective baseline endpoint return occurred between 100 000 and 350 000 nominal interactions.

For the baseline approach, one checkpoint selected by the highest mean validation return occurred at 950 000 and four at 1 000 000 nominal online interactions. For the pretrained approach, the checkpoints were selected at 500 000, 550 000, 650 000, 800 000, and 950 000. Table 20 in Appendix E lists the individual selection counts and corresponding mean validation returns.

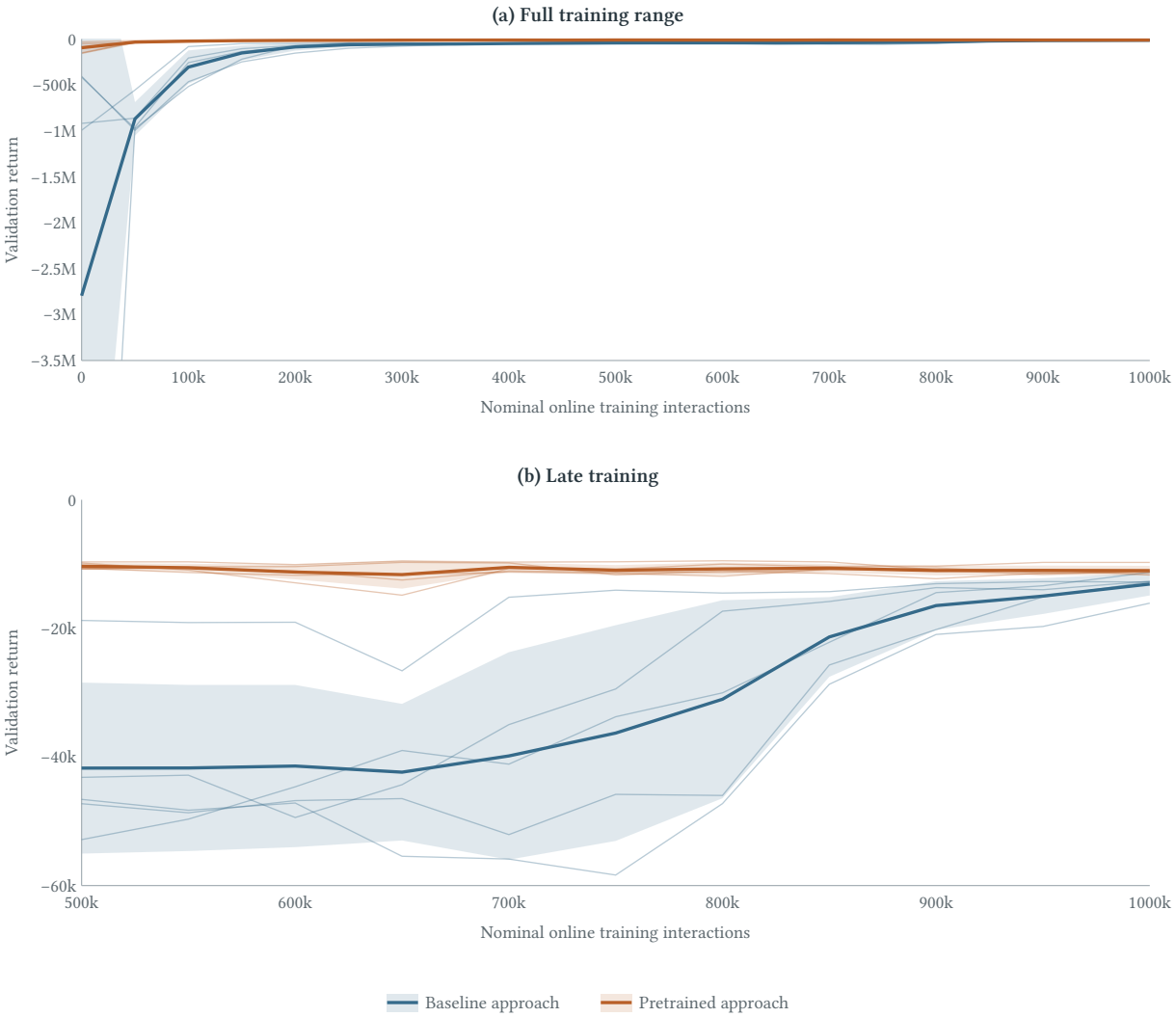


Figure 10: Validation returns over the full training range (a) and from 500 000 to one million nominal online interactions (b), shown against nominal interaction thresholds. Each checkpoint is evaluated on the same 50 validation cases. The panels use separate vertical scales. Thick lines show mean validation returns across five training seeds, shaded bands show one sample standard deviation, and thin lines show individual seeds. Panel (a) uses a vertical range based on the mean validation returns, with seed trajectories and standard deviation bands clipped at its boundaries.

7.2 Final Physical Control Performance

This section compares the selected baseline policy and selected pretrained policy regarding normal operation, contamination response and clearance, and chlorine use. Overall, the selected pretrained policy had lower arsenic burden and greater clearance lead, alongside higher normal-operation chlorine excess and chlorine use. Table 11 summarizes the six physical metrics across five training seeds, with paired differences and 95 % confidence intervals, following Section 6.4. Table 21 in Appendix E lists the paired differences for the individual training seeds.

7.2.1 Normal Operation

Both normal-operation fractions were applicable to all 500 rollouts per approach. Mean reference-normal chlorine shortfall was small for both policies, at 0.0009 % for the selected baseline policy and 0.0074 % for the selected pretrained policy. The paired 95 % confidence interval spanned zero, leaving the direction of the mean difference uncertain.

Table 11: Test control metrics for the selected baseline and selected pretrained policies. The normal-operation chlorine shortfall and excess fractions are $f^{\mathrm{N}}_{\mathrm{low}}$ and $f^{\mathrm{N}}_{\mathrm{high}}$, respectively. The remaining metrics are transported arsenic burden B^{E}, mean arsenic clearance lead Δ_{clear} relative to the zero-chlorine reference, the event-time ceiling-violation fraction $f^{\mathrm{E}}_{\mathrm{cap}}$, and chlorine use U_{eval}. Policy values are means with sample standard deviations across five training seeds. Differences are selected pretrained policy minus selected baseline policy, with fraction differences and their confidence intervals expressed in percentage points. The last column counts applicable evaluations across the five seeds per approach.

Metric	Baseline	Pretrained	Difference	95 % CI	Rollouts
$f^{\mathrm{N}}_{\mathrm{low}}$ (%)	0.0009 ± 0.0014	0.0074 ± 0.0083	0.0065	−0.0041 to 0.0171	500
$f^{\mathrm{N}}_{\mathrm{high}}$ (%)	0.159 ± 0.061	0.268 ± 0.087	0.108	0.040 to 0.177	500
B^{E} (mg h/L)	46.3 ± 3.8	38.1 ± 1.2	−8.3	−13.2 to −3.3	500
Δ_{clear} (h)	2.94 ± 0.26	3.36 ± 0.32	0.41	0.01 to 0.81	475
$f^{\mathrm{E}}_{\mathrm{cap}}$ (%)	0.000 ± 0.000	0.000 ± 0.000	0.000	0.000 to 0.000	480
U_{eval} (t)	2.770 ± 0.032	2.814 ± 0.017	0.043	0.010 to 0.076	500

Reference-normal chlorine excess increased by 0.108 percentage points under the selected pretrained policy. Although small in absolute terms, this increase occurred in all five matched seed pairs, and its 95 % confidence interval excluded zero.

In the exploratory decomposition defined in Section 6.3.3, the first 12 h after reference clearance accounted for 99.88 % of the selected baseline policy's mean chlorine excess and 99.92 % of the selected pretrained policy's mean.

The difference in total excess was almost entirely accounted for by the post-clearance intervals, whose contribution to mean $f^{\mathrm{N}}_{\mathrm{high}}$ was 0.108 percentage points higher under the selected pretrained policy. In the remaining reference-normal periods, the contributions were 0.00019 % for the selected baseline policy and 0.00022 % for the selected pretrained policy, a difference of 0.00003 percentage points.

7.2.2 Contamination Response and Clearance

Mean transported arsenic burden was 17.9 % lower under the selected pretrained policy, with lower burdens in all five matched seed pairs. The paired difference was −8.3 mg h/L, with a 95 % confidence interval from −13.2 to −3.3 mg h/L.

Across the test cases, the policy-independent zero-chlorine reference had a mean case-level burden of 209.1 mg h/L. Relative to this reference mean, transported arsenic burden was reduced by 77.8 % under the selected baseline policy and 81.8 % under the selected pretrained policy. Nearly all of this burden occurred in the main window.

Mean arsenic clearance lead was about 2 hours and 57 minutes for the selected baseline policy and about 3 hours and 21 minutes for the selected pretrained policy. The paired mean difference corresponded to persistent clearance about 25 minutes earlier under the selected pretrained policy. These time values are rounded to the nearest minute. Clearance lead was higher in all five matched seed pairs, but the 95 % confidence interval for the paired difference ranged from about 1 minute to about 49 minutes, indicating uncertainty in the size of this improvement.

This clearance comparison was conditional on 475 eligible paired rollouts per approach. Of the 25 excluded rollouts, 20 had no reference-event windows because the zero-chlorine reference never exceeded 0.01 mg/L. The remaining five concerned `scenario-96` with arsenic schedule `arsenic-5`,

where at least one policy never exceeded the threshold. Neither policy exceeded it for seeds 17, 42, 123, and 71026. For seed 1337, the selected baseline policy stayed at or below the threshold while the selected pretrained policy exceeded it, leaving no eligible paired clearance-time comparison.

The event-time chlorine ceiling-violation fraction was zero for both policies in all applicable event windows. The ceiling-violation fraction excluded only the 20 rollouts without reference-event samples and was therefore applicable to 480 rollouts per approach, following Section 6.3.1.

Figure 11 compares performance across test cases and training seeds. Averaged across the five seeds, the selected pretrained policy's burden was lower in 78 of the 100 fixed test cases, equal in 4, and higher in 18. Across all five matched seed pairs, lower mean burden coincided with earlier persistent clearance on average over the eligible events.

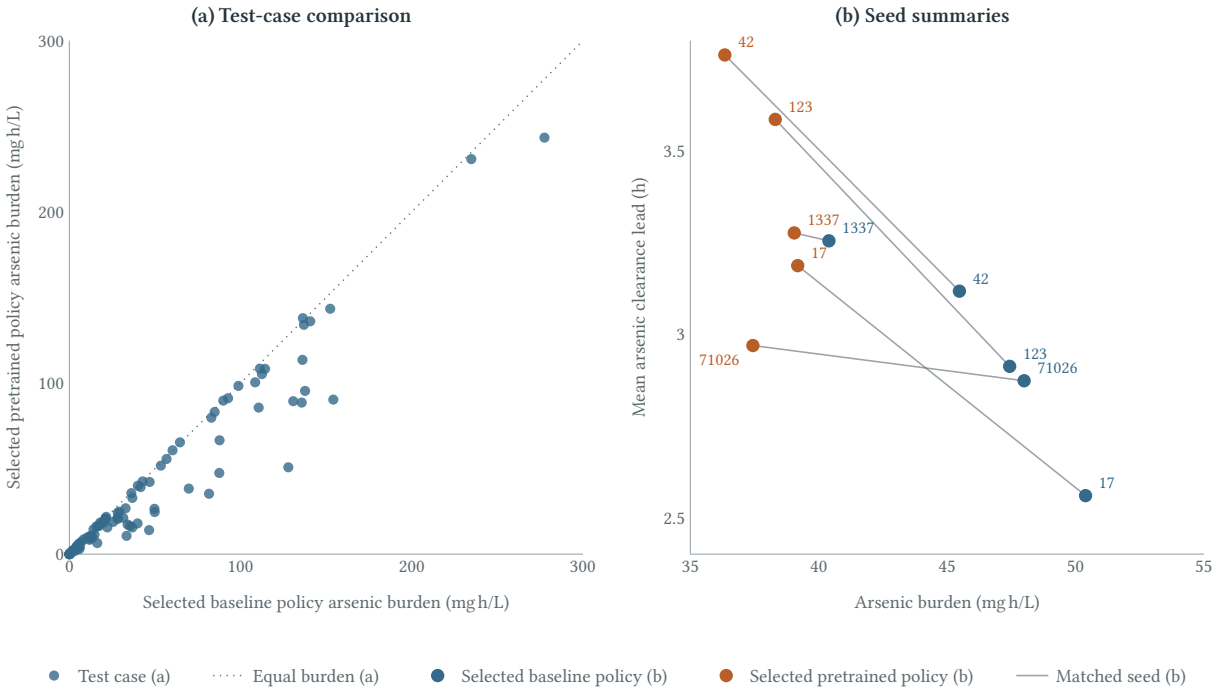


Figure 11: Physical test performance of the selected baseline and selected pretrained policies. In (a), each of the 100 points pairs the selected baseline and selected pretrained arsenic burdens for one test case, averaged across five training seeds. Both axes use the same scale. The dotted line marks equal burdens. Points below it indicate lower burdens for the selected pretrained policy, and points above it indicate higher burdens. In (b), each of the ten points summarizes arsenic burden across all 100 test cases and mean arsenic clearance lead relative to the zero-chlorine reference across the applicable cases for one policy and training seed. Colors identify the policies, labels identify the seeds, and lines connect matched seed pairs.

7.2.3 Chlorine Use

The selected pretrained policy used 1.6 % more chlorine on average, alongside the 17.9 % reduction in arsenic burden. Chlorine use was higher in all five matched seed pairs, and the paired 95 % confidence interval excluded zero. This additional chlorine represents greater treatment effort, with about 80 % of the increase occurring during reference-event periods in the main window.

In the main window, reference-normal operation accounted for about 2.18 t for the selected baseline policy and 2.19 t for the selected pretrained policy. The corresponding reference-event components were about 0.59 t and 0.62 t.

7.3 Computational Cost

Summing the recorded training-pipeline stage runtimes across the five seeds gives about 10 hours and 47 minutes for the baseline approach and about 52 hours and 6 minutes for the pretrained approach, including checkpoint validation and test evaluation. The pretrained pipeline took about 4.8 times as long. Its shorter PPO phase did not offset the offline preparation cost. Table 12 summarizes the runtimes of data generation, model training, and evaluation.

For the baseline approach, the mean total runtime per seed was about 2 hours and 9 minutes and about 10 hours and 15 minutes for the pretrained approach. Across the five training seeds, mean PPO training time was about 1 hour and 27 minutes for the baseline approach and about 1 hour for the pretrained approach. Validation and test evaluation added about 43 minutes per seed for both approaches.

Simulating the offline trajectories used to train the forecasters and to generate the MPC teacher dataset took about 53 minutes with 8 CPU workers. The trajectories were generated once and reused across the five pretrained runs. Their generation time is counted once in the global pretrained total and excluded from the per-seed means.

The seed-specific offline preparation of the forecasting models, the MPC teacher demonstrations, and behavior cloning required a mean of about 8 hours and 32 minutes per pretrained seed. Training the five fold-specific forecasting models required a mean of about 7 hours and 9 minutes per seed, or about 1 hour and 26 minutes per fold-specific forecaster. Generating the MPC demonstrations required a mean of about 1 hour and 16 minutes per seed. The mean times were about 49 minutes for the behavior cloning training dataset and about 27 minutes for the separate validation dataset. Mean generation time per training fold was about 10 minutes. Mean behavior cloning time was about 7 minutes per seed.

The wall-clock times refer to the training computer described in Section 5.8 and the stage-specific worker counts in Table 12. The learning stages used the GPU through CUDA. Trajectory generation and MPC demonstration generation used the CPU.

Table 12: Wall-clock costs per training seed, reported as mean ± sample standard deviation across the five seeds in hours. Forecaster training includes all five folds per seed. Control evaluation includes checkpoint validation and test evaluation. Totals sum the recorded pipeline stages for each seed before calculating the mean and standard deviation. Shared trajectory generation is shown separately from the per-seed totals.

Stage	Device / workers	Baseline (h)	Pretrained (h)
Shared offline trajectory generation (once)	CPU / 8	–	0.878
Forecaster training	GPU / 8	–	7.147 ± 0.761
MPC demonstrations (training)	CPU / 8	–	0.821 ± 0.005
MPC demonstrations (validation)	CPU / 8	–	0.450 ± 0.020
Behavior cloning	GPU / 8	–	0.117 ± 0.021
PPO training	GPU / 4	1.446 ± 0.015	0.999 ± 0.071
Control evaluation	GPU / 8	0.711 ± 0.004	0.712 ± 0.001
Total per seed		2.157 ± 0.015	10.246 ± 0.753

7.4 Additional Variant Comparisons

The two variants defined in Section 5.7.1 were compared through validation return following Section 6.6. Table 13 summarizes their endpoint comparisons with the pretrained approach. Checkpoint selection follows Section 6.1.2.

At 1 550 000 nominal interactions, the extended-budget baseline mean first reached or exceeded the pretrained mean at the one-million-interaction endpoint. It later fell below this reference again before ending slightly above it at 4 000 000 interactions. At this endpoint, the paired difference was positive in 4 of the five seed pairs, and its 95 % confidence interval spanned zero.

The selected policies of the extended-budget baseline had a mean validation return of −10 340, compared with −10 030 for the selected pretrained policies. The checkpoints were selected between 1 800 000 and 4 000 000 nominal interactions for the extended-budget baseline and between 500 000 and 950 000 for the pretrained approach. The mean paired difference between the selected policies was −309, with a paired 95 % confidence interval from −1280 to 658.

Without actor weights, the mean validation return exceeded the ordinary baseline mean from 50 000 through 950 000 nominal interactions, but remained below the pretrained mean at every checkpoint. Its mean first reached or exceeded the ordinary baseline's one-million endpoint return at 750 000 nominal interactions. At the one-million endpoint, its mean was again slightly below the ordinary baseline's. Relative to the pretrained approach, its endpoint return was lower in all five matched seed pairs, with a paired 95 % confidence interval below zero.

The selected policies without actor weights likewise had a mean validation return of −12 470, compared with −10 030 for the selected pretrained policies. This selected-policy difference was negative in all five matched seed pairs, with a paired 95 % confidence interval below zero.

Mean PPO training time per seed was about 4 hours and 10 minutes ± 6 minutes for the extended-budget baseline and about 1 hour and 18 minutes ± 2 minutes for the variant without actor weights. Checkpoint validation for the extended-budget baseline took a mean of about 2 hours and 24 minutes per seed.

Table 13: Endpoint validation returns for the additional variants. The 'Interactions' column gives the nominal online training interaction counts. Returns are means with sample standard deviations across five training seeds. Differences are variant minus pretrained at the pretrained approach's one-million-interaction endpoint, with paired 95 % confidence intervals.

Variant	Interactions	Validation return	Difference	95 % CI
Extended-budget baseline	4 000 000	−10 660 ± 290	355	−514 to 1220
Without actor weights	1 000 000	−13 180 ± 770	−2160	−3870 to −455

7.5 Supporting Case-Level Evidence

Test scenario `scenario-90` with arsenic schedule `arsenic-8` and training seed 17 were selected to illustrate a pronounced reduction in transported arsenic burden under the selected pretrained policy alongside differences in the responses to individual contamination events. The plotted policy rollouts are part of the aggregate test results above. The 25-day episode contains four arsenic source-injection events at different nodes. Two of these overlap in time, resulting in three reference-event windows, as defined in Section 6.3.1. Figure 12 shows the complete episode, including chlorine control between the contamination events.

The largest separation occurs in the third reference-event window, where mean nodal chlorine concentrations under the selected pretrained policy rise above the normal upper bound, while those under the selected baseline policy fall towards the normal lower bound. The transported As(III) concentration sum peaks at 9.79 mg/L in the zero-chlorine reference, compared with 6.96 mg/L for the selected baseline policy and 3.34 mg/L for the selected pretrained policy. In this window, both the zero-

chlorine reference and the selected baseline policy reached persistent clearance at 332 hours and 30 minutes of simulation time, while the selected pretrained policy reached it 1 hour earlier.

For the complete case, transported arsenic burden was 383.7 mg h/L in the zero-chlorine reference, while it was 200.1 mg h/L for the selected baseline policy and 91.5 mg h/L for the selected pretrained policy. Figure 13 places this comparison in the context of all 5 training seeds for the same test case. Arsenic burden was lower for the selected pretrained policy than for the selected baseline policy for seeds 17, 42, and 71026. It was higher for seeds 123 and 1337. Each group compares the two policy burdens with the same zero-chlorine reference.

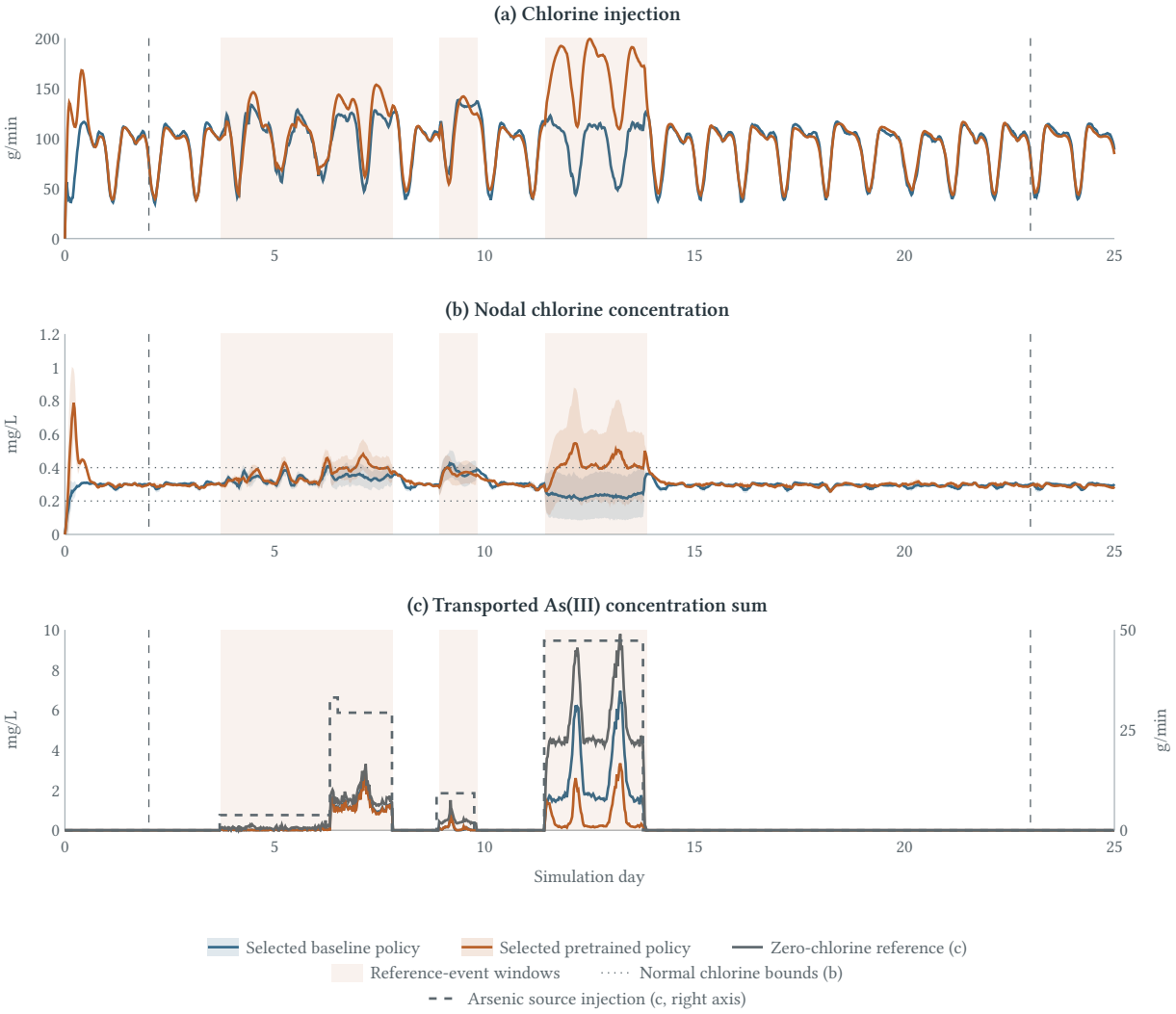


Figure 12: Complete trajectories for the selected illustrative case, `scenario-90` with arsenic schedule `arsenic-8`, and training seed 17. Panels show (a) chlorine injection, (b) mean nodal chlorine concentration with one population standard deviation across the 31 nodes, and (c) the transported As(III) concentration sum (left axis) with total arsenic source injection as dashed steps (right axis). Values are reported every 30 minutes over 25 days. Lower band edges in (b) are clipped at zero. Horizontal dotted lines mark the normal chlorine bounds of 0.2 mg/L and 0.4 mg/L. Vertical dashed lines mark the end of the warm-up at simulation day 2 and the start of the clearance tail at simulation day 23. Shaded intervals mark the reference-event windows.



Figure 13: Transported arsenic burden for the selected illustrative case, test scenario `scenario-90` with arsenic schedule `arsenic-8`, for each training seed. Separate bars show the actual burdens under the selected baseline policy, selected pretrained policy, and zero-chlorine reference.

7.6 Supporting Forecasting Results

Forecasting accuracy was evaluated on the ten forecasting test scenarios, with equal weight assigned to each scenario. For each training seed, MAE and RMSE were averaged across the five fold-specific models. The reported values summarize these five seed means through their mean and sample standard deviation.

Over the full forecasting horizon of eight hours, MAE was 0.02519 ± 0.00042 mg/L and RMSE was 0.05036 ± 0.00039 mg/L. The full-horizon errors for all 25 models are shown by training seed in Figure 15 and listed by seed and fold in Table 22, both in Appendix E.

Both mean errors increased at every step of the forecast horizon, as shown in Figure 14. At the first step, 30 minutes ahead, mean MAE was 0.01954 mg/L and mean RMSE was 0.03576 mg/L. At the final step, 8 hours ahead, the corresponding errors were 0.02754 mg/L and 0.05674 mg/L. Values for each forecast step are listed in Table 23 in Appendix E.

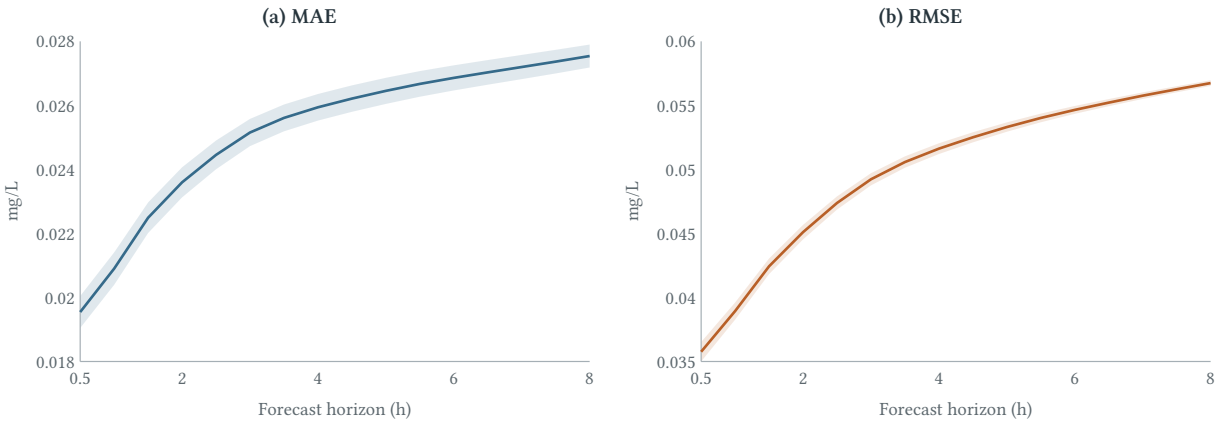


Figure 14: Chlorine-forecast errors over the eight-hour horizon, with (a) MAE and (b) RMSE. Errors are evaluated on the ten forecasting test scenarios under prescribed injection profiles. Lines show the mean and shaded bands show one sample standard deviation across the five seed means.

8 Discussion

This chapter interprets the results in relation to the research question. It first examines how pretraining affects online learning and final physical control, including normal-operation chlorine excess alongside arsenic burden, clearance timing, and chemical use. The discussion then weighs these outcomes against the simulator interactions and computational cost of preparation, and considers opportunities for reuse and parallel execution. Finally, it assesses the supporting forecasting evidence and the limitations that bound the interpretation and practical relevance of the findings.

8.1 Answer to the Research Question

The results support an early online learning advantage from MPC-guided pretraining in the evaluated Hanoi scenarios under the common online interaction budget. The pretrained approach achieves a higher mean validation return at every checkpoint in Figure 10, including before any PPO update. The advantage becomes much smaller as the baseline approach improves, but remains positive in four of the five matched seed pairs at the budget endpoint. For the online sample-efficiency part of the research question, pretraining therefore supplies useful initial behavior and maintains better mean performance throughout the evaluated training range.

The shortfall difference remains uncertain. For final physical control, the selected pretrained policies have lower transported As(III) burden and greater mean clearance lead on the paired eligible events than the selected baseline policies, alongside higher normal-operation chlorine excess and chlorine use. No event-time chlorine ceiling violations occur in the applicable test observations. These four paired effects have confidence intervals excluding zero in Table 11, and their directions agree across all five seeds in Table 21 in Appendix E. The exploratory decomposition of the chlorine excess metric locates nearly the entire excess difference in the first 12 h after reference clearance. Pretraining thus improves two complementary aspects of contamination treatment under the modeled reaction, while the additional chlorine exceedances occur primarily during the subsequent return to normal operation. Its benefit depends on the balance between treatment, residual concentrations, and chemical use required by the intended control application.

8.2 Online Learning and Final Control

The zero-interaction comparison isolates what the initialization provides before reward-based fine-tuning. At this point, the behavior-cloned policy already controls the network more effectively under the validation objective than the ordinarily initialized policy. Because the forecaster and optimizer are absent during policy evaluation, this result shows that the recurrent actor can use the transferred initialization in closed-loop control. It does not establish which transferred component produces the improvement.

The initial mean difference should nevertheless be interpreted alongside the individual seed trajectories. One baseline seed begins with a particularly large negative return, which strongly influences the mean and its spread in Figure 10. The advantage is therefore less uniform in magnitude than the mean curve alone suggests. Its continuation beyond the first few checkpoints matters more than the extreme initial gap, because it shows that the comparison is not determined only by one poor ordinary initialization.

The initial cloned policy also provides a concrete reference for the amount of learning required by the baseline approach. Its mean validation return is approximately −95 000. The baseline approach first exceeds this value at 204 800 actual interactions, corresponding to the nominal 200 000-interaction checkpoint. Within individual seed pairs, the first observed crossing of the respective pretrained zero-interaction return occurs between 100 000 and 850 000 nominal interactions. These are retrospective comparisons with the cloned initialization, rather than interaction savings against a predefined control

target. Checkpoint spacing limits their temporal resolution, and equality of return would not imply equality of the six physical metrics. Nevertheless, they show that the offline initialization supplies behavior that the baseline approach otherwise acquires only after substantial online learning.

Fine-tuning remains useful after cloning. The pretrained approach's mean return improves substantially from its initial value and reaches a much better range during PPO. Meanwhile, the baseline approach continues to improve, especially late in training, narrowing the mean difference to approximately 2090 at the endpoint. This pattern supports an initialization advantage within the investigated budget. The remaining gap does not establish different asymptotic performance.

The endpoint return and the final physical comparison answer different questions. The former describes both approaches at the same cumulative interaction count. The latter evaluates the checkpoints selected by their highest mean validation return, following Section 6.1.2. As Table 20 in Appendix E shows, these checkpoints can occur at different stages of training. Earlier selection of several pretrained checkpoints suggests that useful policies become available sooner, but all runs completed the common budget. Training after the selected checkpoint therefore remains part of the measured cost.

The additional comparisons in Section 7.4 qualify the return advantage at the common online interaction budget. At 1 550 000 nominal interactions, the extended-budget baseline first reaches the pretrained approach's mean validation return at the one-million-interaction endpoint. It ends slightly above this reference after four million nominal interactions. Its selected policies have a lower mean validation return than the selected pretrained policies, although the paired confidence interval for this difference spans zero. The selected pretrained checkpoints occur earlier in online training. Additional baseline training therefore narrows the difference in validation return, but the comparison establishes neither a return advantage after checkpoint selection nor equivalence.

The variant without actor weights has higher mean return than the ordinary baseline during most of training, but this advantage is absent at the endpoint. Full pretraining has higher validation return both at the endpoint and after checkpoint selection, with differences in the same direction across all five seed pairs and confidence intervals excluding zero. This supports a contribution from the cloned actor weights beyond the learning advantage provided by prepared normalization and initial action standard deviation. However, their contribution without these two transferred components remains untested.

Finally, return summarizes the particular reward objective defined in Section 5.2. It combines chlorine tracking, concentration movement, and action changes, with relaxed upper-concentration weighting during recovery. It does not directly minimize transported arsenic burden or injected chlorine mass. The physical test metrics are consequently needed to determine what the return advantage means for treatment and dosing.

8.3 Physical Control Trade-offs

When assessing the final physical control performance, we give priority to lower transported As(III) burden and earlier persistent clearance when weighing these gains against chlorine use and post-clearance chlorine excess.

The final metrics indicate that the selected baseline policies and selected pretrained policies generally maintain the normal-operation lower residual bound in the evaluated cases. The shortfall fractions are small in absolute terms, but the paired confidence interval includes both improvement and deterioration. This uncertainty does not establish equivalence. The selected pretrained policy also has a higher normal-operation excess fraction in every seed pair. Since this difference is measured in percentage points, its absolute magnitude in Table 11 is more informative than a large relative change from a small baseline value.

The exploratory decomposition reported in Results locates almost all chlorine excess under both policies in the first 12 h after reference clearance and nearly the entire pretrained-minus-baseline difference in these same periods. Other reference-normal periods contribute very little under either policy. The metric defined in Section 6.3.3 therefore includes the return to normal operation after contamination. Its higher value under pretraining does not by itself establish generally less stable or poorer control during normal operation. The exceedances remain departures from the normal chlorine bound and are relevant when assessing the residual concentrations left after treatment.

The clearest treatment benefit is the reduction in transported As(III) burden. The mean burden of the selected pretrained policies is about 17.9 % lower than that of the selected baseline policies. The lower burden is consistent across seed summaries, while Panel (a) of Figure 11 shows lower pretrained mean burden in 78 of the 100 test cases, equal burden in four, and higher burden in 18. The benefit is therefore broad across the fixed test catalog but does not apply to every hydraulic scenario and contamination schedule.

Clearance timing complements the integrated burden by showing whether relevant transported As(III) persists for less time. On average, the selected pretrained policies reach persistent clearance about 25 minutes earlier than the selected baseline policies across the paired eligible events. Panel (b) of Figure 11 shows that lower burden and greater clearance lead occur together in every matched seed summary. The timing advantage is modest, and its confidence interval extends close to zero.

Neither approach exceeded the event-time chlorine ceiling in the eligible test observations. The treatment advantage under pretraining is therefore not accompanied by additional event-time ceiling violations. Both approaches share a reward that continues to penalize above-target chlorine during recovery, although with substantially reduced weighting. The actuator bounds in Section 5.1.2 constrain source injection, but do not impose hard limits on every nodal chlorine concentration.

These treatment gains are accompanied by about 1.6 % greater chlorine use. The use decomposition places most of the additional chlorine in the main-window reference-event periods, while the excess decomposition places nearly all additional exceedances in the period after reference clearance. Under matched hydraulic conditions and identical reaction parameters, the higher chlorine use, lower transported As(III) burden, and earlier clearance suggest a stronger treatment response under the modeled oxidation reaction in Equation 60 in Appendix B. The higher post-clearance excess is consistent with elevated residual chlorine remaining after treatment. However, the comparison does not quantify how much of the additional injected chlorine is consumed by oxidation rather than lost through decay or remaining as residual chlorine. It also does not isolate the contribution of the larger injected mass from that of injection timing and the resulting chlorine availability in the network.

Under the treatment priority adopted above, we favor the selected pretrained policies because their lower transported As(III) burden and earlier persistent clearance outweigh the modest increase in chlorine use in the evaluated setting. The absence of additional event-time ceiling violations supports this preference within the observed test cases. The additional normal-operation chlorine excess remains a cost, but its absolute increase is small and occurs almost entirely in the first 12 h after reference clearance. With very little additional excess in other reference-normal periods, this pattern supports accepting the higher post-clearance residuals in favor of more effective contamination treatment. Reducing this excess while preserving the treatment gains would improve the balance further.

8.4 Interaction Cost and Computational Cost

The learning advantage uses an additional source of training information. The precomputed offline trajectories across the training, validation, and test partitions require three million simulator steps before online learning. Forecaster training and MPC teacher generation reuse these trajectories, so repeated processing does not add to this trajectory-generation cost. Together with one million

nominal online interactions, this defines the extended-budget baseline's online interaction budget of four million. Separate simulator checks and control evaluations are excluded from this comparison. The experiment supports better performance per online interaction, but does not establish lower total simulator use.

The stage times in Table 12 give a similarly qualified result. Mean PPO training time is about 1 hour and 27 minutes per baseline run and about 1 hour per pretrained run, despite their matched interaction counts. This shorter fine-tuning stage does not offset preparation. Across all five seeds, forecasting takes about 35 hours and 44 minutes, teacher generation about 6 hours and 21 minutes, and cloning about 35 minutes. Shared trajectory generation adds about 53 minutes. Including the separately reported evaluations, the summed stage runtimes are about 10 hours and 47 minutes for the baseline approach and about 52 hours and 6 minutes for the pretrained approach.

The variant runtimes in Section 7.4 reinforce the distinction between online sample efficiency and total training effort. The extended-budget baseline takes about 4 hours and 10 minutes per seed for PPO and another about 2 hours and 24 minutes for checkpoint validation. Even this longer training remains less costly in measured runtime than fresh pretraining and fine-tuning. Without actor weights, PPO takes about 1 hour and 18 minutes per seed. This variant still uses normalization and action standard deviation obtained through the offline pipeline, so its online runtime does not describe an alternative without preparation cost.

Forecasting is the dominant preparation cost. The five-fold design makes each training scenario contribute to fitting four models, while its teacher labels are generated using the remaining model. This exchanges additional model fitting for reuse of one simulator dataset and scenario-level separation between forecasting inputs and teacher queries. It is appropriate when simulator interactions are the scarce resource, but the measured stage times show why minimizing interactions need not minimize preparation time. On the recorded setup, generating the complete offline collection takes less than one hour with eight CPU workers, whereas fitting the 25 forecasters accounts for most of the offline runtime. Fewer folds could reduce this fitting cost while preserving scenario-level separation. However, larger held-out folds would leave fewer training scenarios for each forecaster, which could impair forecast accuracy and the quality of teacher actions. Comparisons should therefore assess whether fewer folds preserve forecast accuracy and final control performance at lower preparation cost.

An alternative would train one forecaster on 80 scenarios, select it using ten validation scenarios, and generate a separate teacher-query collection under new contamination schedules. Cloning could then use the new training demonstrations and their own validation demonstrations. This would require more simulator trajectories but avoid fitting five forecasters per policy seed. With the same hydraulic scenarios, the two datasets differ only in their contamination schedules. It would not preserve the current exclusion of each teacher scenario from forecaster fitting, which was introduced to reduce the risk of overly optimistic MPC action evaluations on scenarios seen during forecaster training. To preserve that stronger condition with one forecaster, the teacher queries would need a separate hydraulic scenario collection. Either design would also require validation data kept out of fitting and a separate final test set.

The measured costs permit an illustrative estimate of the change in preparation time. One fold-specific model takes about 1 hour and 26 minutes on average and trains on 64 scenarios. If runtime scaled with the number of training windows at unchanged epoch count and throughput, fitting one model on 80 scenarios would take about 1 hour and 47 minutes. Under this assumption, the forecasting training volume would fall by 75 % relative to five 64-scenario fits. Keeping teacher-generation and cloning times at their measured means gives about 3 hours and 10 minutes per run for these three stages, compared with about 8 hours and 32 minutes in the current pipeline. This estimate excludes new

trajectory generation and evaluation. It is a cost projection, since early stopping, model generalization, and the resulting policy quality could change under the alternative design.

Teacher generation would still label the complete training collection. In the current design, each training query is already evaluated using only one fold-specific forecaster, so replacing five folds with one model does not remove four fifths of these queries. A single model would reduce repeated prediction work for validation, where the current teacher uses a five-model ensemble. The resulting total speedup cannot be inferred directly from the number of folds. The estimate above therefore conservatively keeps measured teacher runtime unchanged.

Preparation costs can also be amortized. A published simulation dataset could be reused for further pretraining studies on the same network and compatible demand, reaction, and dosing conditions. Reuse would remove fresh trajectory generation, while changed forecasting models or teacher objectives would still require fitting or relabeling. If an already fitted forecaster and a compatible teacher dataset were reused as well, a new policy would mainly incur cloning and subsequent fine-tuning costs. At the measured means, these two stages together take about 1 hour and 7 minutes per run, excluding evaluation. The present experiment deliberately refits the forecasters and regenerates labels for every seed to assess variability in the complete pipeline. Sharing these artifacts across future policies would be a different replication design.

Historical records from a real drinking-water network could also replace simulated offline trajectories if they provide synchronized chlorine measurements, source injections, and the temporal context required by Section 5.5.3. This would remove the cost of generating those trajectories in a simulator. Its feasibility depends on whether the records cover the relevant hydraulic conditions, dosing sequences, and disturbances. Routine operation alone may offer insufficient evidence for predicting contamination responses or unfamiliar candidate injections. Operational-data reuse is therefore a possible preparation strategy whose forecast and control validity would need separate evaluation. It does not remove the remaining forecaster fitting, teacher-generation, and cloning stages, or the simulator fine-tuning used in this study.

Parallel execution offers another way to shorten elapsed preparation time. Fold-specific forecasting runs are independent and could be trained simultaneously on separate devices. Teacher labeling can be distributed across scenarios or trajectories because one trajectory's generated actions do not determine another trajectory's query states. Each worker still needs the appropriate trained forecaster, observation history, and reconstructed reward state. Thus, independent task allocation is possible even though the reward calculation within a trajectory is stateful. The measured teacher runs already use eight workers. Assigning more independent trajectory groups to additional machines could shorten elapsed time, but would retain the prediction and optimization work and introduce data-transfer and resource costs.

These alternatives change the practical value of the initialization without changing the evidence from the principal comparison. For a single policy with fresh preparation, the current pipeline is far more expensive than the baseline approach. However, for repeated policy training from reusable artifacts, or under a design that uses one forecaster, the remaining preparation cost could be substantially smaller. In all cases, policies trained with either approach use the same recurrent architecture and do not require forecasting or MPC at inference time.

8.5 Meaning of the Supporting Forecast Evaluation

The forecast results assess the predictive basis of the teacher, rather than the quality of its selected actions. Full-horizon mean MAE is about 0.025 mg/L and RMSE about 0.050 mg/L on the ten forecasting test scenarios, as shown in Figure 15 in Appendix E. These values are smaller than the 0.2 mg/L width of the adopted normal-operation band, but remain relevant near its boundaries. An average error below

the band width does not ensure correct classification of residual shortfalls or excesses. The higher RMSE also makes clear that a single mean absolute error does not describe all prediction deviations.

Both errors increase with the prediction horizon in Figure 14. This suggests greater uncertainty in the concentration predictions used for the teacher's longer-term action comparisons. The measured errors nevertheless suggest that the forecasters approximate the multistep chlorine response reasonably well from the limited inputs defined in Section 5.5.3. They use chlorine and injection histories, planned future injections, and calendar features, without direct information about nodal demand or contamination events.

The scope of this evidence also depends on the injection sequences used for evaluation. Both forecaster training and the supporting test use trajectories generated with prescribed injection profiles that expose the models to varied chlorine responses. During teacher generation, however, the optimizer searches for injection sequences with high predicted returns. These sequences may differ from those represented in the offline data. Accuracy on the test trajectories therefore does not establish how accurately the forecasters predict the effects of the candidate sequences used to select teacher actions.

The higher validation return before online training and the subsequent physical test outcomes are therefore stronger evidence for the usefulness of the complete initialization than forecasting errors alone. They show that the pipeline can transfer useful behavior despite imperfect prediction. They do not demonstrate teacher optimality, rule out exploitation of prediction errors, or establish a causal relationship between a fold's forecast error and its policy outcome. Those questions would require evaluating selected candidate sequences in the simulator and relating their prediction and action errors to closed-loop control.

8.6 Limitations

The statistical comparison uses only five matched pairs of training runs. This limits how precisely the average difference between the approaches can be estimated. The confidence intervals describe this uncertainty for the shared offline trajectories and fixed evaluation cases used in this study. They do not capture possible variation from newly generated offline data or different evaluation cases. Within this setting, the confidence intervals for four physical metrics exclude zero, and all five seed pairs agree on the direction of these differences. However, the size of the differences varies across seeds and individual test cases.

The dataset limits how broadly the findings apply. Its hydraulic scenarios differ in demands and uncertain network parameters, but all use the Hanoi topology, the same chlorine source location, and the same actuator constraints. All scenarios also cover the same 25-day period, which captures only a limited part of the seasonal demand patterns provided by LeakDB [52]. The contamination schedules vary, but all follow the same event-generation rules in Section 4.2. Performance on other networks, in other seasons, or under different contamination settings therefore remains untested.

The physical interpretation depends on the transport, mixing, and reaction assumptions in Section 4.3. The model tracks free chlorine and As(III) and omits longitudinal dispersion. It represents the oxidation of As(III) by chlorine without explicitly tracking As(V). A lower transported As(III) burden therefore indicates improved treatment under the modeled oxidation reaction. It does not demonstrate a reduction in total arsenic or establish drinking-water safety.

The reported clearance advantage covers only events in which both policies exceed the transported-arsenic threshold, as defined in Section 6.3.5. If either policy stays at or below this threshold, the event is excluded from the clearance comparison. The result therefore does not describe all contamination events, whereas integrated arsenic burden covers all test cases. Clearance times are determined in half-

hour steps within the fixed observation period, which limits the timing resolution and the duration of follow-up.

The learning curves and checkpoint selection use the same fixed control-validation cases, as defined in Section 6.1.2. The selected validation return therefore does not provide an independent assessment of performance on new cases. Separate test cases provide the final physical comparison. Before fine-tuning, the cloned initialization already achieves a higher mean return on the validation cases. However, a test comparison using the six physical metrics is not reported for this initial policy. The results therefore do not show whether the cloned initialization already achieves the treatment and dosing balance observed after fine-tuning.

The principal comparison evaluates the complete initialization, so its effects cannot be attributed to actor weights alone. Actor weights, observation normalization, and the initial action standard deviation change together, while the critic remains ordinarily initialized. The variant without actor weights in Section 7.4 assesses the contribution of actor weights to validation return. Prepared normalization and action standard deviation are retained in this comparison. The individual contributions of these two components and their interactions therefore remain unclear. The additional variants were compared using validation return, without assessing their physical treatment and dosing outcomes. It also remains unclear whether cross-fitting and the choice of folds, the chosen forecasting architecture, or the particular teacher configuration are necessary for the observed advantages in online learning and final contamination treatment.

The teacher and cloning stages do not directly optimize the cloned policy's performance over complete control episodes. The MPC teacher evaluates actions over eight hours and uses no terminal value to account for later consequences. Cloning minimizes action MSE at sampled query states, without giving extra weight to errors that have greater effects on later control. These query states come from prescribed injection trajectories. The teacher's action labels do not change how those trajectories continue. During closed-loop control, the actor's own actions shape the later states, so it can encounter a different mix of states from those used for cloning. This risk of sequential distribution shift is a known limitation of offline imitation and motivates procedures such as DAgger [41]. Here, PPO allows adaptation through reward feedback. However, the comparison does not show how much of the improvement comes from correcting imitation errors, covering more states, or changing the inherited control strategy.

The findings apply to the observation setting in Table 5, where the controller receives chlorine measurements at all 31 demand nodes. Its performance with fewer sensors, measurement noise, missing values, or sensor delays was not tested. The results therefore do not establish how well the controller would perform under different sensor coverage or measurement conditions.

8.7 Overall Interpretation

The experiment supports the use of offline predictive control to initialize a recurrent policy for the examined partially observable chlorination task. It extends the model-based initialization and model-free fine-tuning approach demonstrated in other control domains [29] through a fixed cross-fitted teacher dataset. The learning and physical test results jointly support useful initialization, with higher mean validation return during online learning and more effective contamination treatment under the adopted priority. These gains are accompanied by modest additional chlorine use and post-clearance excess, while fresh preparation takes more time than is saved during fine-tuning. The practical value of pretraining therefore depends on the relative priority given to treatment and dosing, and on whether preparation can be reused or reduced without losing control quality.

9 Conclusion

This thesis examined whether MPC-guided policy pretraining provides a useful initialization for recurrent chlorination control. This chapter summarizes the findings on online learning, final physical control performance, and preparation cost. We then discuss the contribution and its scope, followed by further work needed to assess the approach beyond the evaluated setting.

9.1 Answer to the Research Question

MPC-guided pretraining improved online sample efficiency in the evaluated Hanoi setting. The pretrained approach had a higher mean validation return at every evaluated checkpoint, including before online training. At 300 000 nominal interactions, its mean validation return first reached the baseline's return at the one-million-interaction endpoint and remained above this level at all subsequent checkpoints. This comparison shows how much earlier the pretrained approach reached this return level. The crossing was identified retrospectively from validation returns and does not establish equal physical control at that point. The mean return difference became much smaller as the baseline improved. The results therefore support a learning advantage within the investigated budget but do not establish differences in asymptotic performance.

The final test comparison also favored pretraining for contamination treatment. After both approaches completed the common online budget, the policies selected by validation return were evaluated separately on the fixed test cases. The selected pretrained policies had 17.9 % lower mean transported As(III) burden and reached persistent clearance about 25 minutes earlier on average over the paired eligible events. Both differences favored pretraining in all five matched seed pairs. The clearance comparison covers only events in which both policies exceeded the operational threshold, whereas the burden comparison covers all test cases.

These treatment gains were accompanied by higher chlorine use and greater normal-operation chlorine excess. Mean chlorine use increased by 1.6 %, and the reference-normal excess fraction increased by 0.108 percentage points. Together with the lower transported As(III) burden and earlier clearance, the higher chlorine use suggests a stronger treatment response under the modeled oxidation reaction. The comparison does not determine how much of the additional chlorine was consumed by oxidation. In the exploratory decomposition, almost all of the excess difference occurred in the first 12 h after reference clearance. The direction of the shortfall difference remained uncertain, and neither approach exceeded the 5.0 mg/L event-time ceiling in the applicable test observations. Under the treatment priority adopted in Section 8.3, the lower burden and earlier clearance favor the pretrained policies despite their higher chlorine use and post-clearance excess. Pretraining therefore improved the treatment balance under this priority but did not improve every control metric.

The online learning advantage did not reduce total computational effort. Across the five seeds, the recorded pipeline stages, shared offline trajectory generation, and control evaluations took a total of about 52 hours and 6 minutes for the pretrained approach and about 10 hours and 47 minutes for the baseline approach. With fresh preparation, the pretrained approach therefore took about 4.8 times as long, despite a shorter PPO stage. It also required offline simulator trajectories in addition to the matched online interactions. Pretraining therefore improved online sample efficiency and final contamination treatment but required additional preparation effort.

9.2 Main Contribution and Scope

The main contribution is an evaluated training pipeline that combines multistep chlorine forecasting, cross-fitted MPC teacher generation, behavior cloning, and RecurrentPPO fine-tuning. Scenario-wise cross-fitting reuses the offline trajectories but generates teacher actions for each scenario using a forecaster fitted without that scenario. Behavior cloning trains the recurrent actor to reproduce the teacher

actions before it is fine-tuned through reward-based interaction. The forecaster and optimizer are used only during offline preparation. The resulting controller observes only chlorine measurements, its own dosing, and time records.

Both approaches use the same deployed architecture, observations, actuator constraints, reward, and online training conditions. The principal comparison therefore assesses the complete initialization, which includes actor weights, observation normalization, and initial action standard deviation. The comparison with the variant without actor weights supports a contribution from cloned actor weights to validation return, conditional on the other transferred components. The extended-budget baseline shows that additional online training can narrow the return difference, but these additional comparisons do not establish equal physical control. The forecast evaluation provides supporting evidence for the prediction model, while the parameter screening provides practical settings. The final treatment benefit is supported by the replicated policy comparison on separate test cases.

These findings are based on five matched pairs of training runs using the fixed Hanoi scenario partition, the specified contamination process, and chlorine measurements at all 31 demand nodes without added sensor noise. Final control performance refers to the checkpoints selected by validation return after the nominal one-million-interaction budget was completed, following Section 6.1.2. Because the offline trajectories and evaluation cases are shared, the repeated runs describe training variability under these conditions. Transfer to other networks, operating periods, contamination processes, or measurement conditions remains untested.

The physical conclusions are limited to the modeled chlorine–arsenite reaction. A lower transported As(III) burden indicates improved oxidation in the simulator. It does not demonstrate total arsenic removal or drinking-water safety. The source-injection bounds and reward-based concentration control do not guarantee that nodal chlorine limits will always be satisfied. No event-time ceiling violations were observed in the tested cases, but this does not constitute a hard concentration constraint.

9.3 Future Work

Further comparisons should determine which initialization components are needed to preserve the treatment gains. Separating the effects of actor weights, observation normalization, and initial action standard deviation would clarify their roles in the early learning advantage. Evaluating the cloned policy before fine-tuning and the additional variants with the six physical metrics would also show how their return differences relate to treatment and dosing. More training seeds and independently generated offline datasets would test how strongly the findings depend on the current replication design. Changes to the reward or checkpoint-selection criterion could then be evaluated for their ability to produce a more favorable balance between post-clearance excess, chlorine use, arsenic burden, and earlier clearance.

Reducing preparation cost is another priority, since forecasting accounts for most of the recorded offline runtime. Fewer cross-fitting folds, a single forecaster with separate teacher-query scenarios, or reuse of compatible trajectories, forecasters, and demonstrations could reduce repeated preparation. The total simulator interactions and measured stage runtimes of these alternatives should be evaluated alongside final physical control performance. Such comparisons would show whether the treatment advantage is preserved with a smaller preparation budget and with reused artifacts instead of fresh preparation.

DAgger provides a way to investigate the difference between the states in prescribed offline trajectories and those visited by the cloned actor [41]. The actor would generate new simulator trajectories, and the teacher would label the query states visited along them. Combining these demonstrations and repeating behavior cloning would train the actor on histories produced by its own earlier decisions.

Any improvement would need to be weighed against the additional simulator interactions and MPC queries, which the current fixed-dataset design avoids.

The quality of teacher actions should also be assessed when the optimizer searches for sequences with high predicted returns. A teacher inspired by PETS could compare candidate actions using probabilistic ensembles and sampled trajectories to account for uncertainty [12]. This would require validating the uncertainty estimates for the partially observable multistep forecasting task. Comparing candidate forecasts with simulator responses could then test whether accounting for uncertainty reduces reliance on optimistic prediction errors and improves the demonstrations. However, this extension would not by itself guarantee that the optimizer cannot exploit model errors.

Further work could assess whether spatial and temporal representations improve forecasting and policy learning. A graph neural network could encode the network topology before the forecasting model or recurrent actor processes the measurements. Node-specific estimates of injection response or arrival time, conditioned on the time of day and recent dosing, could provide more explicit information about delayed effects. Under the current policy-input restrictions, these estimates would need to be derived from available measurements and dosing histories. They should be evaluated through final control performance as well as prediction accuracy. Improved forecasting accuracy alone does not establish better teacher actions or policy behavior.

Application beyond the evaluated setting requires tests on other topologies, seasonal demand periods, and contamination settings, as well as reduced sensor coverage, noise, missing measurements, and delays. Tests with calibrated transport and reaction models would help assess how much of the treatment balance depends on the synthetic chemistry used here. If concentration limits must be guaranteed, a separate safety mechanism would need to account for delayed network responses and model uncertainty. These evaluations would determine whether the observed gains in online learning and contamination treatment can support chlorination control beyond the simulated setting studied here.

References

[1] Talgat Abzanov and Erik Haße. 2025. *Three Reinforcement Learning Approaches for Optimal Chlorine Injection Control in Water Distribution Networks.* Retrieved from https://github.com/WaterFutures/AI-for-Drinking-Water-Chlorination-Challenge-IJCAI-25/blob/main/AI4DWC-25_paper_2.pdf

[2] Kwangjun Ahn, Zakaria Mhammedi, Horia Mania, Zhang-Wei Hong, and Ali Jadbabaie. 2023. Model Predictive Control via On-Policy Imitation Learning. In *Proceedings of the 5th Annual Learning for Dynamics and Control Conference (Proceedings of Machine Learning Research 211)*, 2023. 1493–1505. Retrieved from https://proceedings.mlr.press/v211/ahn23a.html

[3] André Artelt, Luca Hermes, Janine Strotherm, Barbara Hammer, Stelios G. Vrachimis, Marios S. Kyriakou, Demetrios G. Eliades, Marios M. Polycarpou, Sotirios Paraskevopoulos, Stefanos Vrochidis, Riccardo Taormina, Dragan Savic, and Phoebe Koundouri. 2025. 1st AI for Drinking Water Chlorination Challenge @ IJCAI-2025 (Competition/Challenge). Retrieved from https://github.com/WaterFutures/AI-for-Drinking-Water-Chlorination-Challenge-IJCAI-25

[4] André Artelt, Marios S. Kyriakou, Stelios G. Vrachimis, Demetrios G. Eliades, Barbara Hammer, and Marios M. Polycarpou. 2026. EPyT-Flow 0.17.0. Retrieved from https://github.com/WaterFutures/EPyT-Flow/releases/tag/v0.17.0

[5] André Artelt. 2026. EPANET-PLUS 0.3.1. Retrieved from https://github.com/WaterFutures/EPANET-PLUS/releases/tag/v0.3.1

[6] Samy Bengio, Oriol Vinyals, Navdeep Jaitly, and Noam Shazeer. 2015. Scheduled Sampling for Sequence Prediction with Recurrent Neural Networks. In *Advances in Neural Information Processing Systems 28*, 2015. 1171–1179. Retrieved from https://papers.nips.cc/paper/5956-scheduled-sampling-for-sequence-prediction-with-recurrent-neural-networks

[7] Christoph Bergmeir, Rob J. Hyndman, and Bonsoo Koo. 2018. A Note on the Validity of Cross-Validation for Evaluating Autoregressive Time Series Prediction. *Computational Statistics & Data Analysis* 120, (2018), 70–83. https://doi.org/10.1016/j.csda.2017.11.003

[8] Dominic L. Boccelli, Michael E. Tryby, James G. Uber, Lewis A. Rossman, Michael L. Zierolf, and Marios M. Polycarpou. 1998. Optimal Scheduling of Booster Disinfection in Water Distribution Systems. *Journal of Water Resources Planning and Management* 124, 2 (1998), 99–111. https://doi.org/10.1061/(ASCE)0733-9496(1998)124:2(99)

[9] Pieter-Tjerk de Boer, Dirk P. Kroese, Shie Mannor, and Reuven Y. Rubinstein. 2005. A Tutorial on the Cross-Entropy Method. *Annals of Operations Research* 134, 1 (2005), 19–67. https://doi.org/10.1007/s10479-005-5724-z

[10] Zdravko I. Botev, Dirk P. Kroese, Reuven Y. Rubinstein, and Pierre L'Ecuyer. 2013. The Cross-Entropy Method for Optimization. In *Handbook of Statistics, Vol. 31: Machine Learning: Theory and Applications*. Elsevier, 35–59. https://doi.org/10.1016/B978-0-444-53859-8.00003-5

[11] Kyunghyun Cho, Bart van Merriënboer, Caglar Gulcehre, Dzmitry Bahdanau, Fethi Bougares, Holger Schwenk, and Yoshua Bengio. 2014. Learning Phrase Representations Using RNN Encoder–Decoder for Statistical Machine Translation. In *Proceedings of the 2014 Conference on Empirical Methods in Natural Language Processing*, 2014. Association for Computational Linguistics, 1724–1734. https://doi.org/10.3115/v1/D14-1179

[12] Kurtland Chua, Roberto Calandra, Rowan McAllister, and Sergey Levine. 2018. Deep Reinforcement Learning in a Handful of Trials Using Probabilistic Dynamics Models. In *Advances in Neural Information Processing Systems 31*, 2018. Retrieved from https://proceedings.neurips.cc/paper_files/paper/2018/hash/3de568f8597b94bda53149c7d7f5958c-Abstract.html

[13] Michael C. Dodd, Ngoc Duy Vu, Adrian Ammann, Van Chieu Le, Reinhard Kissner, Hung Viet Pham, The Ha Cao, Michael Berg, and Urs von Gunten. 2006. Kinetics and Mechanistic Aspects of As(III) Oxidation by Aqueous Chlorine, Chloramines, and Ozone: Relevance to Drinking Water Treatment. *Environmental Science & Technology* 40, 10 (2006), 3285–3292. https://doi.org/10.1021/es0524999

[14] Rousslan Fernand Julien Dossa, Shengyi Huang, Santiago Ontañón, and Takashi Matsubara. 2021. An Empirical Investigation of Early Stopping Optimizations in Proximal Policy Optimization. *IEEE Access* 9, (2021), 117981–117992. https://doi.org/10.1109/ACCESS.2021.3106662

[15] Farama Foundation. 2025. Gymnasium 1.2.3. Retrieved from https://github.com/Farama-Foundation/Gymnasium/releases/tag/v1.2.3

[16] Noé Ferreira-Rodríguez, Sebastian Beggel, Juergen P. Geist, Vanessa Modesto, Martin Österling, Nicoletta Riccardi, Ronaldo Sousa, and Maria Urbańska. 2023. Freshwater Mussels as Sentinels for Safe Drinking Water Supply in Europe. *ACS ES&T Water* 3, 12 (2023), 3730–3735. https://doi.org/10.1021/acsestwater.3c00012

[17] Okitsugu Fujiwara and Do Ba Khang. 1990. A Two-Phase Decomposition Method for Optimal Design of Looped Water Distribution Networks. *Water Resources Research* 26, 4 (1990), 539–549. https://doi.org/10.1029/WR026i004p00539

[18] Felix A. Gers, Jürgen Schmidhuber, and Fred Cummins. 2000. Learning to Forget: Continual Prediction with LSTM. *Neural Computation* 12, 10 (2000), 2451–2471. https://doi.org/10.1162/089976600300015015

[19] Ian Goodfellow, Yoshua Bengio, and Aaron Courville. 2016. *Deep Learning*. MIT Press. Retrieved from https://www.deeplearningbook.org/

[20] Sepp Hochreiter and Jürgen Schmidhuber. 1997. Long Short-Term Memory. *Neural Computation* 9, 8 (1997), 1735–1780. https://doi.org/10.1162/neco.1997.9.8.1735

[21] Shengyi Huang, Rousslan Fernand Julien Dossa, Antonin Raffin, Anssi Kanervisto, and Weixun Wang. 2022. The 37 Implementation Details of Proximal Policy Optimization. Retrieved from https://iclr-blog-track.github.io/2022/03/25/ppo-implementation-details/

[22] Rob J. Hyndman and George Athanasopoulos. 2021. *Forecasting: Principles and Practice* (3rd ed.). OTexts. Retrieved from https://otexts.com/fpp3/

[23] Rob J. Hyndman and Anne B. Koehler. 2006. Another Look at Measures of Forecast Accuracy. *International Journal of Forecasting* 22, 4 (2006), 679–688. https://doi.org/10.1016/j.ijforecast.2006.03.001

[24] Leslie Pack Kaelbling, Michael L. Littman, and Anthony R. Cassandra. 1998. Planning and Acting in Partially Observable Stochastic Domains. *Artificial Intelligence* 101, 1–2 (1998), 99–134. https://doi.org/10.1016/S0004-3702(98)00023-X

[25] Christoph Kowalski and Jonas Schilin. 2025. *1st AI for Drinking Water Chlorination Challenge IJCAI 2025: Challenge Report.* Retrieved from https://github.com/WaterFutures/AI-for-Drinking-Water-Chlorination-Challenge-IJCAI-25/blob/main/AI4DWC-25_paper_1.pdf

[26] Solomon Kullback and Richard A. Leibler. 1951. On Information and Sufficiency. *The Annals of Mathematical Statistics* 22, 1 (1951), 79–86. https://doi.org/10.1214/aoms/1177729694

[27] Miejskie Przedsiębiorstwo Wodociągów i Kanalizacji w m.st. Warszawie S.A. 2019. Małże i ryby testerami wody. Retrieved from https://www.mpwik.com.pl/view/malze-i-ryby-testerami-wody

[28] Rivaaj Monsia, Daniel Young, Olivier Francon, and Risto Miikkulainen. 2026. Optimizing Chlorination in Water Distribution Systems via Surrogate-assisted Neuroevolution. In *Genetic and Evolutionary Computation Conference (GECCO '26)*, 2026. Association for Computing Machinery. https://doi.org/10.1145/3795095.3805109

[29] Anusha Nagabandi, Gregory Kahn, Ronald S. Fearing, and Sergey Levine. 2018. Neural Network Dynamics for Model-Based Deep Reinforcement Learning with Model-Free Fine-Tuning. In *2018 IEEE International Conference on Robotics and Automation*, 2018. IEEE, 7559–7566. https://doi.org/10.1109/ICRA.2018.8463189

[30] Ashvin Nair, Bob McGrew, Marcin Andrychowicz, Wojciech Zaremba, and Pieter Abbeel. 2018. Overcoming Exploration in Reinforcement Learning with Demonstrations. In *2018 IEEE International Conference on Robotics and Automation*, 2018. IEEE, 6292–6299. https://doi.org/10.1109/ICRA.2018.8463162

[31] National Research Council. 2006. *Drinking Water Distribution Systems: Assessing and Reducing Risks*. The National Academies Press, Washington, DC. https://doi.org/10.17226/11728

[32] Tianwei Ni, Benjamin Eysenbach, and Ruslan Salakhutdinov. 2022. Recurrent Model-Free RL Can Be a Strong Baseline for Many POMDPs. In *Proceedings of the 39th International Conference on Machine Learning (Proceedings of Machine Learning Research 162)*, 2022. 16691–16723. Retrieved from https://proceedings.mlr.press/v162/ni22a.html

[33] NumPy Developers. 2026. NumPy 2.4.4. Retrieved from https://github.com/numpy/numpy/releases/tag/v2.4.4

[34] Razvan Pascanu, Tomas Mikolov, and Yoshua Bengio. 2013. On the Difficulty of Training Recurrent Neural Networks. In *Proceedings of the 30th International Conference on Machine Learning (Proceedings of Machine Learning Research 28)*, 2013. 1310–1318. Retrieved from https://proceedings.mlr.press/v28/pascanu13.html

[35] Gal Perelman and Avi Ostfeld. 2025. Data Enabled Predictive Control for Water Distribution Systems Optimization. *Water Resources Research* 61, 4 (2025), e2024WR039059. https://doi.org/10.1029/2024WR039059

[36] Marco Pleines, Matthias Pallasch, Frank Zimmer, and Mike Preuss. 2022. Generalization, Mayhems and Limits in Recurrent Proximal Policy Optimization. *arXiv preprint arXiv:2205.11104* (2022). https://doi.org/10.48550/arXiv.2205.11104

[37] Python Software Foundation. 2025. Python 3.13.3. Retrieved from https://www.python.org/downloads/release/python-3133/

[38] PyTorch Team. 2026. PyTorch 2.11.0. Retrieved from https://github.com/pytorch/pytorch/releases/tag/v2.11.0

[39] Antonin Raffin, Ashley Hill, Adam Gleave, Anssi Kanervisto, Maximilian Ernestus, and Noah Dormann. 2021. Stable-Baselines3: Reliable Reinforcement Learning Implementations. *Journal of Machine Learning Research* 22, 268 (2021), 1–8. Retrieved from https://jmlr.org/papers/v22/20-1364.html

[40] James B. Rawlings, David Q. Mayne, and Moritz M. Diehl. 2020. *Model Predictive Control: Theory, Computation, and Design* (2nd ed.). Nob Hill Publishing. Retrieved from https://sites.engineering.ucsb.edu/~jbraw/mpc/

[41] Stéphane Ross, Geoffrey J. Gordon, and J. Andrew Bagnell. 2011. A Reduction of Imitation Learning and Structured Prediction to No-Regret Online Learning. In *Proceedings of the Fourteenth International Conference on Artificial Intelligence and Statistics (Proceedings of Machine Learning Research 15)*, 2011. 627–635. Retrieved from https://proceedings.mlr.press/v15/ross11a.html

[42] Lewis A. Rossman, Hyoungmin Woo, Michael Tryby, Feng Shang, Robert Janke, and Terranna Haxton. 2020. *EPANET 2.2 User Manual*. Retrieved from https://usepa.github.io/EPANET2.2/

[43] John Schulman, Philipp Moritz, Sergey Levine, Michael I. Jordan, and Pieter Abbeel. 2016. High-Dimensional Continuous Control Using Generalized Advantage Estimation. In *Proceedings of the 4th International Conference on Learning Representations*, 2016. Retrieved from https://arxiv.org/abs/1506.02438

[44] John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. 2017. Proximal Policy Optimization Algorithms. *arXiv preprint arXiv:1707.06347* (2017). https://doi.org/10.48550/arXiv.1707.06347

[45] Feng Shang, Lewis A. Rossman, and James G. Uber. 2023. *EPANET-MSX 2.0 User Manual*. Retrieved from https://nepis.epa.gov/Exe/ZyPURL.cgi?Dockey=P10188CK.TXT

[46] Stable-Baselines Team. 2026. SB3-Contrib 2.8.0. Retrieved from https://github.com/Stable-Baselines-Team/stable-baselines3-contrib/releases/tag/v2.8.0

[47] Stable-Baselines Team. 2026. Stable-Baselines3 2.8.0. Retrieved from https://github.com/DLR-RM/stable-baselines3/releases/tag/v2.8.0

[48] Ilya Sutskever, Oriol Vinyals, and Quoc V. Le. 2014. Sequence to Sequence Learning with Neural Networks. In *Advances in Neural Information Processing Systems 27*, 2014. 3104–3112. Retrieved from https://papers.nips.cc/paper/5346-sequence-to-sequence-learning-with-neural-networks

[49] Richard S. Sutton and Andrew G. Barto. 2018. *Reinforcement Learning: An Introduction* (2nd ed.). MIT Press, Cambridge, MA.

[50] U.S. Environmental Protection Agency. 2018. *Guidance for Responding to Drinking Water Contamination Incidents*. Retrieved from https://www.epa.gov/sites/default/files/2018-12/documents/responding_to_dw_contamination_incidents.pdf

[51] Stelios G. Vrachimis, Demetrios G. Eliades, and Marios M. Polycarpou. 2024. Disinfection Scheduling in Water Distribution Networks Considering Input Time-Delay Uncertainty. *Journal of Hydroinformatics* 26, 2 (2024), 386–396. https://doi.org/10.2166/hydro.2024.099

[52] Stelios G. Vrachimis, Marios S. Kyriakou, Demetrios G. Eliades, and Marios M. Polycarpou. 2018. LeakDB: A Benchmark Dataset for Leakage Diagnosis in Water Distribution Networks. In *Proceedings of the 1st International WDSA/CCWI Joint Conference*, 2018. Kingston, Ontario, Canada. https://doi.org/10.5281/zenodo.1313116

[53] Shen Wang, Ahmad F. Taha, and Ahmed A. Abokifa. 2021. How Effective Is Model Predictive Control in Real-Time Water Quality Regulation? State-Space Modeling and Scalable Control. *Water Resources Research* 57, 5 (2021), e2020WR027771. https://doi.org/10.1029/2020WR027771

[54] World Health Organization. 2023. *Water Safety Plan Manual: Step-by-Step Risk Management for Drinking-Water Suppliers* (2nd ed.). World Health Organization, Geneva. Retrieved from https://www.who.int/publications/i/item/9789240067691

[55] World Health Organization. 2026. *Guidelines for Drinking-Water Quality: Fourth Edition Incorporating the First, Second and Third Addenda*. World Health Organization, Geneva. Retrieved from https://www.who.int/publications/i/item/9789240121225

[56] Jason Yosinski, Jeff Clune, Yoshua Bengio, and Hod Lipson. 2014. How Transferable Are Features in Deep Neural Networks? In *Advances in Neural Information Processing Systems 27*, 2014. 3320–3328. Retrieved from https://proceedings.neurips.cc/paper_files/paper/2014/hash/532a2f85b6977104bc93f8580abbb330-Abstract.html

[57] Zakład Wodociągów i Kanalizacji Sp. z o.o. w Skawinie. 2023. Małże testerami wody. Retrieved from https://zwik.skawina.pl/news%2Cmalze_testerami_wody

Appendix

A Dataset Reproduction Artifacts

This appendix provides the additional information needed to reproduce the dataset described in the Dataset chapter.

For a base scenario with numeric index i, the demand-pattern seed is $1000 + i$. For network parameter uncertainties, a scenario-specific seed is derived deterministically from the master seed 42 and i using the first 32-bit output of NumPy's `SeedSequence`.

For the offline dataset, fixed arsenic schedules use seeds 6103, 6104, and 6105 in the training, validation, and test partitions, respectively. Chlorine profiles use seeds 7103, 7104, and 7105 in the same order. Within each base scenario, an index determines the individual schedule or profile.

For online policy training, seed 8128 is combined with the environment index, the scenario identifier, and the number of earlier episodes generated for that scenario to generate arsenic schedules following Section 5.4.2. For the fixed control-validation and control-test catalogs, seeds 8129 and 8130, respectively, are combined with the base scenario and the schedule index within that scenario.

A.1 Arsenic-Schedule Generation

The scheduler generates the events in each schedule sequentially. For each event, it samples the duration uniformly from the half-hour grid between 2 and 72 h, including both endpoints. It then samples a weekly start position uniformly from the 336 half-hour positions of a week.

For that start position, the scheduler selects uniformly among the eligible junction–week combinations described in Section 4.2. If none is eligible, it samples another weekly position. This procedure is intended to give each weekday the same probability of an arsenic event starting. The probability of an event being active nevertheless differs across weekdays.

The feasibility of an injection rate is determined from the scenario's hydraulic outflow time series. For a candidate event, let $Q_{\min}$ and $Q_{\max}$ be the minimum and maximum positive outflow at its source junction, in L/min. Let $a_{\min}$ = 0.1 mg/L be the lower concentration limit, $a_{\max}$ = 10 mg/L the upper concentration limit, and $\dot{m}_{\mathrm{As,max}}$ = 50 000 mg/min the per-event mass-rate limit. The lowest and highest feasible targets for the maximum immediate local As(III) contribution are

$$\begin{aligned} a_{\mathrm{peak,min}} &= a_{\min}\,\frac{Q_{\max}}{Q_{\min}}, \\ a_{\mathrm{peak,max}} &= \min\left(a_{\max}, \frac{\dot{m}_{\mathrm{As,max}}}{Q_{\min}}\right). \end{aligned} \tag{58}$$

A candidate is rejected when $a_{\mathrm{peak,min}} > a_{\mathrm{peak,max}}$. Otherwise, the target a^{*}_{peak} is sampled log-uniformly between $a_{\mathrm{peak,min}}$ and $a_{\mathrm{peak,max}}$. Its constant arsenic mass rate is

$$\dot{m}_{\mathrm{As}} = a^{*}_{\mathrm{peak}}\,Q_{\min}. \tag{59}$$

This choice accounts for the change in outflow during the event while satisfying the per-event limits given in Section 4.2.

A.2 Chlorine-Profile Generation

The order of the five profile segments is sampled uniformly from their possible permutations. A sampling band is chosen independently for the initial rate and each segment. The bands 0–200 000, 200 000–350 000, and 350 000–500 000 mg/min have selection probabilities of 70 %, 20 %, and 10 %,

respectively. Values and signal parameters are sampled uniformly within the selected band or the ranges given below.

The constant segment maintains a single injection rate. The periodic segment is a clipped sinusoid with a period of 2 to 240 reporting steps, a phase from 0 to 2π, a center within its band, and an amplitude up to half the band's width. The piecewise-constant segment samples a new rate every 12 steps. The ramp interpolates from the preceding segment's final value to a sampled endpoint. The random walk also starts at that final value and changes by up to one tenth of its band's width per step, with values clipped to the band.

At each segment boundary, a linear transition from the preceding value replaces the first quarter of the next segment with 25 % probability. A final limit on rate changes also applies between the initial zero-injection state and the first profile value.

B Simulation Configuration

This appendix specifies the hydraulic settings and reaction equations used for the simulations described in Section 4.3. It also defines the local source correction used to calculate transported As(III) for the evaluation.

EPyT-Flow's LeakDB loader sets pressure-driven demand for the Hanoi scenarios, with minimum and required pressures of 0 m and 0.1 m and a pressure exponent of 0.5. The loaded Hanoi network uses the Hazen–Williams head-loss relation [4]. These settings are used unchanged in the scenario files for this study. The required pressure is a synthetic study setting.

Let c and a denote the chlorine and As(III) concentrations in mg/L. The pipe reaction rates are

$$r_{\mathrm{Cl}} = -k_{\mathrm{B}}\, c - k_2\, \frac{c\, a}{M_{\mathrm{As}}}, \qquad r_{\mathrm{As}} = -k_2\, \frac{c\, a}{M_{\mathrm{Cl}}}. \tag{60}$$

The first-order chlorine-decay coefficient is $k_{\mathrm{B}} = 0.3\,\mathrm{d}^{-1}$, and the chlorine–As(III) reaction coefficient is $k_2 = 100\,000\,\mathrm{L/(mmol\,d)}$. The factor $M_{\mathrm{Cl}} = 70.9\,\mathrm{mg/mmol}$ converts the chlorine concentration from its $\mathrm{Cl_2}$-equivalent mass basis to a molar concentration. For As(III), reported as arsenic mass, the corresponding factor is $M_{\mathrm{As}} = 74.92\,\mathrm{mg/mmol}$. These factors make the coupled reaction terms consume equal molar amounts of chlorine and As(III). The rate coefficients are synthetic assumptions rather than values calibrated to measured concentrations.

B.1 Local Source Correction for Transported Arsenic

The correction described in Section 6.3.1 starts with the reported As(III) concentration $a^{\mathrm{raw}}_{v,t}$ at monitored node v and reporting step t. Let $\dot{m}_{\mathrm{As},v}(t)$ be the local injection rate in mg/min. The nodal outflow is

$$Q_v(t) = \max\bigl(0, d_v(t)\bigr) + \sum_{e \in \mathcal{E}_v} \max\Bigl(0, q_{v,e}(t)\Bigr). \tag{61}$$

Here $d_v(t)$ is nodal demand, $\mathcal{E}_v$ is the set of connected links, and $q_{v,e}(t)$ is positive for flow leaving the node. For the model's flow unit of cubic meters per hour, $\kappa = \frac{1000}{60}$ converts Q_v to L/min.

Because concentrations are reported at the end of each interval Δt, the estimated local contribution at step t uses the injection rate and outflow at the preceding step $t-1$.

$$\hat{a}^{\mathrm{local}}_{v,t} = \begin{cases} \frac{\dot{m}_{\mathrm{As},v}(t-1)}{\kappa\, Q_v(t-1)} & \text{if } Q_v(t-1) > \delta_Q \\ 0 & \text{otherwise} \end{cases}. \tag{62}$$

Here $\delta_Q = 10^{-9}$ in the hydraulic flow unit separates positive outflow from numerical zero. The initial sample at $t = 0$ has no preceding interval, so its estimated local contribution is zero.

An exception applies to a complete, contiguous source interval if no other monitored source is active and no other monitored node has detectable As(III) at any time during that interval. In this case, the raw source-node concentration is treated as entirely local. Source activity and As(III) detection use thresholds of 1×10^{-9} mg/min and 1×10^{-9} mg/L, respectively. In all other cases, the hydraulic estimate applies. With $a_{v,t}^{\mathrm{local}}$ denoting the selected local contribution, the transported concentration is

$$a_{v,t} = \max\left(0, a_{v,t}^{\mathrm{raw}} - a_{v,t}^{\mathrm{local}}\right). \tag{63}$$

C Network-Delay Diagnostic for Temporal-Window Selection

The diagnostic examines how long the effect of a single chlorine action remains visible. It compares two simulations of the same hydraulic scenario. Both use a constant injection rate of 100 000 mg/min, except that one raises the rate to 125 000 mg/min for a single 30-minute step. The resulting difference in chlorine concentration is recorded for 48 h at each of the 31 monitored junctions. No arsenic is injected.

The start times cover every hour of the day and every phase of the weekly demand cycle. For each combination of weekday and hour, the diagnostic considers three occurrences during days 2 to 22 of each of the 80 forecasting-training scenarios. Within each scenario, it selects the occurrence with the lowest source outflow over the following 24 h. It then selects the ten scenarios with the lowest outflow. This procedure yields 1680 pulse trials across 168 weekday-hour combinations. A scenario can be selected for more than one combination, so the trials cover 15 distinct scenarios.

At each junction, a positive difference greater than 0.0001 mg/L counts as a visible pulse effect. This threshold is an analysis rule, not a measured sensor limit. An effect is visible in 51 879 of the 52 080 junction-pulse cases. The other 201 cases have no sample above the threshold. The same cases are used to check the history and horizon chosen in Section 5.5.2.

For the history comparison, the diagnostic identifies the last sample above the threshold at each junction. A case has settled within a candidate window if this sample occurs by the end of that window. The mean response seen is the average, across visible cases, of the share of each case's 48-hour above-threshold response recorded within the window. Table 14 also reports the number of visible junction-pulse cases with a later sample above the threshold and the share of pulse trials in which all 31 junctions have settled. Junctions without a visible effect count as settled for this last measure.

At 12 h, 168 visible junction-pulse cases still have a later effect, and all junctions have settled in 90.83 % of pulse trials. At 20 h, at least one junction has a later effect in 13 of the 1680 pulse trials. At 24 h, only one trial still has a later effect, with its last visible sample at 29.5 h. The mean response seen rounds to 100 % because very little of this case's effect remains after 24 h. The reduction in late effects and coverage of a complete daily cycle support 24 h as a practical history length under the study's feasibility constraints. They do not establish that this history length gives the best forecasts.

The horizon comparison examines when the same pulse effect begins, reaches its peak, and settles. For each visible junction-pulse case, the effect begins at the first sample above the threshold. The peak is the largest positive concentration difference during the 48-hour observation. The effect has settled by a candidate horizon if no later sample within those 48 h exceeds the threshold. Table 15 reports the share of cases that have reached each point by the candidate horizon.

By the selected eight-hour horizon, the effect has begun in 99.85 % of visible cases, reached its peak in 99.70 %, and settled in 98.41 %. At four hours, 90.30 % have settled. Longer horizons cover only small additional shares of cases while requiring longer forecasts and evaluation of more predicted steps

for each MPC plan. These observations support eight hours as a practical choice under the study's feasibility constraints. They do not establish an empirically optimal horizon, since the diagnostic does not measure forecasting or control performance.

Table 14: Response timing by candidate history length. The mean response seen is the average, across visible cases, of the share of each case's 48-hour response above 0.0001 mg/L recorded within the window. A trial is fully settled when none of its 31 junctions has a later sample above this threshold within the 48-hour observation. The mean and trial share are percentages. Values are rounded to at most two decimal places.

History (h)	Mean response seen (%)	Cases still visible	Trials fully settled (%)
12	99.94	168	90.83
16	99.99	31	98.15
20	99.99	13	99.23
24	100	1	99.94
28	100	1	99.94
32	100	0	100

Table 15: Timing of the pulse response. Values are percentages of 51 879 visible junction-pulse cases, rounded to at most two decimal places. A pulse effect is visible above 0.0001 mg/L and has settled by a horizon if no later sample within the 48-hour observation exceeds this threshold.

Horizon (h)	Effect begun (%)	Peak reached (%)	Effect settled (%)
4	97.32	96.10	90.30
6	99.53	99.14	96.57
8	99.85	99.70	98.41
10	99.97	99.91	99.20
12	100	99.98	99.68
14	100	99.99	99.86
16	100	99.99	99.94

D Policy and Reward Parameter Selection

This appendix documents the search summarized in Section 5.4.3 and the subsequent reward-parameter comparison. All candidates used ordinarily initialized recurrent policies and training seed 42, with the architecture fixed throughout the search. The search first varied policy-training parameters while keeping the reward fixed. It then varied selected reward parameters with the chosen PPO configuration held fixed.

Each candidate was evaluated on the same 50 validation cases, comprising five fixed arsenic schedules for each of ten validation scenarios. Validation took place at intervals of approximately 50 000 training interactions. The budgets reported below are nominal totals across the four parallel environments. Because runs completed whole rollouts, their actual interaction counts could slightly exceed these budgets. Longer runs started from ordinary initialization rather than continuing the shorter screening checkpoints. The test scenarios were excluded from parameter selection.

D.1 Policy-Training Parameters

The initial reference used a learning rate of 1×10^{-4}, 1024 rollout steps per environment, a minibatch size of 512, and an entropy coefficient of zero. The discount factor was $\gamma = 0.99$ and the GAE parameter was $\lambda = 0.95$. The remaining settings were held at the values in Table 6. The first stage compared this reference with eight single-parameter variants, each trained for 500 000 interactions. Table 16 lists the alternatives, with all unlisted parameters unchanged.

The second stage compared selected settings and targeted combinations using the budgets in Table 17. It included a reference run at one million interactions so that the longer candidates could be assessed at the same training budget.

After selecting 2048 rollout steps per environment and a minibatch size of 512, the third stage compared all six combinations of $\gamma \in \{0.98, 0.99, 0.995\}$ and $\lambda \in \{0.95, 0.98\}$ at a budget of one million interactions. The five new combinations were compared with the existing run using $\gamma = 0.99$ and $\lambda = 0.95$. The final configuration kept these discount and GAE parameters together with the learning rate of 1×10^{-4} and zero entropy coefficient.

Table 16: Single-parameter alternatives in the initial PPO screening.

Parameter	Reference	Alternatives
Learning rate	1×10^{-4}	5×10^{-5}, 3×10^{-4}
Rollout steps per environment	1024	512, 2048
Minibatch size	512	256, 1024
Entropy coefficient	0	1×10^{-3}, 1×10^{-2}

Table 17: Follow-up PPO runs. Interaction counts refer to training. Changes are relative to the initial reference in Table 16.

Configuration change	Nominal interactions
None	1 000 000
2048 rollout steps per environment	1 000 000
Minibatch size 256	1 000 000
2048 rollout steps and minibatch size 256	1 000 000
Learning rate 3×10^{-4}	1 000 000
2048 rollout steps and learning rate 3×10^{-4}	500 000

D.2 Reward Parameters

The reward comparison started from the parameters in Table 4 and used the selected PPO configuration in Table 6. Four additional candidates each changed one reward parameter, as shown in Table 18. Each was trained for one million interactions and compared with the existing run at the same budget using the reference reward. The target, concentration thresholds, reference scales, movement and action-change weights, and startup timings remained fixed.

The comparison supported keeping the four-step hold, twelve-step release, and minimum upper-tracking factor of 1×10^{-4} as a compromise between contamination response and normal-operation control.

Table 18: Reward-parameter alternatives, each applied separately. Each control step corresponds to 30 min.

Parameter	Reference	Alternatives
Hold period	4 steps / 2 h	8 steps / 4 h
Release duration	12 steps / 6 h	6 steps / 3 h
Minimum upper-tracking factor	1×10^{-4}	1×10^{-3}, 1×10^{-5}

D.3 Selection Criteria and Supporting Diagnostic

Selection considered the validation learning curves, final performance, and averages over the last three evaluations. Validation return was assessed alongside physical control performance, including arsenic burden and chlorine excess during normal operation as defined in Evaluation. The reward variants were compared using physical control performance because their returns correspond to different reward weightings.

An additional diagnostic measured chlorine shortfalls during reference-event operation. Using the notation and reference windows in Section 6.3.1, it is defined for each validation case as

$$f^{\mathrm{E}}_{\mathrm{low}} = \frac{\sum_{t \in \mathcal{T}_{\mathrm{M}}} \sum_{v=1}^{N} I^{\mathrm{E}}(t)\, \mathbf{1}\left[c_{v,t} < L\right]}{N \sum_{t \in \mathcal{T}_{\mathrm{M}}} I^{\mathrm{E}}(t)}. \tag{64}$$

Here, $L = 0.2$ mg/L is the lower chlorine bound, and $\mathcal{T}_{\mathrm{M}}$ contains the timesteps of the main evaluation window. The diagnostic is the fraction of monitored node–time observations below this threshold during reference-event periods. Values equal to the threshold do not count as shortfalls. A case without eligible observations is recorded as not applicable. Applicable case values are averaged first over schedules within each scenario and then over scenarios. The clearance tail is excluded. This supporting diagnostic differs from the primary chlorine-shortfall metric in Equation 49, which measures shortfalls during reference-normal operation.

These comparisons guided the choice of practical settings for the principal experiment. The limited candidate sets and single training seed do not establish a global optimum or robustness across training seeds.

E Numerical Results

This appendix provides the numerical values underlying the learning curves, selected policy checkpoints, physical control comparison, and forecasting results reported in Results. The tables follow the aggregation defined in Evaluation.

E.1 Validation Return and Selected Policy Checkpoints

Table 19 lists mean validation returns and sample standard deviations across the five training seeds for the baseline approach and pretrained approach. Mean paired differences are calculated as pretrained minus baseline. Alongside each nominal interaction threshold, the table reports the actual interaction count of the evaluated checkpoint.

For each approach and training seed, the checkpoint with the highest mean validation return is selected following Section 6.1.2. Table 20 lists the selected nominal interaction counts and corresponding mean validation returns.

Table 19: Validation return at all nominal checkpoints. The sample standard deviation is calculated across five seeds. Paired differences are calculated as pretrained minus baseline. Rounding follows Section 6.4.

Interactions		Baseline	Pretrained	Difference
Nominal	Actual			
0	0	−2 800 000 ± 4 700 000	−95 000 ± 58 000	2 700 000
50 000	57 344	−870 000 ± 180 000	−34 000 ± 10 000	835 000
100 000	106 496	−310 000 ± 180 000	−22 700 ± 7000	283 000
150 000	155 648	−152 000 ± 82 000	−17 200 ± 3000	134 000
200 000	204 800	−86 000 ± 40 000	−14 800 ± 3100	71 300
250 000	253 952	−62 000 ± 24 000	−14 100 ± 2900	47 900
300 000	303 104	−56 000 ± 13 000	−12 100 ± 1100	43 900
350 000	352 256	−54 100 ± 5100	−11 550 ± 720	42 600
400 000	401 408	−48 600 ± 3800	−10 920 ± 710	37 700
450 000	450 560	−45 000 ± 8000	−11 600 ± 890	33 400
500 000	507 904	−42 000 ± 13 000	−10 320 ± 530	31 400
550 000	557 056	−42 000 ± 13 000	−10 580 ± 620	31 100
600 000	606 208	−41 000 ± 13 000	−11 200 ± 1100	30 200
650 000	655 360	−42 000 ± 11 000	−11 600 ± 2200	30 700
700 000	704 512	−40 000 ± 16 000	−10 510 ± 710	29 300
750 000	753 664	−36 000 ± 17 000	−10 960 ± 800	25 300
800 000	802 816	−31 000 ± 15 000	−10 740 ± 980	20 300
850 000	851 968	−21 300 ± 6200	−10 640 ± 670	10 700
900 000	901 120	−16 500 ± 3800	−11 010 ± 750	5450
950 000	950 272	−15 000 ± 2800	−10 990 ± 750	3990
1 000 000	1 007 616	−13 100 ± 1800	−11 010 ± 750	2090

Table 20: Nominal online interaction counts and mean validation returns at the selected checkpoints for each training seed. Returns follow the scenario weighting defined in Section 6.1.2 and the rounding in Section 6.4.

Training seed	Baseline		Pretrained	
	Interactions	Return	Interactions	Return
17	1 000 000	−12 700	500 000	−10 600
42	950 000	−12 700	950 000	−9740
123	1 000 000	−12 700	650 000	−9730
1337	1 000 000	−11 300	550 000	−10 600
71026	1 000 000	−16 100	800 000	−9490

E.2 Paired Physical Metric Differences

Table 21 lists the individual paired differences underlying the aggregate test results in Table 11. For each physical metric, the selected baseline policy's value is subtracted from the selected pretrained policy's value within the same training seed.

Table 21: Paired test-metric differences for the individual training seeds, calculated as the selected pretrained policy's value minus the selected baseline policy's value in the units shown. The normal-operation chlorine shortfall and excess fractions are $f^{\mathrm{N}}_{\mathrm{low}}$ and $f^{\mathrm{N}}_{\mathrm{high}}$, respectively. The remaining metrics are transported arsenic burden B^{E}, mean arsenic clearance lead Δ_{clear} relative to the zero-chlorine reference, the event-time ceiling-violation fraction $f^{\mathrm{E}}_{\mathrm{cap}}$, and chlorine use U_{eval}. pp = percentage points.

Metric	Seed 17	Seed 42	Seed 123	Seed 1337	Seed 71026
$f^{\mathrm{N}}_{\mathrm{low}}$ (pp)	0.0001	0.0036	−0.0013	0.0110	0.0192
$f^{\mathrm{N}}_{\mathrm{high}}$ (pp)	0.027	0.093	0.151	0.167	0.104
B^{E} (mg h/L)	−11.2	−9.1	−9.1	−1.4	−10.6
Δ_{clear} (h)	0.63	0.64	0.67	0.02	0.10
$f^{\mathrm{E}}_{\mathrm{cap}}$ (pp)	0.000	0.000	0.000	0.000	0.000
U_{eval} (t)	0.050	0.009	0.074	0.023	0.059

E.3 Forecast Errors

Figure 15 and Table 22 report MAE and RMSE over the complete eight-hour horizon for all 25 fold-specific models. Each model's errors give equal weight to the ten forecasting test scenarios. In the table, the mean row for each training seed averages the errors of its five models.

The full-horizon errors are complemented by Table 23, which lists the values shown in Figure 14 for each half-hour horizon step. At each step, the five fold-specific model errors are averaged within each training seed. The reported mean and sample standard deviation are then calculated across the five seed means.

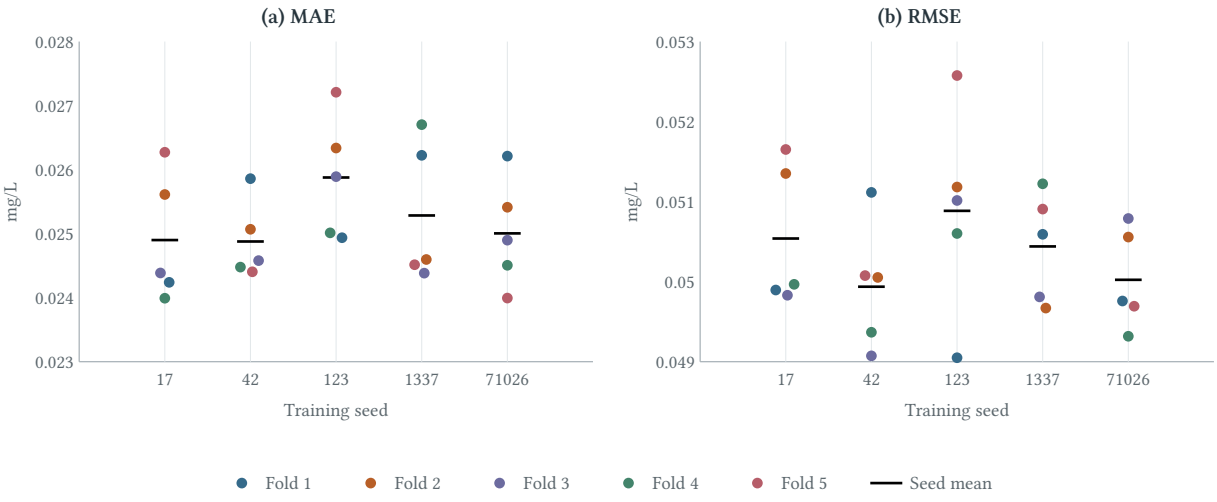


Figure 15: Chlorine-forecast errors over the full eight-hour horizon by training seed, with (a) MAE and (b) RMSE. Errors are evaluated on the ten forecasting test scenarios under prescribed injection profiles. Colors identify the five folds consistently across both panels. Black horizontal bars mark seed means.

Table 22: Forecast errors over the full eight-hour horizon by training seed and fold. Errors are evaluated on the ten forecasting test scenarios under prescribed injection profiles, expressed in mg/L, and rounded to five decimal places.

Training seed	Fold	MAE (mg/L)	RMSE (mg/L)
17	1	0.02424	0.04990
17	2	0.02561	0.05135
17	3	0.02439	0.04983
17	4	0.02399	0.04997
17	5	0.02627	0.05165
17	Mean	0.02490	0.05054
42	1	0.02586	0.05112
42	2	0.02507	0.05005
42	3	0.02458	0.04907
42	4	0.02448	0.04937
42	5	0.02440	0.05008
42	Mean	0.02488	0.04994
123	1	0.02494	0.04905
123	2	0.02634	0.05118
123	3	0.02589	0.05101
123	4	0.02501	0.05060
123	5	0.02721	0.05258
123	Mean	0.02588	0.05088
1337	1	0.02622	0.05059
1337	2	0.02460	0.04967
1337	3	0.02438	0.04981
1337	4	0.02670	0.05122
1337	5	0.02451	0.05091
1337	Mean	0.02528	0.05044
71026	1	0.02621	0.04976
71026	2	0.02541	0.05056
71026	3	0.02490	0.05079
71026	4	0.02451	0.04932
71026	5	0.02399	0.04969
71026	Mean	0.02500	0.05002

Table 23: Forecast errors at each half-hour horizon step in mg/L, evaluated on the ten forecasting test scenarios under prescribed injection profiles. Means and sample standard deviations across seed means are rounded to five decimal places.

Step	Horizon (h)	MAE (mg/L)	RMSE (mg/L)
1	0.5	0.01954 ± 0.00050	0.03576 ± 0.00076
2	1.0	0.02090 ± 0.00051	0.03894 ± 0.00067
3	1.5	0.02249 ± 0.00048	0.04243 ± 0.00061
4	2.0	0.02360 ± 0.00047	0.04510 ± 0.00057
5	2.5	0.02445 ± 0.00045	0.04738 ± 0.00051
6	3.0	0.02515 ± 0.00043	0.04923 ± 0.00048
7	3.5	0.02561 ± 0.00042	0.05057 ± 0.00044
8	4.0	0.02594 ± 0.00042	0.05162 ± 0.00042
9	4.5	0.02621 ± 0.00041	0.05251 ± 0.00040
10	5.0	0.02645 ± 0.00041	0.05331 ± 0.00037
11	5.5	0.02667 ± 0.00040	0.05402 ± 0.00034
12	6.0	0.02686 ± 0.00039	0.05465 ± 0.00031
13	6.5	0.02703 ± 0.00039	0.05521 ± 0.00028
14	7.0	0.02720 ± 0.00038	0.05576 ± 0.00026
15	7.5	0.02737 ± 0.00037	0.05626 ± 0.00024
16	8.0	0.02754 ± 0.00036	0.05674 ± 0.00022

F Submission Repository

The submission repository contains the final thesis, its sources, and the code for dataset generation, training, and evaluation. It also provides the paper's numerical inputs, supporting tools and tests, and reproduction instructions.

`https://github.com/llamadeus/master-thesis`

Eigenständigkeitserklärung

Hiermit erkläre ich, dass ich die vorliegende Masterarbeit selbstständig verfasst und gelieferte Datensätze, Zeichnungen, Skizzen und graphische Darstellungen selbstständig erstellt habe. Ich habe keine anderen Quellen als die angegebenen benutzt und habe die Stellen der Arbeit, die anderen Werken entnommen sind – einschließlich verwendeter Tabellen und Abbildungen – in jedem Fall unter Angabe der Quelle als Entlehnung kenntlich gemacht.

Bielefeld, den _____ _____

(Unterschrift)